

CHATBOT AZ-3016

"So, this whole thing is a **chatbot project** I did for my **SkillUpOnline Module AZ-3016**, which is all about **Microsoft Azure Services**. The big goal was to get the chatbot up and running, or even just access it, right from my computer's terminal (that's like using **VS Code**) by hooking it up with an **Azure Functions app**. Now, I hit some **technical snags** and couldn't quite finish everything, which was a bit of a bummer. But even with those limitations, I still managed to get the **Functions app deployed** in Azure and even in VS Code! So, I definitely made some great progress!"

Problem Statement :

Project Title: Simple FAQ Copilot (CLI Version)

Project Overview

The Simple FAQ Copilot is an AI-powered tool that answers frequently asked questions based on a predefined dataset. Users will interact with the copilot through a command-line interface.

Project Components

1. Define Requirements

Target Audience: Users seeking quick answers to common questions.

Features:

- Users can type questions and receive answers.
- Ability to suggest new questions for future training (optional).
- User feedback mechanism for improving accuracy (optional).

2. Technology Stack

- **Programming Language:** Python (or any language of your choice).
- **Backend:** Azure Functions (for serverless architecture).
- **AI Model:** Azure AI Studio for training the custom FAQ model.
- **Database:** Optional for storing user feedback.

Implementation Steps

Step 1: Set Up Azure AI Studio

- 1 . **Create an Azure Account:**
 - Sign up for an Azure account if you don't have one.
- 2 . **Navigate to Azure AI Studio:**
 - Go to Azure AI Studio and create a new project for your FAQ Copilot.
- 3 . **Gather Training Data:**
 - Create a dataset of frequently asked questions and their corresponding answers. For example:

Question	Answer
What are your operating hours?	We are open from 9 AM to 5 PM, Monday to Friday.
How can I contact support?	You can reach us at support@example.com.
What is your return policy?	You can return items within 30 days of purchase.

- 4 . **Train the Model:**

- Use the dataset to train your FAQ model in Azure AI Studio.
- Define the model parameters, such as language and context.
- Test the model to ensure it generates relevant answers.

Step 2: Build the Backend with Azure Functions

- 1 . **Create an Azure Function:**

- Set up an Azure Function that will serve as the backend API for your application.

- This function will accept user input (questions) and call the Azure AI model to get answers.

```
import azure.functions as func
import requests
import json

def main(req: func.HttpRequest) -> func.HttpResponse:
    try:
        req_body = req.get_json()
        user_question = req_body.get('question')

        # Call the Azure AI model
        response = requests.post(
            'YOUR_AZURE_AI_MODEL_ENDPOINT',
            headers={'Authorization': 'Bearer YOUR_API_KEY'},
            json={'input': user_question}
        )
        answer = response.json().get('answer')
        return func.HttpResponse(json.dumps({'answer': answer}), status_code=200)

    except ValueError:
        return func.HttpResponse("Invalid input", status_code=400)
```

Step 3: Create the Command-Line Interface

1 . Create a Python Script:

Create a Python script (e.g., `faq_copilot.py`) that will serve as the command-line interface.

```
import requests
import json
def get_answer(question):
    try:
        response = requests.post(
            'YOUR_AZURE_FUNCTION_URL',
            headers={'Content-Type': 'application/json'},
            json={'question': question}
        )
        if response.status_code == 200:
            answer = response.json().get('answer')
            return answer
        else:
            return "Error: Unable to get an answer."
    except Exception as e:
        return f"Error: {str(e)}"
def main():
    print("Welcome to the Simple FAQ Copilot!")
    print("Type 'exit' to quit.")
    while True:
        user_question = input("You: ")
        if user_question.lower() == 'exit':
            print("Goodbye!")
            break
        answer = get_answer(user_question)
        print(f"Copilot: {answer}")
if __name__ == "__main__":
    main()
```

Running the Project

1 . Install Required Libraries:

- Ensure you have the **requests** library installed. You can install it using pip:
pip install requests

Run the Script:

- Execute the Python script from the command line:
python faq_copilot.py

Interact with the Copilot:

- Type your questions in the command line and receive answers from

My Approach to the Solution:

Project Rundown

Alright, so even though I couldn't quite finish the whole project, I still got a ton of important stuff done and set up! My main goal was building this smart chatbot, and for that, I grabbed this cool **HDFC Banking FAQs dataset** from Kaggle -> link [<https://www.kaggle.com/datasets/abbbhishekkk/faq-datasets-for-chatbot-training>] for which the owner of the dataset is ["Owner": "<https://www.kaggle.com/abbbhishekkk>"]. The whole thing was totally adaptive, I hope, Just kept pushing forward, even when things got tough!

I. Getting My Data Ready!

- **Setting Up My Spot:** First off, I set up a special folder.
- **Grabbing the Goods:** Then, I got the **HDFC Banking FAQ dataset** from Kaggle. Picked it because it was perfect for my banking chatbot idea.
- **File Format Swap!** After downloading and **unzipping** it, turns out the raw data was just plain text but the content was json array. But I needed it in **.json** format for my azure openai tools, especially my **vector databases**. So, I had to convert it. This wasn't just a quick edit; it was like translating it into a language my tools could actually use! Super crucial!

II. Setting Up My Azure Cloud Stuff!

- **Keeping It Local:** Big rule I stuck to: **all my Azure cloud stuff had to be in the same spot**, geographically.
- Like, 'Sweden Central' was my chosen area. Why? Well, it just makes everything run way smoother and faster, cuts down on annoying lag, and helps with data rules (super important for banking!).
- **Azure Portal Playtime:** I hopped onto the **Azure** and whipped up a **custom Resource Group**. Yep, 'Sweden Central' was the home base again, keeping it consistent!
 - Next, I set up an **Azure Storage Account**. This is where all my project's persistent data would just hang out. I specifically used its '**Blob Storage**' feature, which is great for unstructured files like my JSON data. Again, right there in the same Resource Group and region for top performance.
 - Inside that new Storage Account, I made a **dedicated container**. Kind of like a sub-folder in my storage. And guess what? I securely **uploaded my .json dataset** right into a folder in there. Now, all my other Azure services could totally get to that valuable data. Sweet!

III. Getting Azure OpenAI Ready!

- **Firing Up OpenAI:** Okay, big moves now! Next step was getting the **Azure OpenAI Service resource** deployed in Azure. I went with Azure OpenAI text embedding model to vectorize my Jason array data stored as a blob storage.
 - After that was set up, I bounced over to the AI Foundry.

- And in there, I strategically deployed a **Text Embedding model**.
- They take normal text and turn it into these super clever numbers (**vectors**) that actually capture what the text *means*.
- This is key for **smart, semantic search** – meaning it finds stuff not just by keywords, but by truly understanding what your question means! I made sure it was linked to my Azure Storage Account so it could grab all the data for the embedding process.

IV. Setting Up Azure AI Search!

- **Making My Data Super Searchable!** My next big mission was setting up the **Azure AI Search service**. This thing is a total powerhouse for searching! It hooks up perfectly with those OpenAI text embeddings, giving you wicked fast and super relevant search results from your data.
 - I used its '**Import and vectorize data**' feature. This whole process is pretty slick: it automatically pulls in the data from my Azure Blob Storage container, then uses the Text Embedding model to create those special numerical codes (vectors). Finally, it builds a powerful **search index** that's awesome for both regular keyword searches and those smart semantic ones. I carefully picked the **right settings** here to make sure the search was speedy and on point.
 - Super important: I **double-checked all the extracted fields** (like 'question' and 'answer') to make sure they were mapped correctly for the chatbot. Seriously vital for the bot finding the right info! Also, I **set up semantic configurations** for each data chunk. This just means I helped the search service really *get* the deep meaning of questions and documents. It leads to way better, super relevant search results, and I totally made sure everything was checked and configured for peak performance!

V. Deploying and Testing Our Chatbot!

- **Building the Bot!** Back in the AI Studio, I deployed the **GPT-4o model** – it's amazing for smart, relevant answers! This super advanced AI model was my top pick because it's really good at chatting, understands complex stuff, and just gives solid responses. Perfect for a fancy banking FAQ chatbot!
- **Testing & Tweaking Time:** Then, it was time to hit the **Chat Playground**. This is my dedicated spot to really put the chatbot through its paces, making sure it's intuitive and performs like a champ. Testing is absolutely crucial, you know, to make sure the bot actually answers user questions effectively and gracefully!
 - This serious testing involved **trying it all sorts of test questions** (just like real people would ask!). I also made **small, iterative tweaks to its settings** – like 'temperature' (which makes it more creative or random), 'top_p' (how diverse its responses are), and 'max_tokens' (how long its answers can be). This fine-tuning helps get the bot's output just right. Oh, and also the **custom prompt** I added! That's a super important little instruction manual that tells the AI *how* to act – like being a super helpful banking assistant – and how its answers should look. This step totally supercharges the user experience and keeps the bot on the straight and narrow for the project's goals!

VI. Getting Our App Running Locally and Serverless!

- **Saving My Important Stuff:** For safekeeping (and for any future fixes), I made sure to **securely save all the crucial bits** like endpoints and API keys in a safe spot. Gotta keep that security tight and make it easy to grab them later!
- **VS Code Prep:** I kicked off the development in **Visual Studio Code (VS Code)** – it's just the best! I installed the **Azure Functions extension**, which made working with serverless functions a total breeze. And I got my Azure account connected right in VS Code, so everything just talked to each other smoothly.
- **Azure Function App Go!** Then, I deployed the **Function App through the Azure Portal**. Function Apps are awesome because they're **serverless**, meaning you just write code and Azure handles all the boring server

stuff for you! And yep, it was deployed in the **same region and Resource Group** as everything else – consistent architecture!

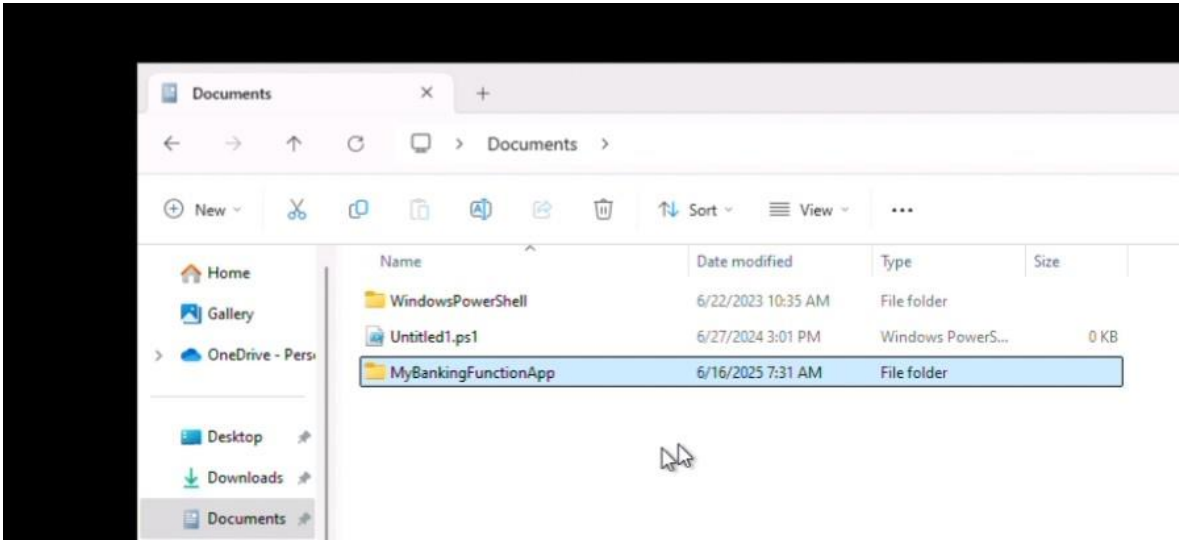
- Once it was out there, I jumped back to VS Code and **started a new project using the Function App template**. This just gives you a head start with the right project structure. Then, I smoothly **connected this local project to the Function App** I'd just deployed in the cloud, setting up a direct line for future code pushes.
- **Local Test Drive!** Before sending it fully to the cloud, I did some **serious testing of the Function App right there in VS Code**. This was super important! I used those API keys and endpoints to pretend I was a real user, making sure everything worked exactly as it should in my own little test zone.
 - This testing even involved writing a **simple Python script** that would **call the Function's web address (URL) and use the API key**. This script was my ultimate **end-to-end test**, confirming the API calls went through, responses came back right, and the whole serverless setup was humming along perfectly. It totally showed my app was ready to go, all serverless and prepped for the big leagues!

Personal Solution

1. Getting Data Ready.

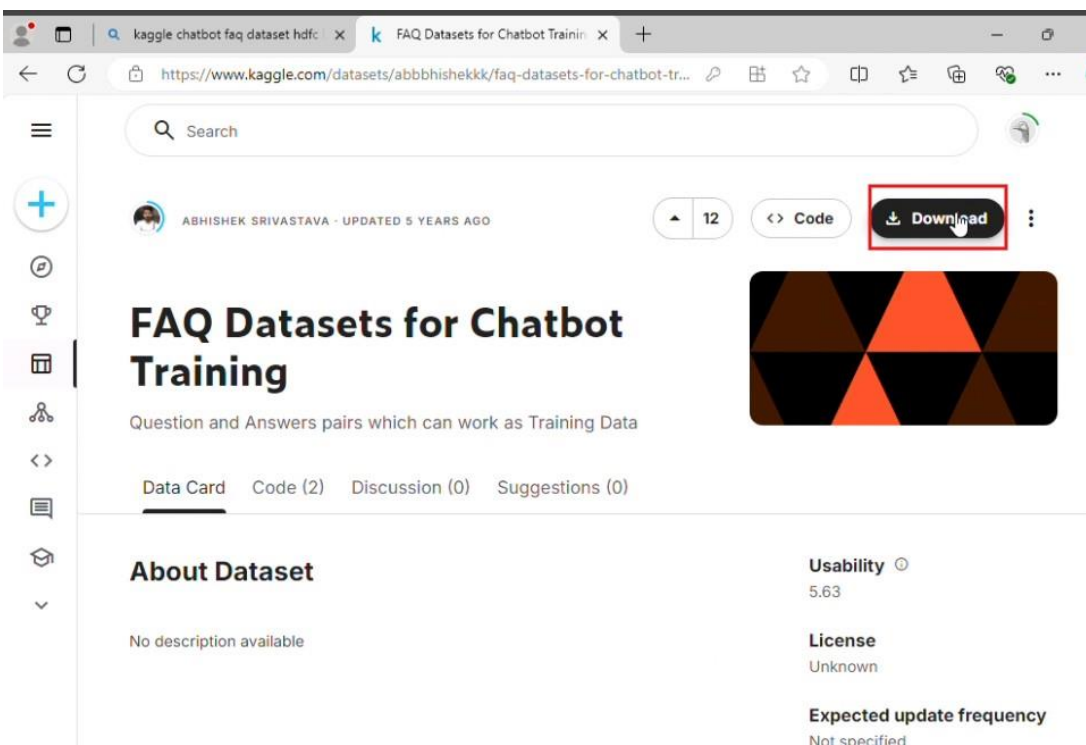
- Step 1 :

Create a new empty directory *"MyBankingFunctionApp"*.



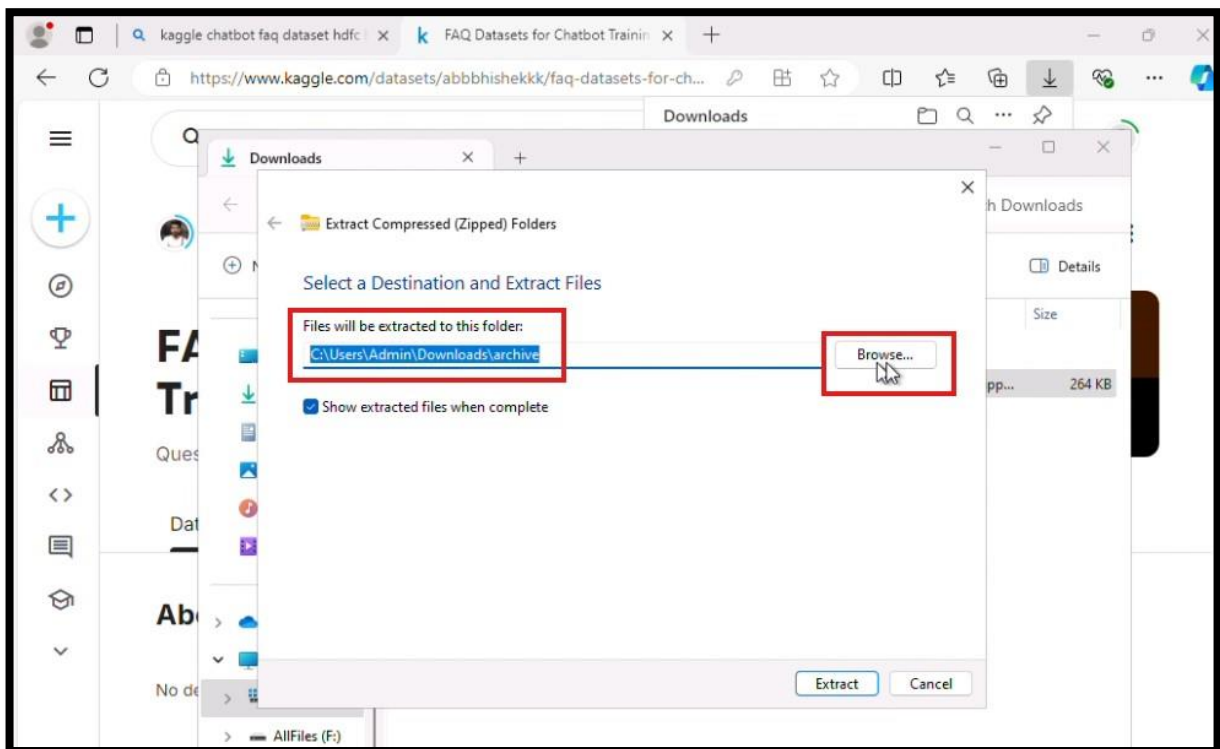
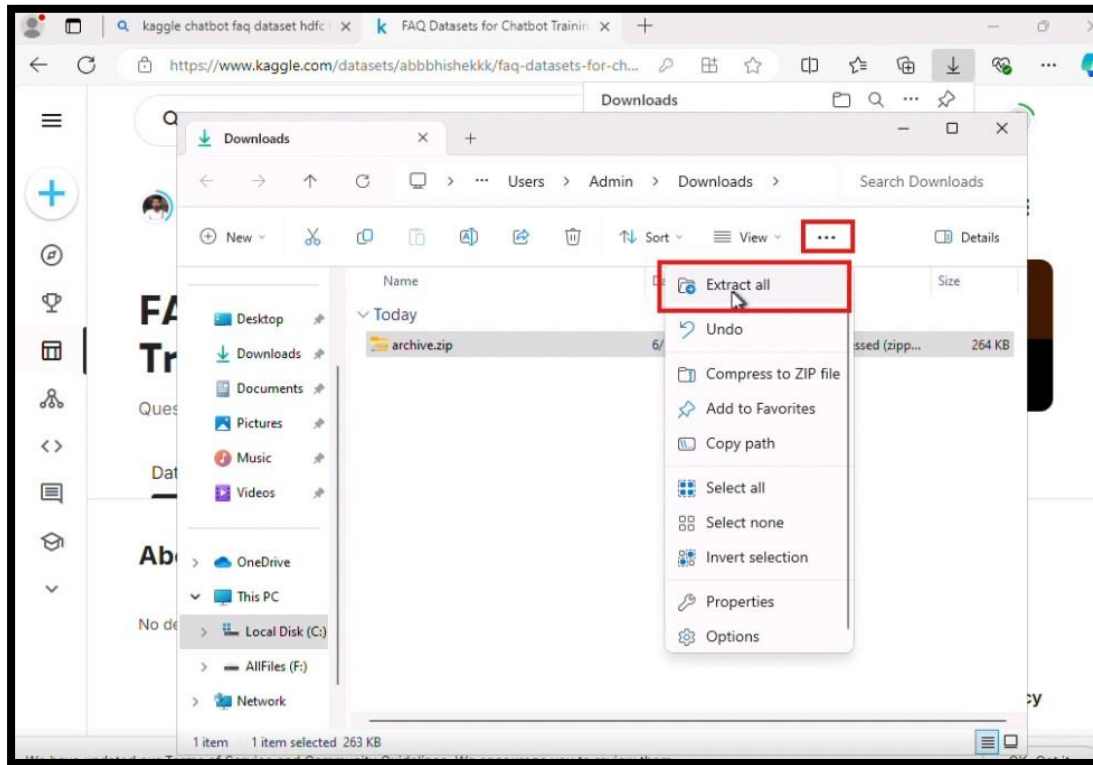
- Step 2 :

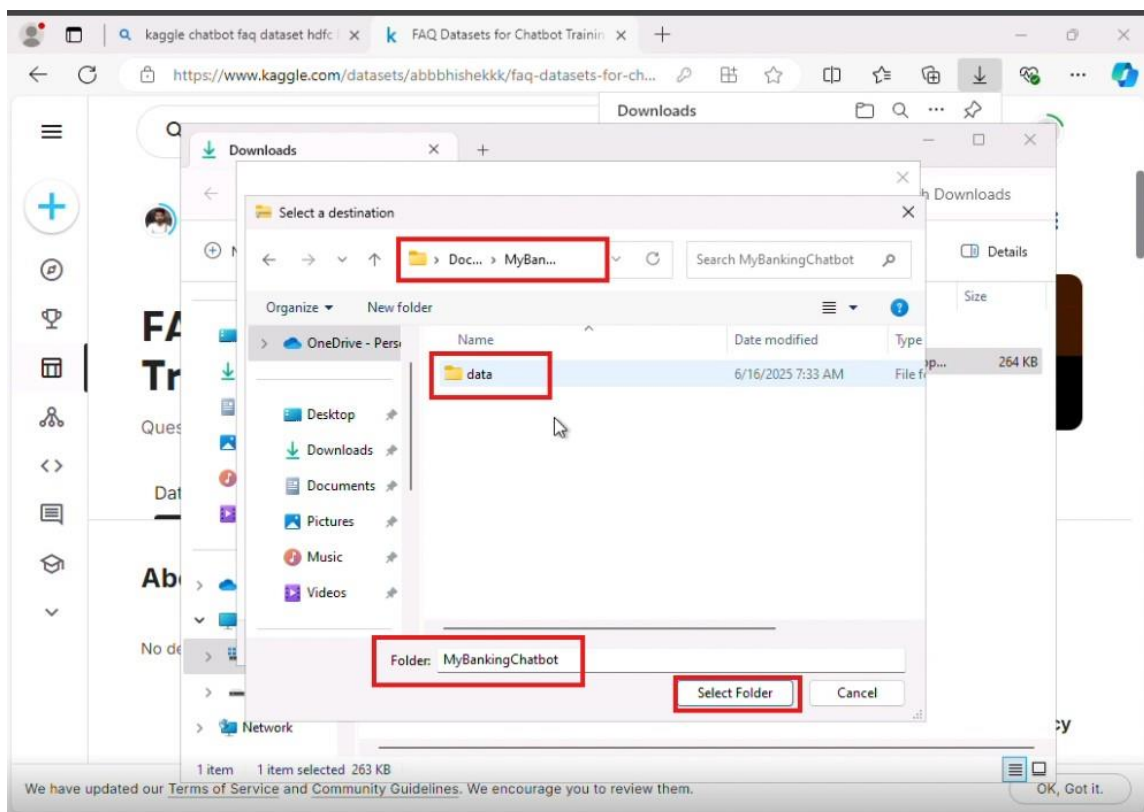
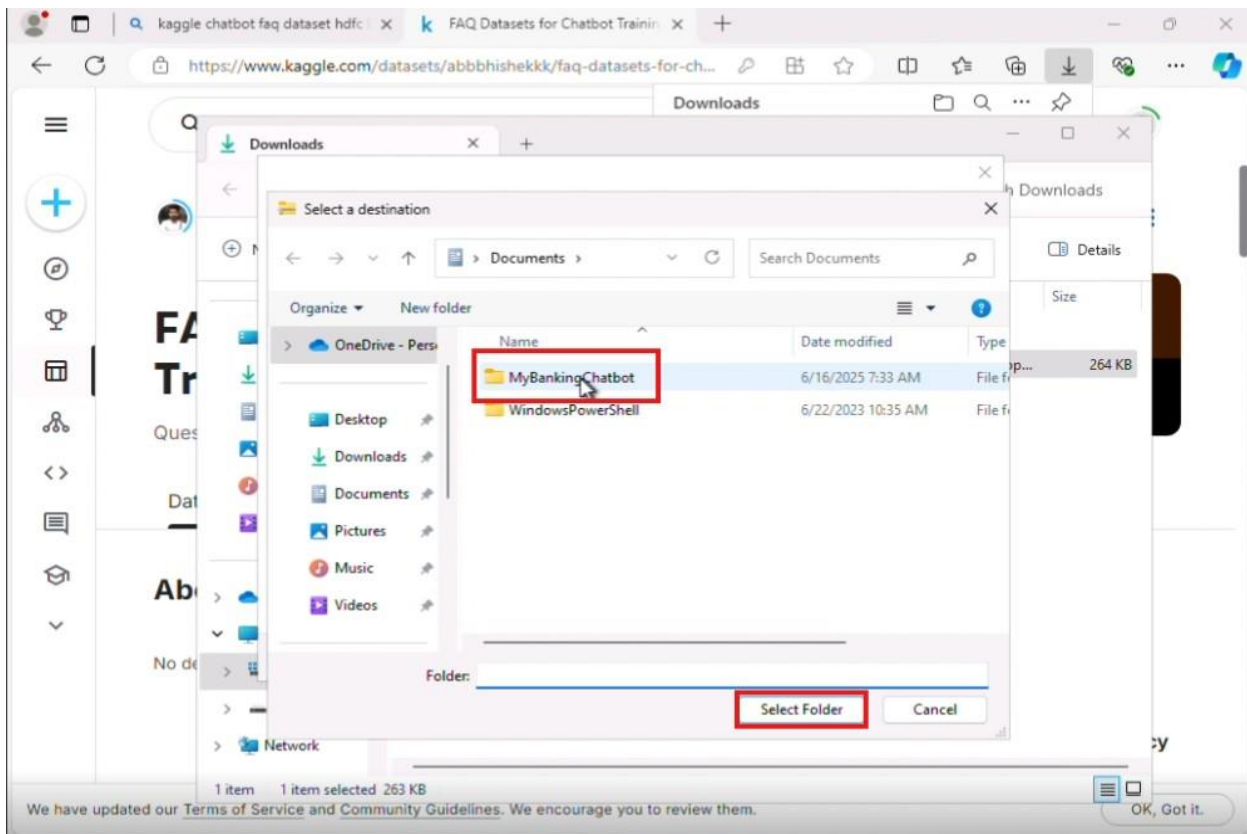
Goto Kaggle and download the data

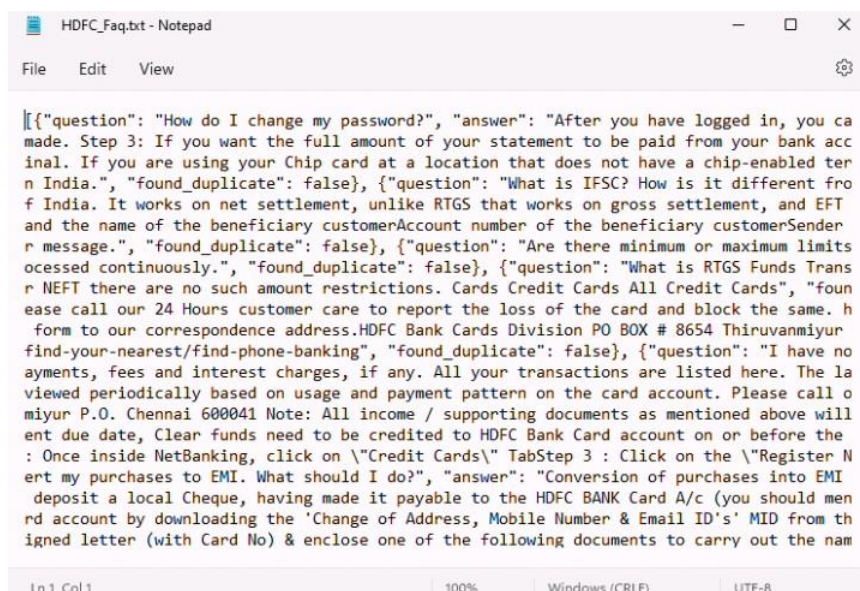
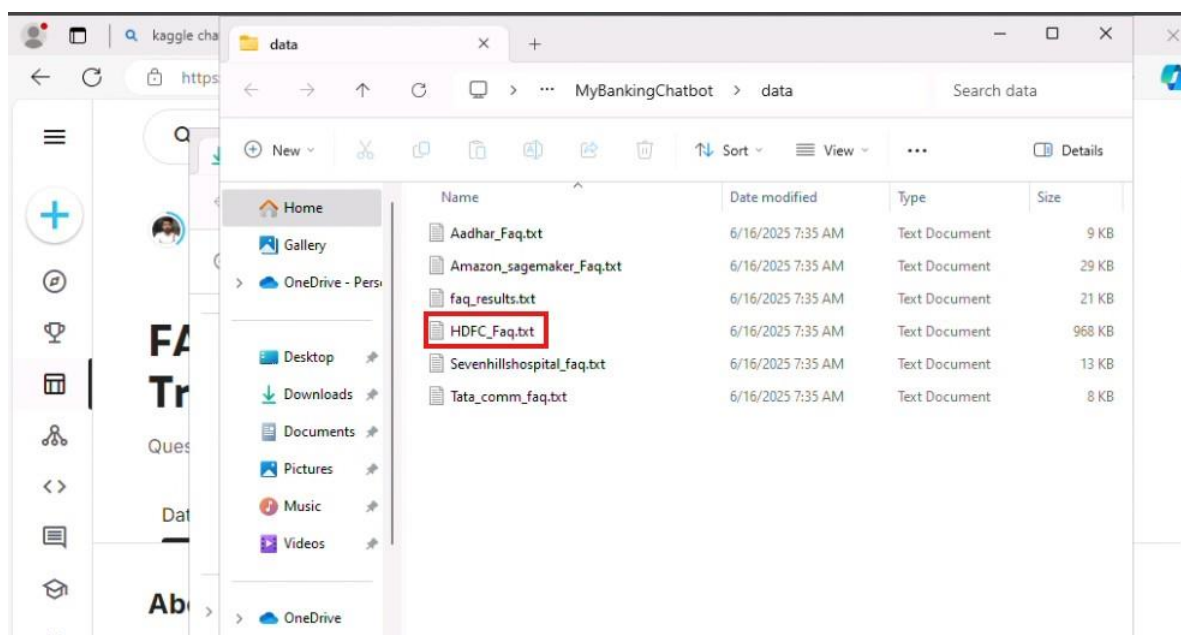
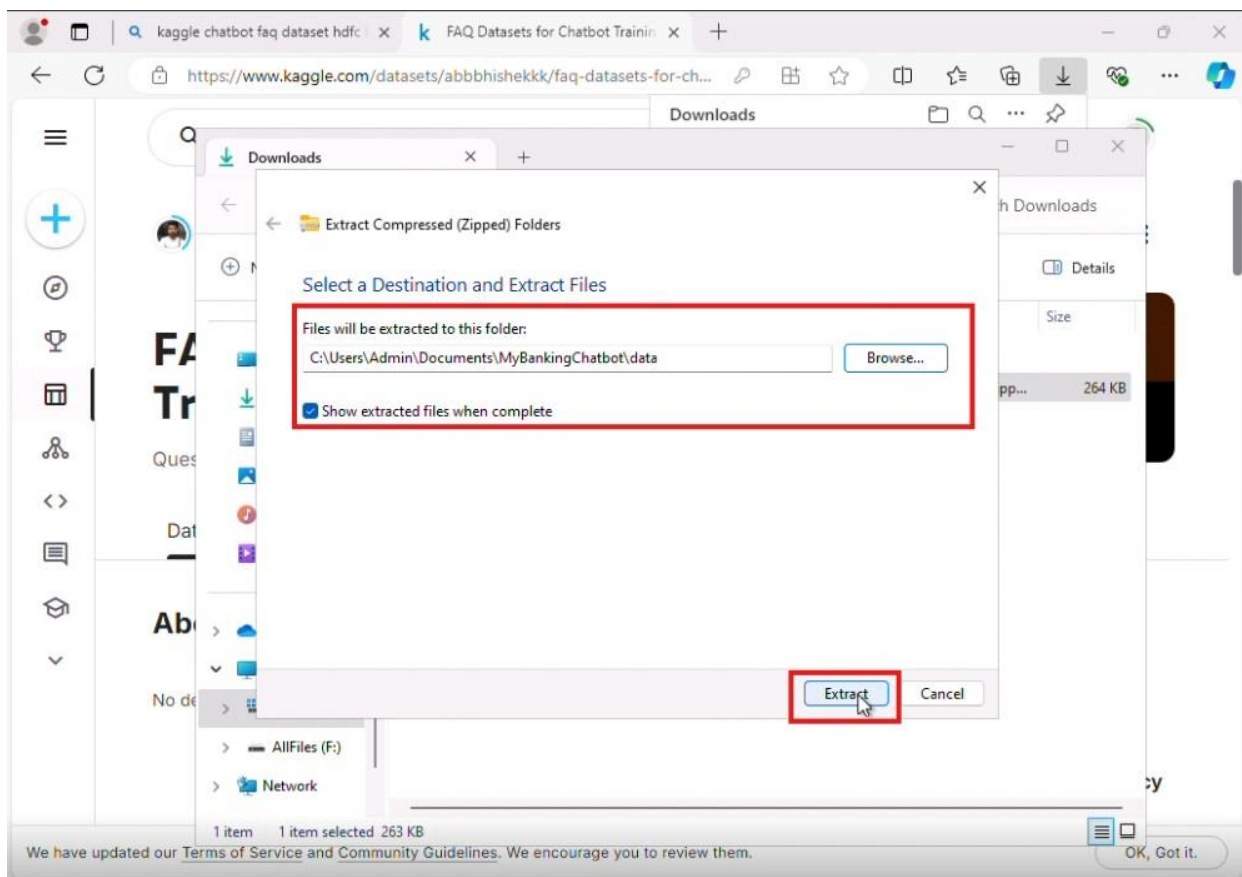


- Step 3 :

Unzip the data to the main to be function app folder

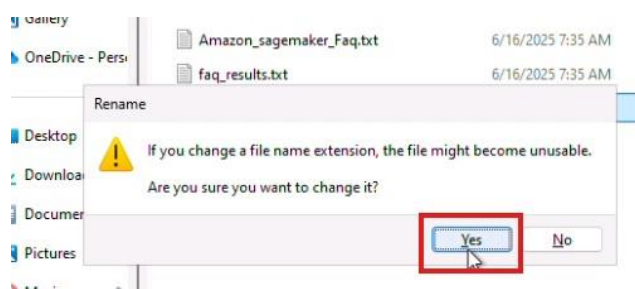
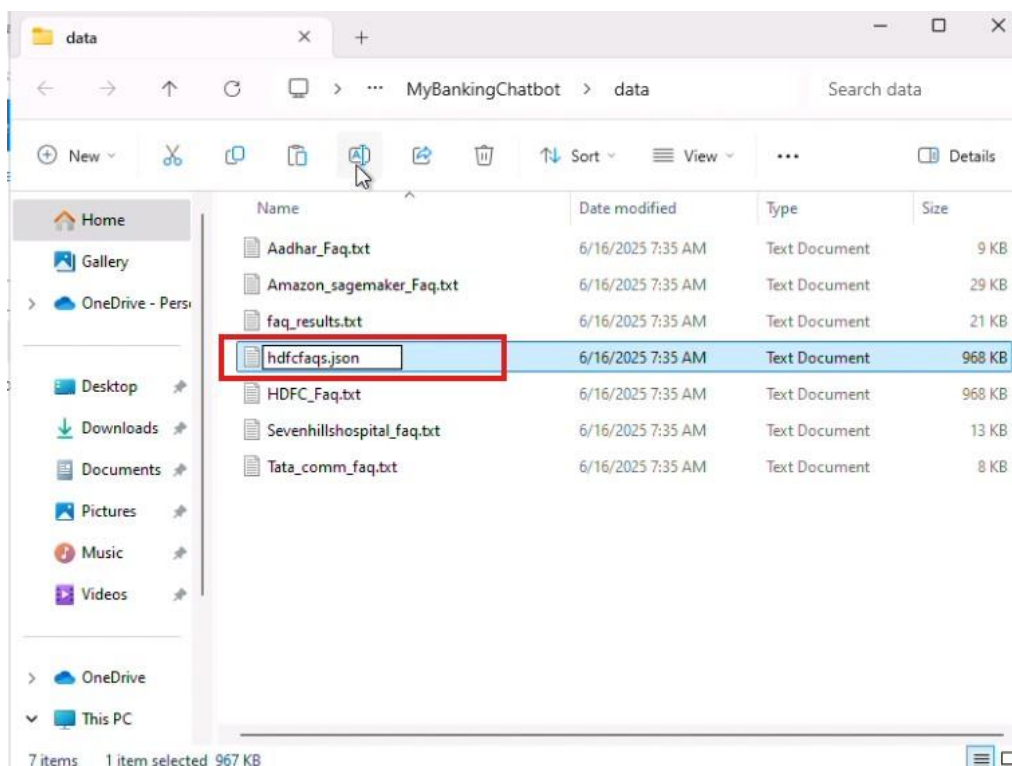
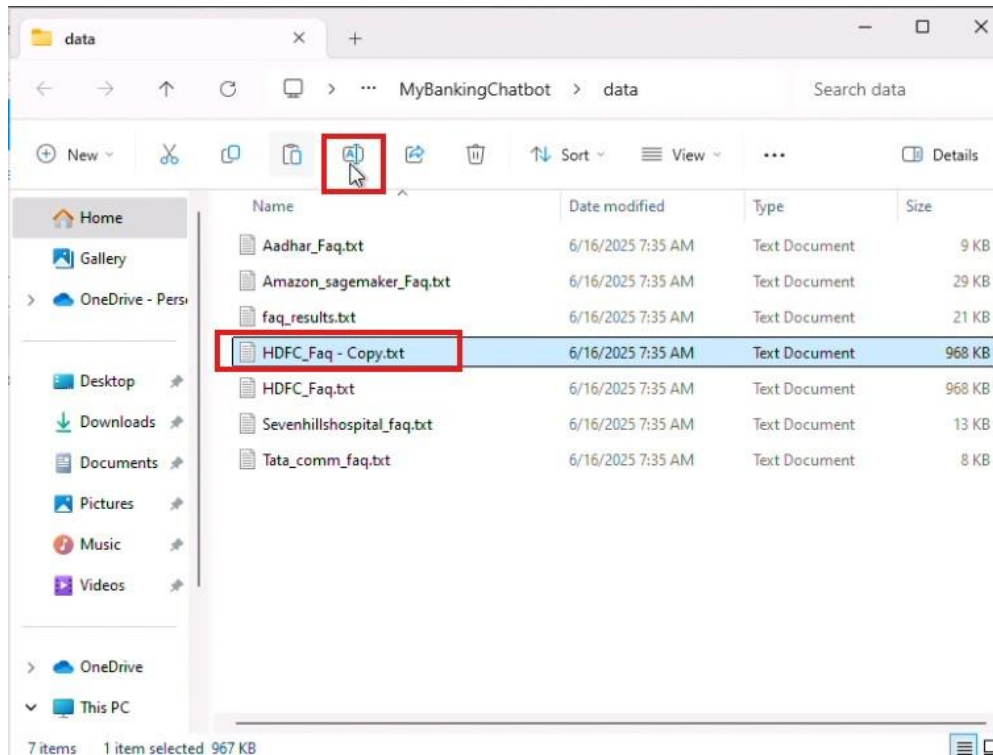


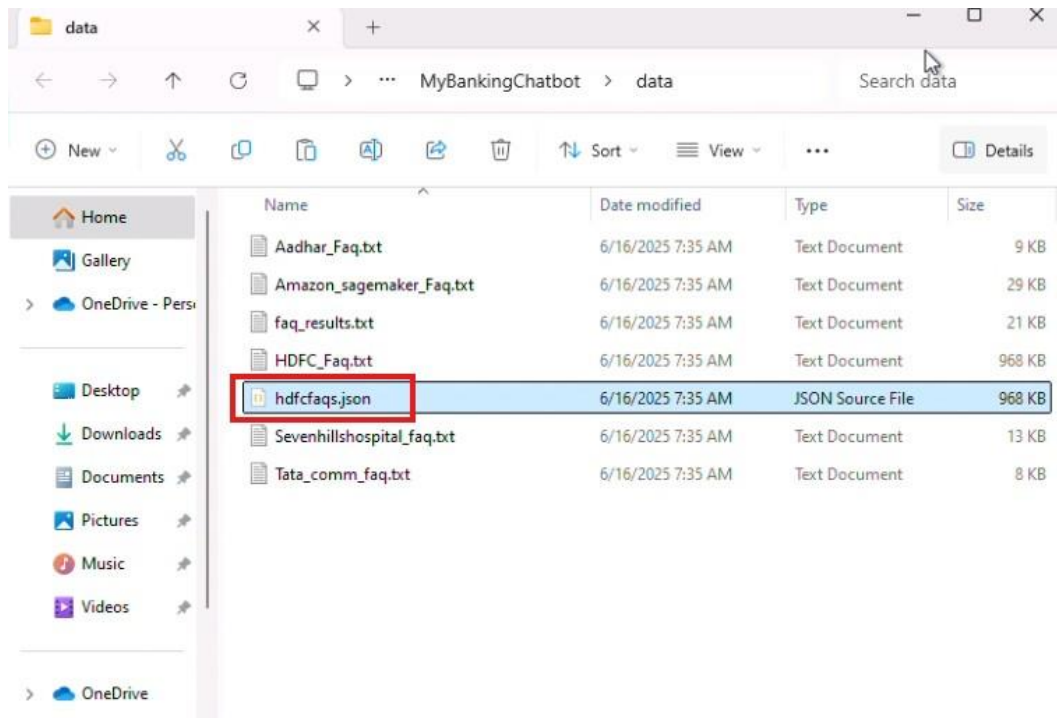




- Step 4 :

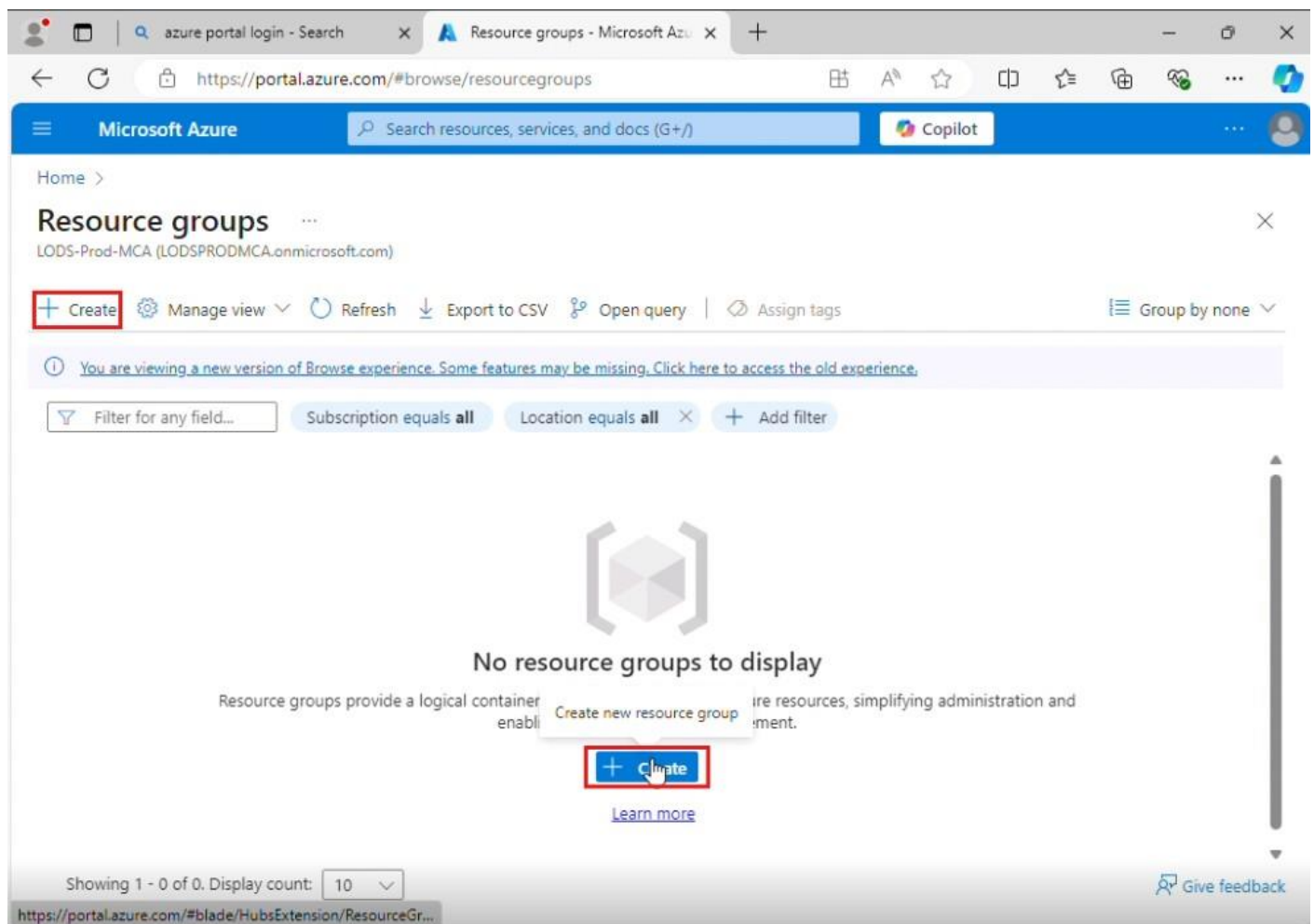
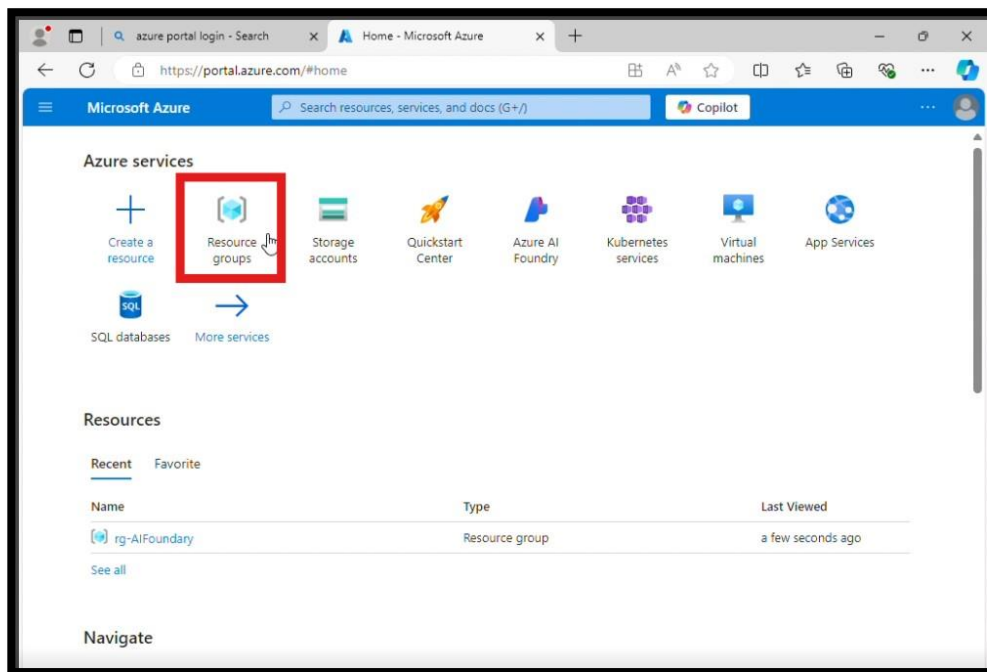
Change the '.txt' to '.json'





2. Setting azure Resources.

- Step 1 : Deploy the Resource Group from the available subscription.



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https://portal.azure.com/#view/HubsExtension/ResourceGroupCreate.React...

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Home > Resource groups >

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ MOC Subscription-lod50723329

Resource group name * ⓘ chatbot-rg

Region * ⓘ (Europe) Sweden Central

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https://portal.azure.com/#view/HubsExtension/ResourceGroupCreate.React...

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[Automation Link](#)

Basics

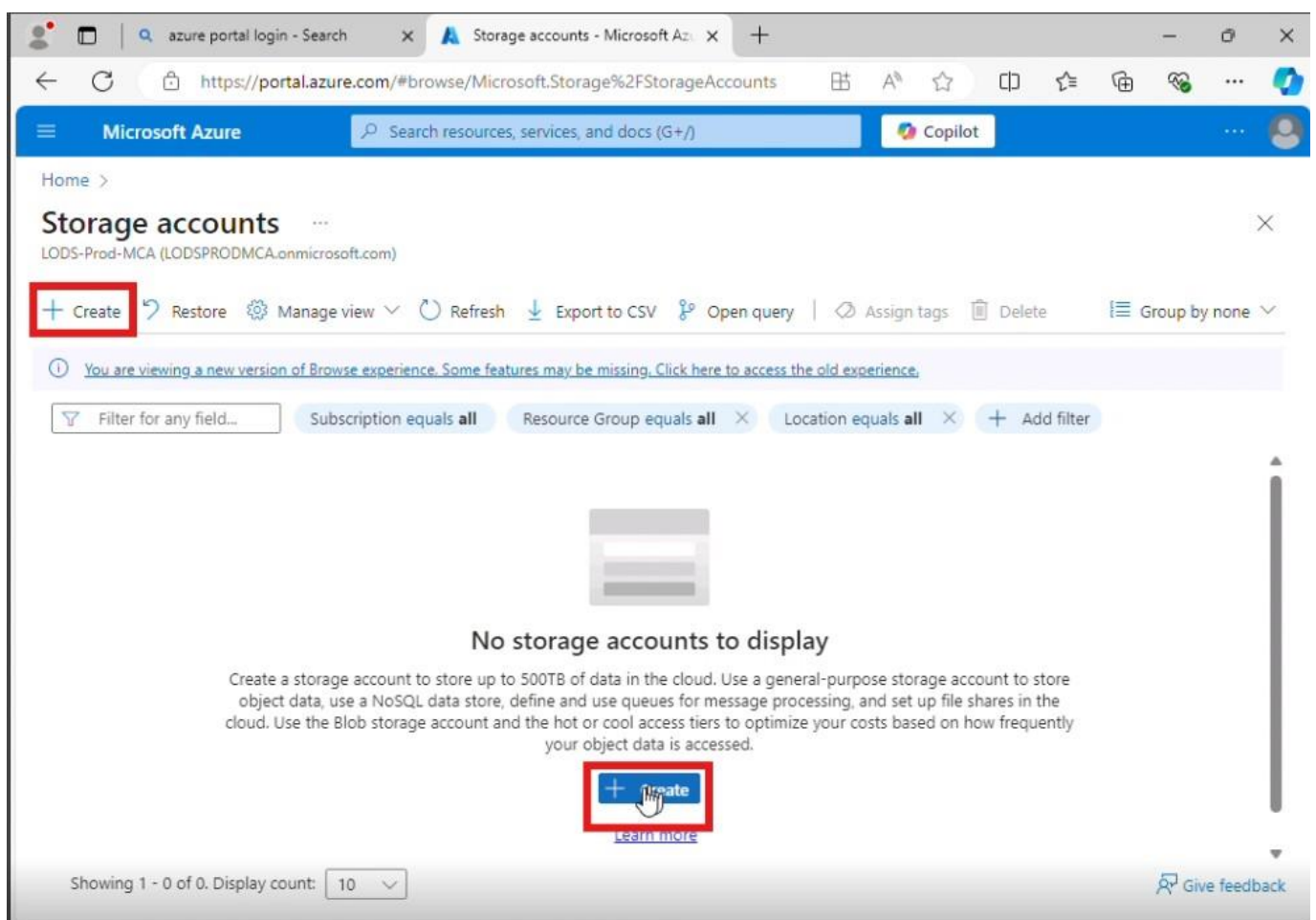
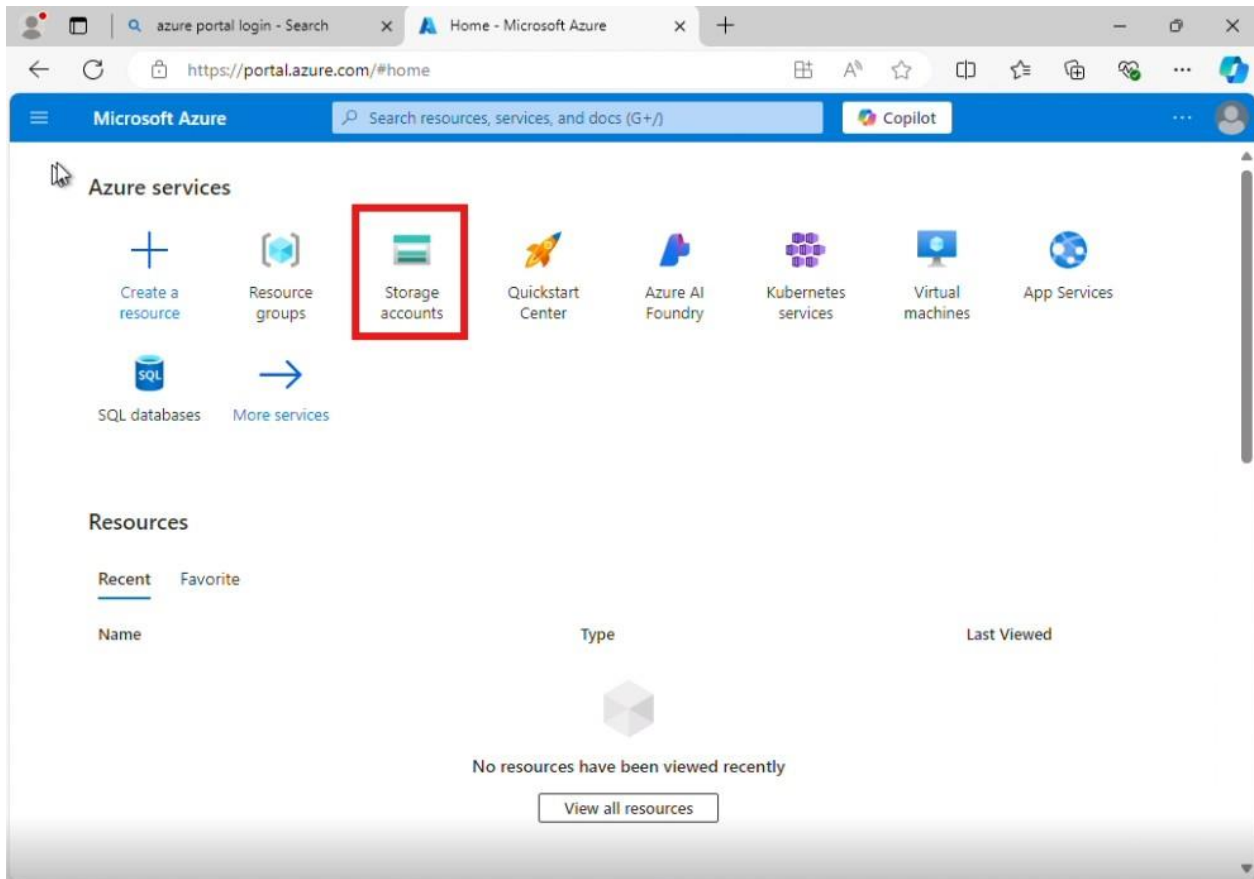
Subscription	MOC Subscription-lod50723329
Resource group name	chatbot-rg
Region	Sweden Central

Tags

None

Previous Next Create

- Step 2 : Deploying the Storage Account Resources



azure portal login - Search x Create a storage account - Micro x +

https://portal.azure.com/#create/Microsoft.StorageAccount

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Storage accounts >

Create a storage account

Subscription * MOC Subscription-lod50723329

Resource group * chatbot-rg
[Create new](#)

Instance details

Storage account name * chatbotstorageacc333

Region * (Europe) Sweden Central
[Deploy to an Azure Extended Zone](#)

Primary service * Azure Blob Storage or Azure Data Lake Storage Gen 2

Performance * ☒ Standard: Recommended for most scenarios (general-purpose v2 account)
☐ Premium: Recommended for scenarios that require low latency.

Redundancy * Locally-redundant storage (LRS)

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https://portal.azure.com/#create/Microsoft.StorageAccount

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Home > Storage accounts >

Create a storage account

Basics **Advanced** Networking Data protection Encryption Tags Review + create

Security

Configure security settings that impact your storage account.

Require secure transfer for REST API operations ☒

Allow enabling anonymous access on individual containers ☒

Enable storage account key access ☒

Default to Microsoft Entra authorization in the Azure portal ☐

Minimum TLS version * Version 1.2

Permitted scope for copy operations (preview) * From any storage account

Hierarchical Namespace

Hierarchical namespace, implemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerator

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https://portal.azure.com/#create/Microsoft.StorageAccount

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Home > Storage accounts >

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Enable SFTP ☐ SFTP can only be enabled for hierarchical namespace accounts

Enable network file system v3 ☐ To enable NFS v3 'hierarchical namespace' must be enabled. [Learn more about NFS v3](#)

Blob storage

Allow cross-tenant replication ☒

Access tier ☒ Hot: Optimized for frequently accessed data and everyday usage scenarios
☐ Cool: Optimized for infrequently accessed data and backup scenarios
☐ Cold: Optimized for rarely accessed data and backup scenarios

Azure Files

Enable large file shares ☒

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https://portal.azure.com/#create/Microsoft.StorageAccount

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Basics Advanced Networking **Data protection** Encryption Tags Review + create

Recovery

When point-in-time restore is enabled, versioning, blob change feed and blob soft delete are also enabled. The retention periods for each of these features must be greater than that of point-in-time restore, if applicable. [Learn more](#)

Protect your data from accidental or erroneous deletion or modification.

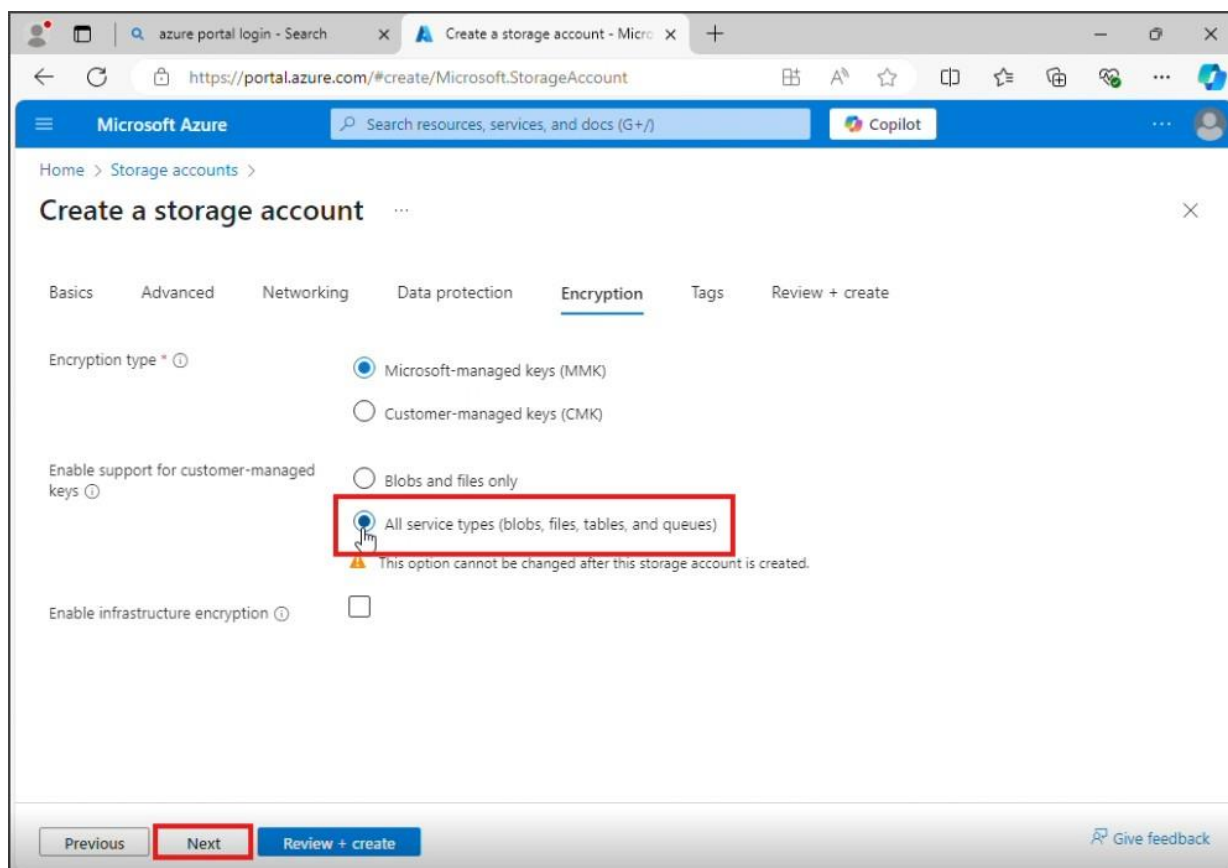
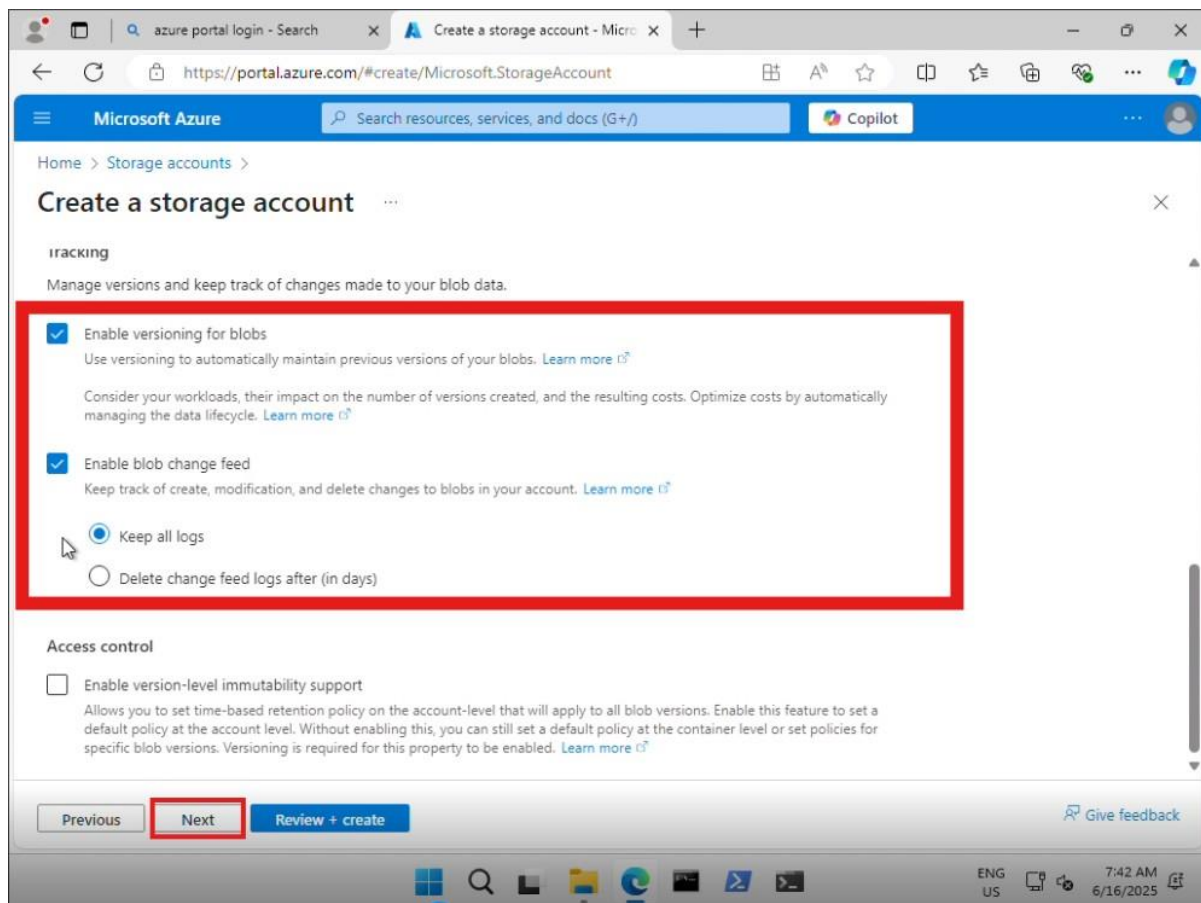
☒ Enable point-in-time restore for containers
Use point-in-time restore to restore one or more containers to an earlier state. If point-in-time restore is enabled, then versioning, change feed, and blob soft delete must also be enabled. [Learn more](#)

Maximum restore point (days ago)

☒ Enable :
Soft delete for blobs
[Learn more](#) enables you to recover blobs that were previously marked for deletion, including blobs that were overwritten. [Learn](#)

Days to

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https://portal.azure.com/#create/Microsoft.StorageAccount

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Home > Storage accounts >

Create a storage account

Basics Advanced Networking Data protection Encryption Tags Review + create

[View automation template](#)

Basics

Subscription	MOC Subscription-Iod50723329
Resource group	chatbot-rg
Location	Sweden Central
Storage account name	chatbotstorageacc333
Primary service	Azure Blob Storage or Azure Data Lake Storage Gen 2
Performance	Standard
Replication	Locally-redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled

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https://portal.azure.com/#home

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Azure services

Create a resource Storage accounts Resource groups Quickstart Center Azure AI Foundry Kubernetes services Virtual machines App Services

SQL databases More services

Resources

Recent Favorite

Name	Type	Last Viewed
chatbotstorageacc333	Storage account	a few seconds ago
chatbot-rg	Resource group	a few seconds ago

See all

Navigate

- Step 3 : Uploading the json file into the storage container.

The screenshot shows the Microsoft Azure portal home page. The 'Resources' section displays a table with the following data:

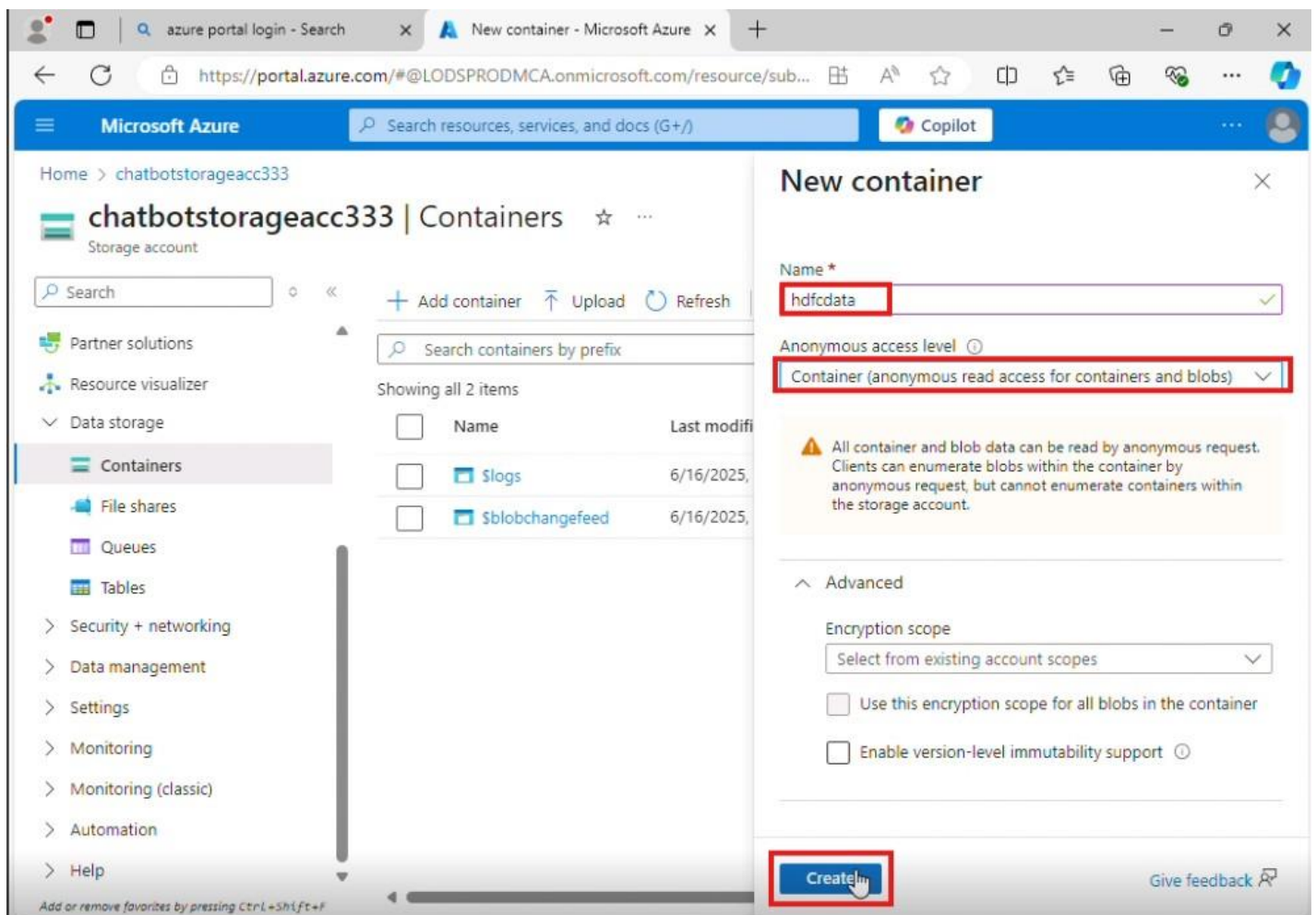
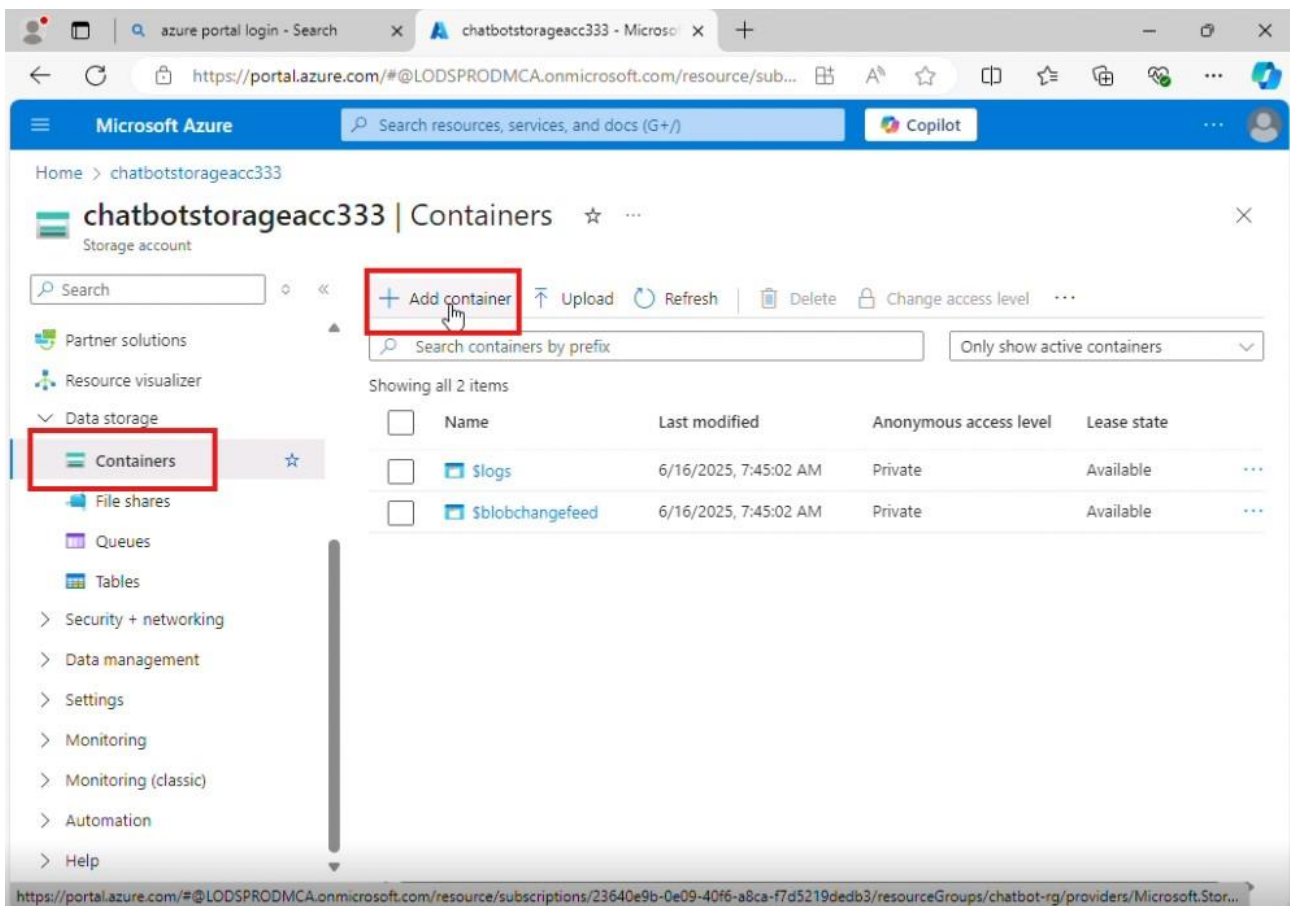
Name	Type	Last Viewed
chatbotstorageacc333	Storage account	a few seconds ago
chatbot-rg	Resource group	a few seconds ago

The 'chatbotstorageacc333' entry is highlighted with a red box. The 'Azure services' section shows various service icons like 'Create a resource', 'Storage accounts', 'Resource groups', 'Quickstart Center', 'Azure AI Foundry', 'Kubernetes services', 'Virtual machines', 'App Services', 'SQL databases', and 'More services'.

The screenshot shows the details of the 'chatbotstorageacc333' storage account. The left sidebar shows the 'Data storage' and 'Containers' sections highlighted with red boxes. The main content area shows the 'Essentials' section with various properties:

- Resource group (move): chatbot-rg
- Location: swedencentral
- Subscription (move): MOC Subscription-lod50723329
- Subscription ID: 23640e9b-0e09-40f6-a8ca-f7d5219dedb3
- Disk state: Available
- Tags (edit): Add tags
- Performance: Standard
- Replication: Locally-redundant storage (LRS)
- Account kind: StorageV2 (general purpose v2)
- Provisioning state: Succeeded
- Created: 6/16/2025, 7:44:35 AM

The 'Data storage' and 'Containers' sections in the left sidebar are highlighted with red boxes. The 'Containers' section is also highlighted with a red box. The 'Blob service' and 'Security' sections are visible at the bottom.



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https://portal.azure.com/#@LODSPRODMCA.onmicrosoft.com/resource/sub...

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Home > chatbotstorageacc333

chatbotstorageacc333 | Containers

Storage account

Search

Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Containers
File shares

+ Add container ↑ Upload Refresh Delete Change access level ...

Search containers by prefix Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state
☐	\$logs	6/16/2025, 7:45:02 AM	Private	Available
☐	\$blobchangefeed	6/16/2025, 7:45:02 AM	Private	Available
☐	hdfcdata	6/16/2025, 7:46:06 AM	Container	Available

azure portal login - Search x hdfcdata - Microsoft Azure x +

https://portal.azure.com/#view/Microsoft_Azure_Storage/ContainerMenuBla...

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Home > chatbotstorageacc333 | Containers >

hdfcdata

Container

Search

Overview
Diagnose and solve problems
Access Control (IAM)
Settings

+ Add Director ↑ Upload Change access level Refresh Delete Copy Paste ...

hdfcdata

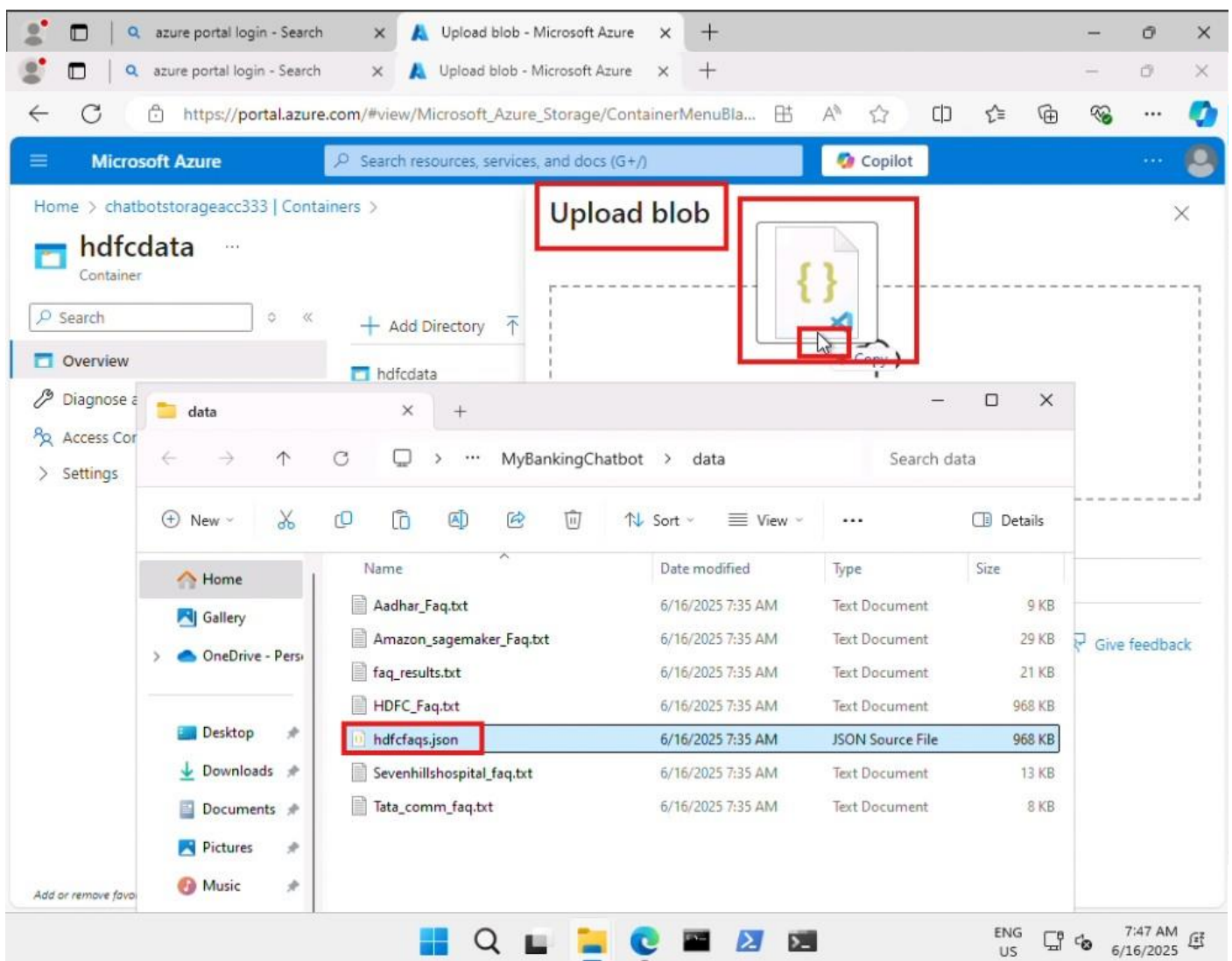
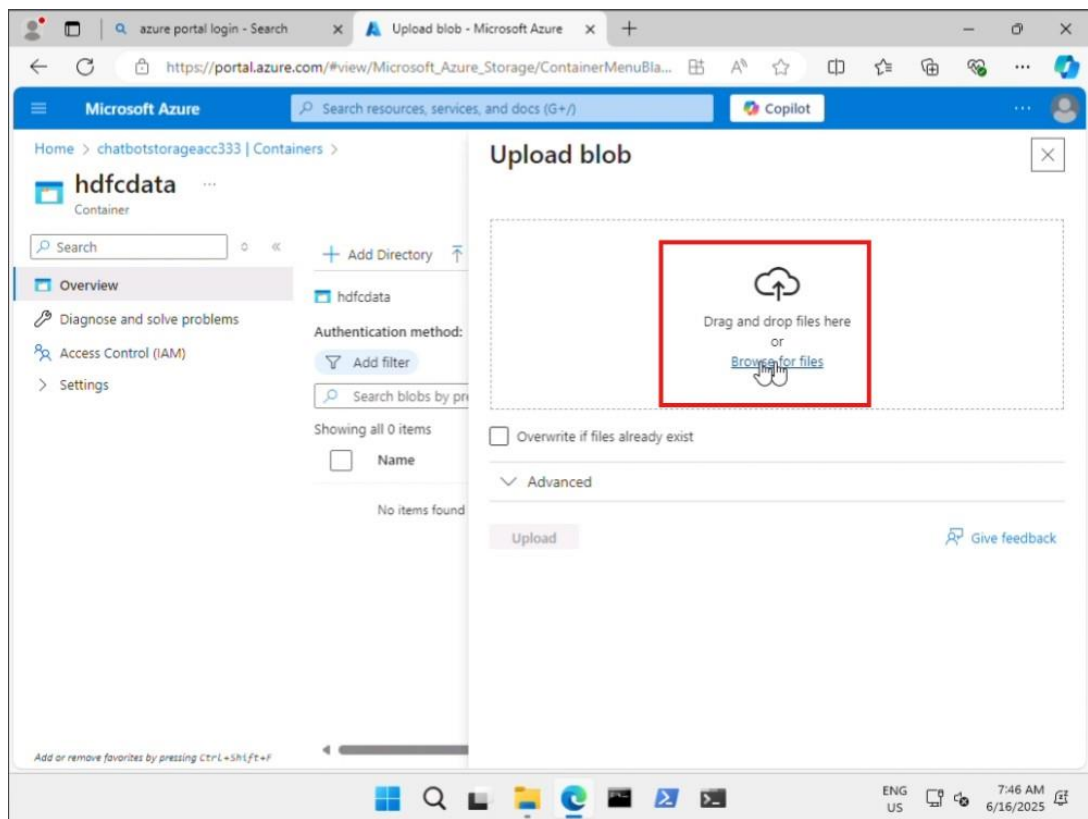
Authentication method: Access key (Switch to Microsoft Entra user account)

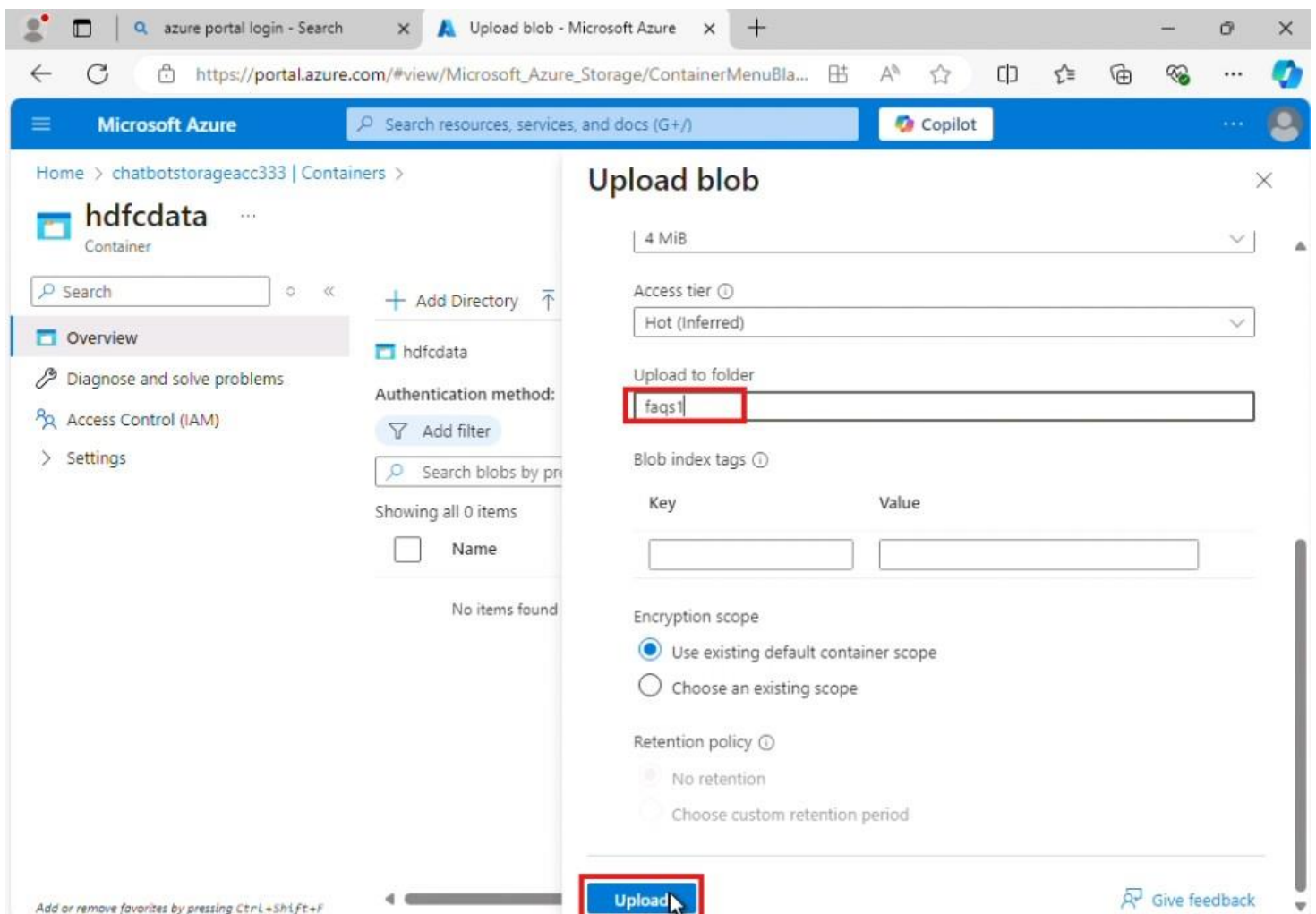
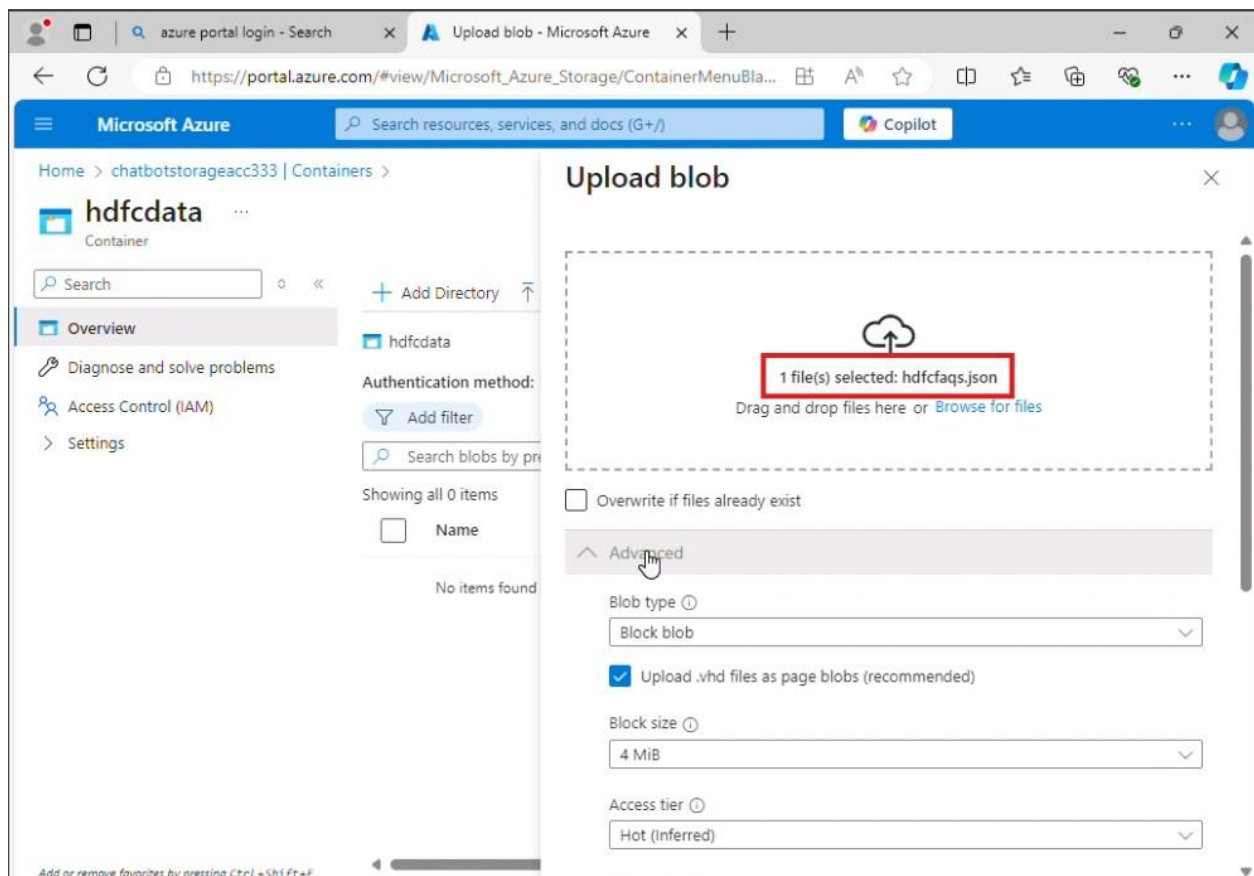
Add filter

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 0 items

☐	Name	Last modified	Access tier	Blob type	Size
No items found					





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https://portal.azure.com/#view/Microsoft_Azure_Storage/ContainerMenuBla...

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Home > chatbotstorageeacc333 | Containers >

hdfcdata Container

Search

+ Add Directory ↑ Upload Change access level Refresh Delete Copy Paste ...

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

hdfcdata

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size
☐	faq1				

Successfully uploaded blob(s)
Successfully uploaded 1 blob(s).

Add or remove favorites by pressing Ctrl+Shift+F

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https://portal.azure.com/#view/Microsoft_Azure_Storage/ContainerMenuBla...

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Home > chatbotstorageeacc333 | Containers >

hdfcdata Container

Search

+ Add Directory ↑ Upload Refresh Delete Copy Paste Rename ...

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

hdfcdata > faq1

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

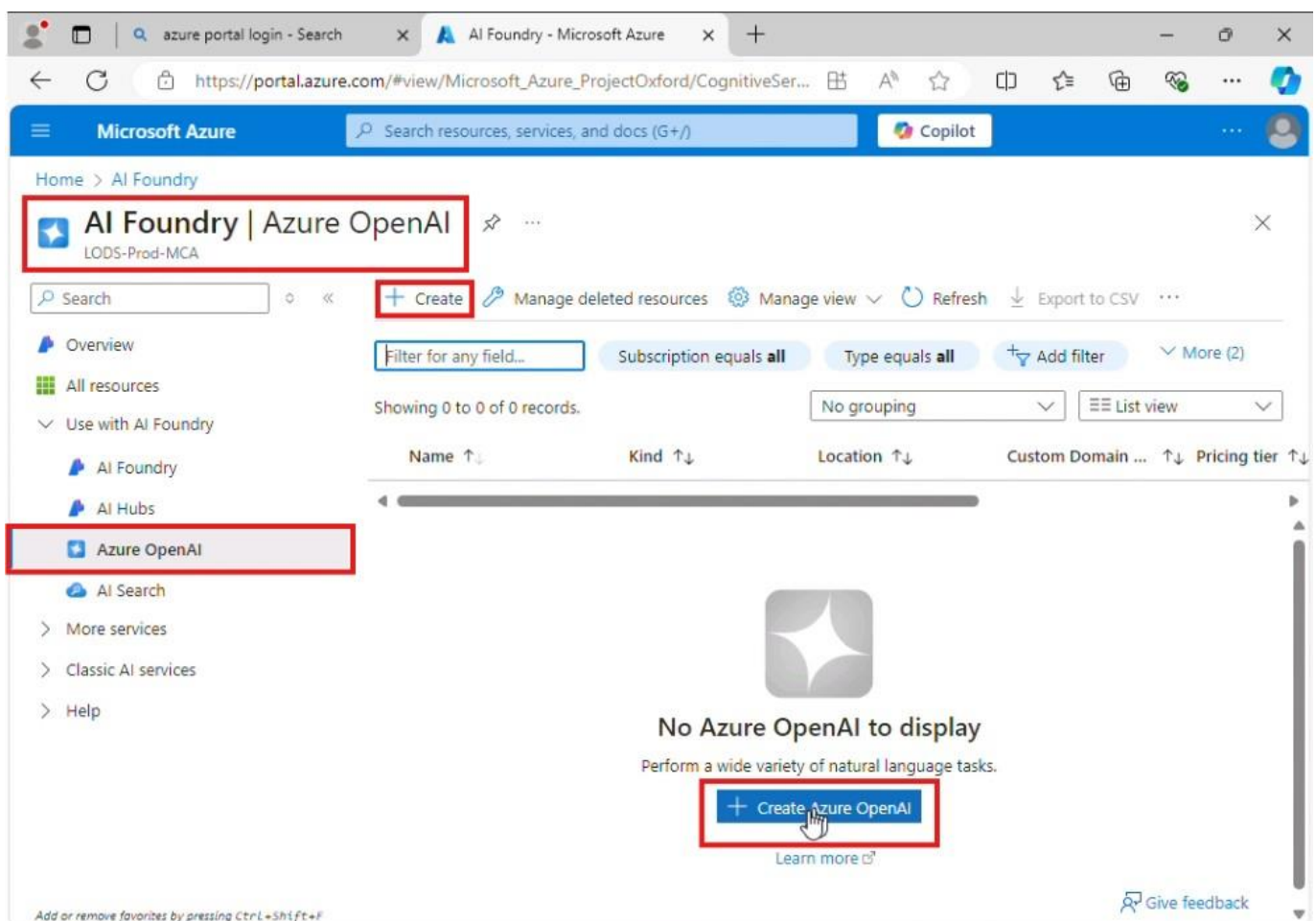
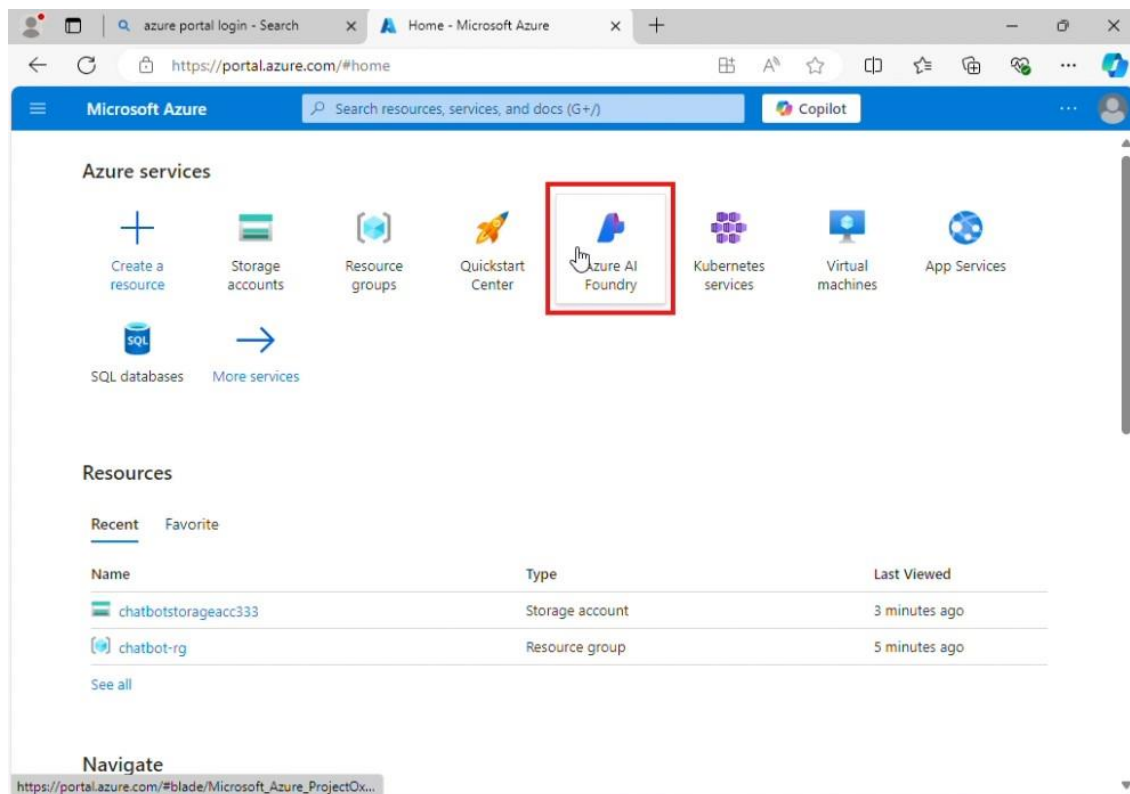
Only show active blobs

Sorting all 1 items

☐	Name ↑	Last modified	Access tier	Blob type	Size
☐	[.]				
☐	hdfcfaq.json	6/16/2025, 7:48:15 AM	Hot (Inferred)	Block blob	967.24

Add or remove favorites by pressing Ctrl+Shift+F

3. Setting up Open AI services.



azure portal login - Search x Create Azure OpenAI - Microsoft x +

https://portal.azure.com/#create/Microsoft.CognitiveServicesOpenAI

Microsoft Azure Search resources, services, and docs (G+/I) Copilot

Home > AI Foundry | Azure OpenAI >

Create Azure OpenAI

Resource group * chatbot-rg Create new

Instance Details

Region Sweden Central

Name * openAIservices333

Pricing tier * Standard S0

[View full pricing details](#)

Content review policy

To detect and mitigate harmful use of the Azure OpenAI Service, Microsoft logs the content you send to the

[Previous](#) [Next](#) [Give feedback](#)

azure portal login - Search x Create Azure OpenAI - Microsoft x +

https://portal.azure.com/#create/Microsoft.CognitiveServicesOpenAI

Microsoft Azure Search resources, services, and docs (G+/I) Copilot

Home > AI Foundry | Azure OpenAI >

Create Azure OpenAI

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Basics

Subscription	MOC Subscription-Iod50723329
Resource group	chatbot-rg
Region	Sweden Central
Name	openAIservices333
Pricing tier	Standard S0

Network

Type All networks, including the internet, can access this resource.

[Previous](#) [Next](#) [Create](#) [Give feedback](#)

azure portal login - Search x AI Foundry - Microsoft Azure x +

https://portal.azure.com/#view/Microsoft_Azure_ProjectOxford/CognitiveSer...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > AI Foundry

AI Foundry | Azure OpenAI

LODS-Prod-MCA

Search

Create Manage deleted resources Manage view Refresh Export to CSV

Overview All resources Use with AI Foundry AI Foundry AI Hubs Azure OpenAI AI Search More services Classic AI services Help

Filter for any field... Add filter More (4)

Showing 1 to 1 of 1 records. No grouping List view

Name	Kind	Location	Custom Domain	Pricing tier
openAIservices333	OpenAI	Sweden Central	openaiservices333	S0

Page 1 of 1

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

azure portal login - Search x openAIservices333 - Microsoft A x +

https://portal.azure.com/#@LODSPRODMCA.onmicrosoft.com/resource/sub...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > AI Foundry | Azure OpenAI >

openAIservices333

Azure OpenAI

Go to Azure AI Foundry portal Delete

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Resource visualizer Resource Management Security Monitoring Automation Help

Get Started Develop Monitor

Build your own secure copilot and generative AI applications with Azure OpenAI Service

Deploy an Azure OpenAI model and start making API calls. Connect your own data, call functions, and improve workflow with Azure OpenAI language, image and speech models. You can access the service through REST APIs, Python SDK, or our web-based interface in the Azure AI Foundry portal.

Learn More

Explore and deploy

Explore and deploy the generative AI models, craft unique prompts for your use cases, and fine-tune select models.

Explore Azure AI Foundry portal

Add or remove favorites by pressing Ctrl+Shift+F

azure portal login - Search | openAIservices333 - Microsoft A... | Model catalog - Azure OpenAI

https://ai.azure.com/resource/models?wsid=/subscriptions/23640e9b-0e09-40f6-a8ca-f7...

Azure AI Foundry | Azure OpenAI | openAIservices333 | Model catalog

Find the right model to build your custom AI solution

Announcements

Sora

Generate up to 20s 1080p videos with Sora and its uniqu...

Check out model

Read blog

Introducing Model Router

Route prompts to different models to help save cost while...

Check out model

Read blog

Introducing gpt-image-

Check out model

Read blog

Get started

Model catalog

Playgrounds

Chat

Assistants PREVIEW

Video PREVIEW

Audio PREVIEW

Images

Completions

Tools

Fine-tuning

Azure OpenAI Evaluation PREVIEW

Stored completions PREVIEW

Batch jobs

Inference tasks

Fine-tuning tasks

Search

Models 37

sora

model-router

https://ai.azure.com/resource/models?wsid=/subscriptions/23640e9b-0e09-40f6-a8ca-f7d5219dedb3/resourceGroups/chatbot-rg/providers/Microsoft.CognitiveServices/accounts/op...

azure portal login - Search | openAIservices333 - Microsoft A... | Model catalog - Azure OpenAI

https://ai.azure.com/resource/models?wsid=/subscriptions/23640e9b-0e09-40f6-a8ca-f7...

Model catalog

text

Models 10

text-embedding-3-large

Embeddings

text-embedding-ada-002

Embeddings

text-embedding-3-small

Embeddings

gpt-4o-mini-tts

Text to speech

gpt-4o-mini-transcribe

Speech to text

gpt-image-1

Text to image

tts-hd

Text to speech

tts

Text to speech

gpt-4o-transcribe

Speech to text

dall-e-3

Text to image

Prev

Next

ai.azure.com/resource/models/.../azure-openai?wsid=/subs...

azure portal login - Search | openAiservices333 - Microsoft A... | text-embedding-ada-002 - Azure

https://ai.azure.com/resource/models/text-embedding-ada-002/version/2/registry/azure...

Azure AI Foundry | Azure OpenAI | openAiservices333 | Models | text-embedding-ada-002

text-embedding-ada-002

[Use this model](#) Fine-tune

Details Existing deployments

text-embedding-ada-002 outperforms all the earlier embedding models on text search, code search, and sentence similarity tasks and gets comparable performance on text classification. Embeddings are numerical representations of concepts converted to number sequences, which make it easy for computers to understand the relationships between those concepts.

Note: this model can be deployed for inference, specifically for embeddings, but cannot be finetuned.

Model variation

text-embedding-ada-002 is part of gpt-3 model family.

Learn more at <https://learn.microsoft.com/azure/cognitive-services/openai/concepts/models#embeddings-models>
[See less](#)

Model Versions

[Learn more about region availability](#)

Quick facts

OpenAI

text-embedding-ada-002

embeddings

Training data last updated
Not available

Pricing
[See pricing](#)

Model ID
Reference this model ID when deploying the model in code

azure portal login - Search | openAiservices333 - Microsoft A... | text-embedding-ada-002 - Azure

https://ai.azure.com/resource/models/text-embedding-ada-002/version/2/registry/azure...

Deploy text-embedding-ada-002

Deployment name *
text-embedding-ada-002

Deployment type
Global Standard

Global Standard: Pay per API call with the highest rate limits. [Learn more about Global deployment types](#).
Data might be processed globally, outside of the resource's Azure geography, but data storage remains in the AI resource's Azure geography. [Learn more about data residency](#).

Deployment details

Model version
2

Capacity
250K tokens per minute (TPM)

Content safety
DefaultV2

[Customize](#)

AI resource
openAiservices333

Resource location
Sweden Central

Version upgrade policy
Model version will not be automatically upgraded

[Deploy](#) [Cancel](#)

Deploy text-embedding-ada-002

Deployment details

Model version upgrade policy

Opt out of automatic model version upgrades

Automatic version update is not available for embedding and finetuned models.

Model version

2 (Default)

AI resource

openAIservices333

10M tokens per minute quota available for your deployment

Tokens per Minute Rate Limit

995K

Corresponding requests per minute (RPM) = 5.97K

Content filter

DefaultV2

Deploy

Cancel

Deployments

Name	Model name	Model version	State
text-embedding-ada-002	text-embedding-ada-002	2	Succeeded

4. Setting up AI search services.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a Copilot button. The breadcrumb trail indicates the user is in the 'AI Foundry' section. The left sidebar lists various resources, with 'AI Search' highlighted under the 'Use with AI Foundry' category. The main content area shows a list of search services, which is currently empty. A 'Create' button is visible in the top toolbar, and a 'Create search service' button is prominently displayed in the center of the page.

The screenshot shows the 'Create a search service' form in the Microsoft Azure portal. The form is divided into two main sections: 'Project details' and 'Instance Details'. In the 'Project details' section, the 'Resource group' dropdown is set to 'chatbot-rg'. In the 'Instance Details' section, the 'Service name' field is set to 'searchservicesproject3016', the 'Location' dropdown is set to '(Europe) Sweden Central', and the 'Pricing tier' dropdown is set to 'Standard'. The 'Next: Scale' button is highlighted at the bottom of the form.

azure portal login - Search x Create a search service - Microsc x Models + endpoints - Azure Op x + - x

https://portal.azure.com/#create/Microsoft.Search

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > AI Foundry | AI Search >

Create a search service

Basics Scale Networking Tags Review + create

SEARCH UNITS ⓘ EST. COST PER MONTH ⓘ

1/36 \$249.98

Replicas

Replicas distribute workloads across the service. We guarantee 99.9% availability for read operations with 2 replicas, and for read and write operations with 3 or more replicas. [Learn more about the SLA for Azure AI Search](#)

1

- 99.9% availability for read operations (choose 2 or more replicas)
- 99.9% availability for read/write operations (choose 3 or more replicas)

Partitions

Partitions allow for scaling of document counts as well as faster data ingestion by spanning your index over multiple units.

1

160 GB Storage

Previous Next: Networking Review + create

azure portal login - Search x Create a search service - Microsc x Models + endpoints - Azure Op x + - x

https://portal.azure.com/#create/Microsoft.Search

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > AI Foundry | AI Search >

Create a search service

Basics Scale Networking Tags Review + create

Basics

Subscription	MOC Subscription-lod50723329
Resource Group	chatbot-rg
Location	Sweden Central
Service name	searchservicesproject3016
Pricing tier	Standard (160 GB/Partition, max 12 replicas, max 12 partitions, max 36 search units)
Estimated cost per month	\$249.98

Scale

Replicas	1
Partitions	1

Networking

Endpoint connectivity (data)	Public
------------------------------	--------

Previous Next Create Download a template for automation

Microsoft Azure portal home page. The "Azure services" section shows various services like Create a resource, Azure AI Foundry, Storage accounts, Resource groups, Quickstart Center, Kubernetes services, Virtual machines, and App Services. The "Resources" section shows a list of recent resources. A red box highlights the resource "searchservicesproject3016" in the list. A tooltip for this resource is shown, displaying the name "searchservicesproject3016" and a star icon. The tooltip also includes "View" and "Delete" links.

Name	Type	Last Viewed
searchservicesproject3016		a few seconds ago
chatbot-rg		a few seconds ago
openAIService333		6 minutes ago
chatbotstorageacc333		12 minutes ago

Microsoft Azure portal page for the "searchservicesproject3016" resource. The page shows the resource details, including the resource group "chatbot-rg", location "Sweden Central", subscription "MOC Subscription-lod50723329", and subscription ID "23640e9b-0e09-40f6-a8ca-f7d5219dedb3". The resource is in the "Running" status and was created on "Jun 16, 2025, 7:58:10 AM". The "Tags" section shows "Add tags". The "Import and vectorize data" button is highlighted with a red box. The "Search explorer" button is also visible. The "JSON View" link is present. The "Get started" button is highlighted.

searchservicesproject3016

Search service

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Search management

Settings

Monitoring

Automation

Help

Add or remove favorites by pressing Ctrl+Shift+F

Get started

Properties

Usage

Monitoring

Import and vectorize data

Search explorer

Upgrade

JSON View

Essentials

Resource group (move)

chatbot-rg

Location (move)

Sweden Central

Subscription (move)

MOC Subscription-lod50723329

Subscription ID

23640e9b-0e09-40f6-a8ca-f7d5219dedb3

Status

Running

Date created

Jun 16, 2025, 7:58:10 AM

Tags (edit)

Add tags

Url

https://searchservicesproject3016.search.window...

Pricing tier

Standard

Replicas

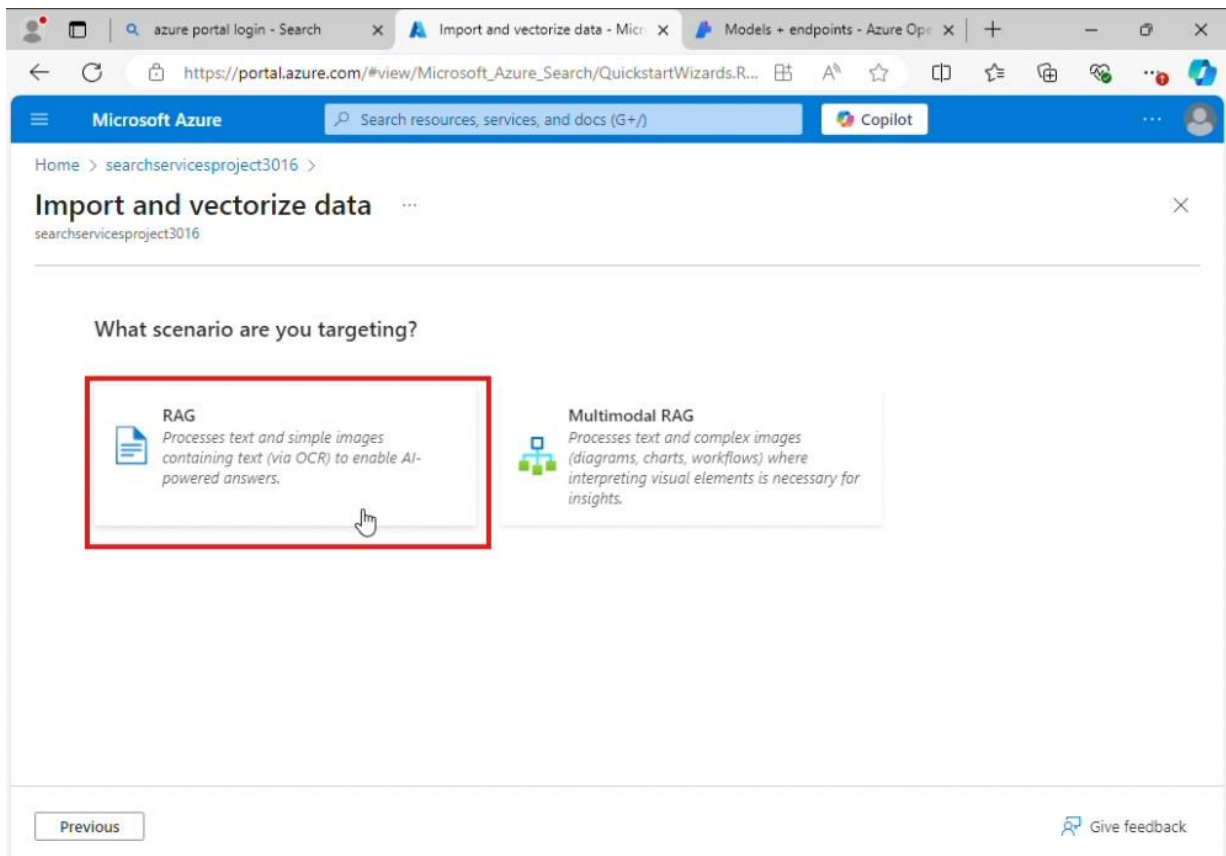
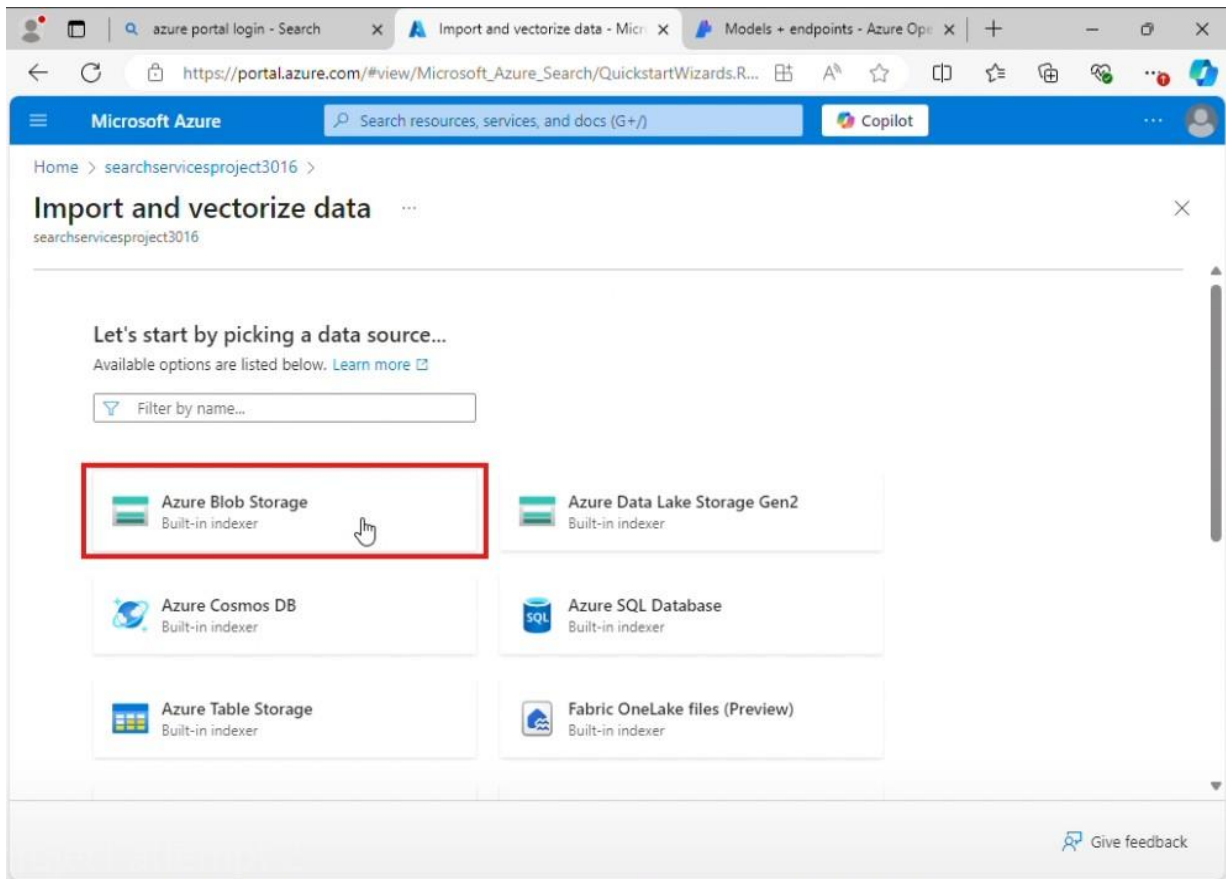
1 (No SLA)

Partitions

1

Search units

1



azure portal login - Search x RAG - Microsoft Azure x Models + endpoints - Azure OpenAI x

https://portal.azure.com/#view/Microsoft_Azure_Search/QuickstartWizards.R...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Home > searchservicesproject3016 >

RAG

searchservicesproject3016

- Connect to your data
- Vectorize your text
- Advanced settings
- Review and create

Storage account *

chatbotstorageacc333

Blob container *

hdfcdata

Blob folder

faqs1

Parsing mode

JSON array

Document Root

/level1/level2

☐ Enable deletion tracking

☐ Authenticate using managed identity. [Learn more](#)

Previous Next

Give feedback

azure portal login - Search x RAG - Microsoft Azure x Models + endpoints - Azure OpenAI x

https://portal.azure.com/#view/Microsoft_Azure_Search/QuickstartWizards.R...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Home > searchservicesproject3016 >

RAG

searchservicesproject3016

- Connect to your data
- Vectorize your text
- Advanced settings
- Review and create

Vectorize your text

Connect to an Azure OpenAI, AI Foundry or an Azure AI service and select an embedding model or multi-service account for vector generation. [Learn more](#)

Column to vectorize *

answer

Kind

Azure OpenAI

Subscription *

MOC Subscription-lod50723329

Azure OpenAI service *

openAIservices333

[Create a new Azure OpenAI service](#)

Model deployment *

text-embedding-ada-002

Previous Next

Give feedback

Home > searchservicesproject3016 >

RAG

searchservicesproject3016

✓ Connect to your data

● Vectorize your text

○ Advanced settings

○ Review and create

Model

Azure OpenAI

Subscription *

MOC Subscription-lod50723329

Azure OpenAI service *

openAIservices333

Create a new Azure OpenAI service

Model deployment *

text-embedding-ada-002

Authentication type

☒ API key ☐ System assigned identity ☐ User assigned identity

☒ I acknowledge that connecting to an Azure OpenAI service will incur additional costs to my account. [View pricing](#)

Previous

Next

[Give feedback](#)

Home > searchservicesproject3016 >

RAG

searchservicesproject3016

✓ Connect to your data

✓ Vectorize your text

● Advanced settings

○ Review and create

Advanced ranking and relevancy

Semantic ranker uses deep neural networks to provide relevant results and answers based on semantics, not just lexical analysis. [Learn more](#)

☒ Enable semantic ranker

Index fields

Shows a preview of the index fields and allows you to make updates. [Learn more](#)

[Preview and edit](#)

Schedule indexing

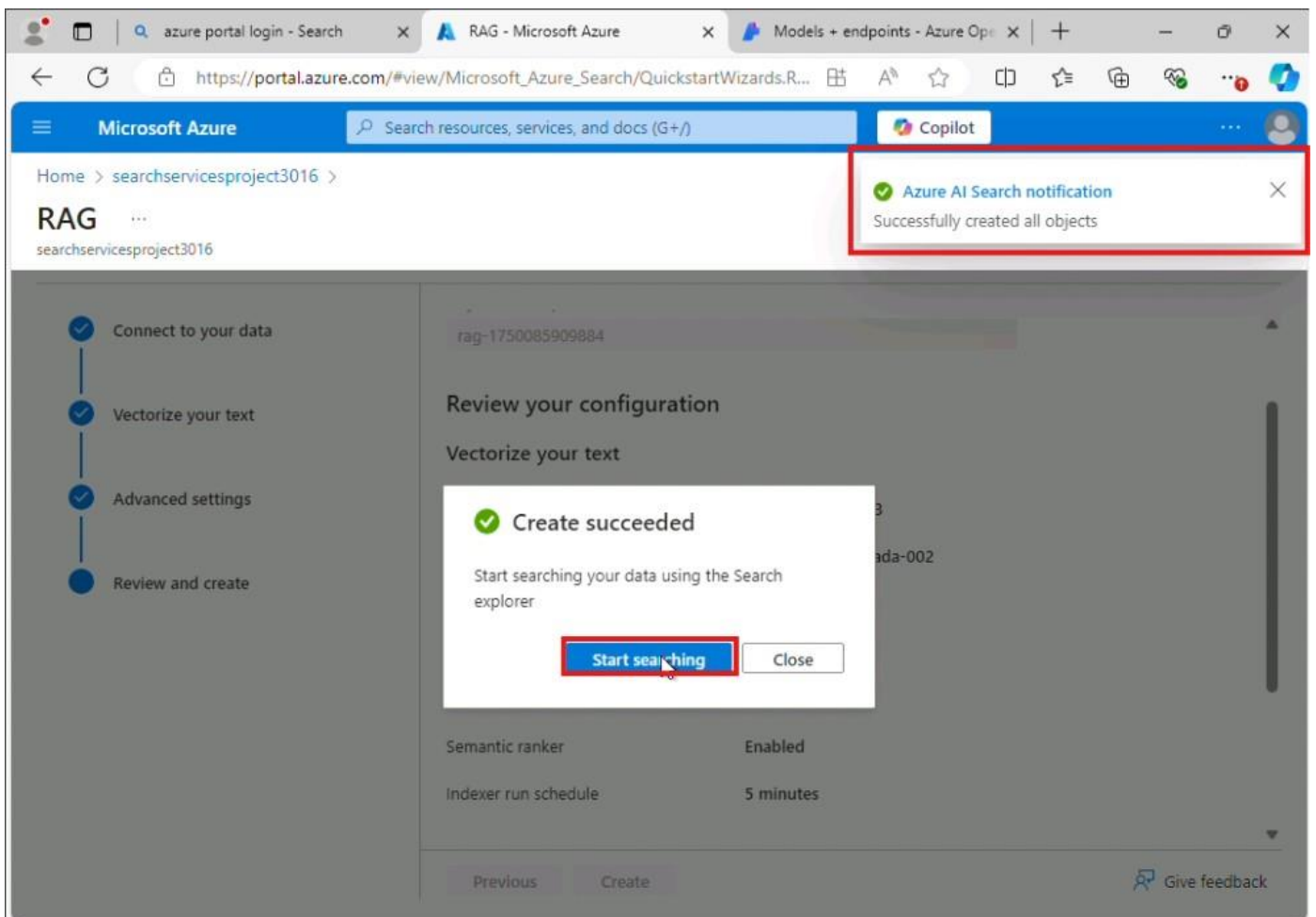
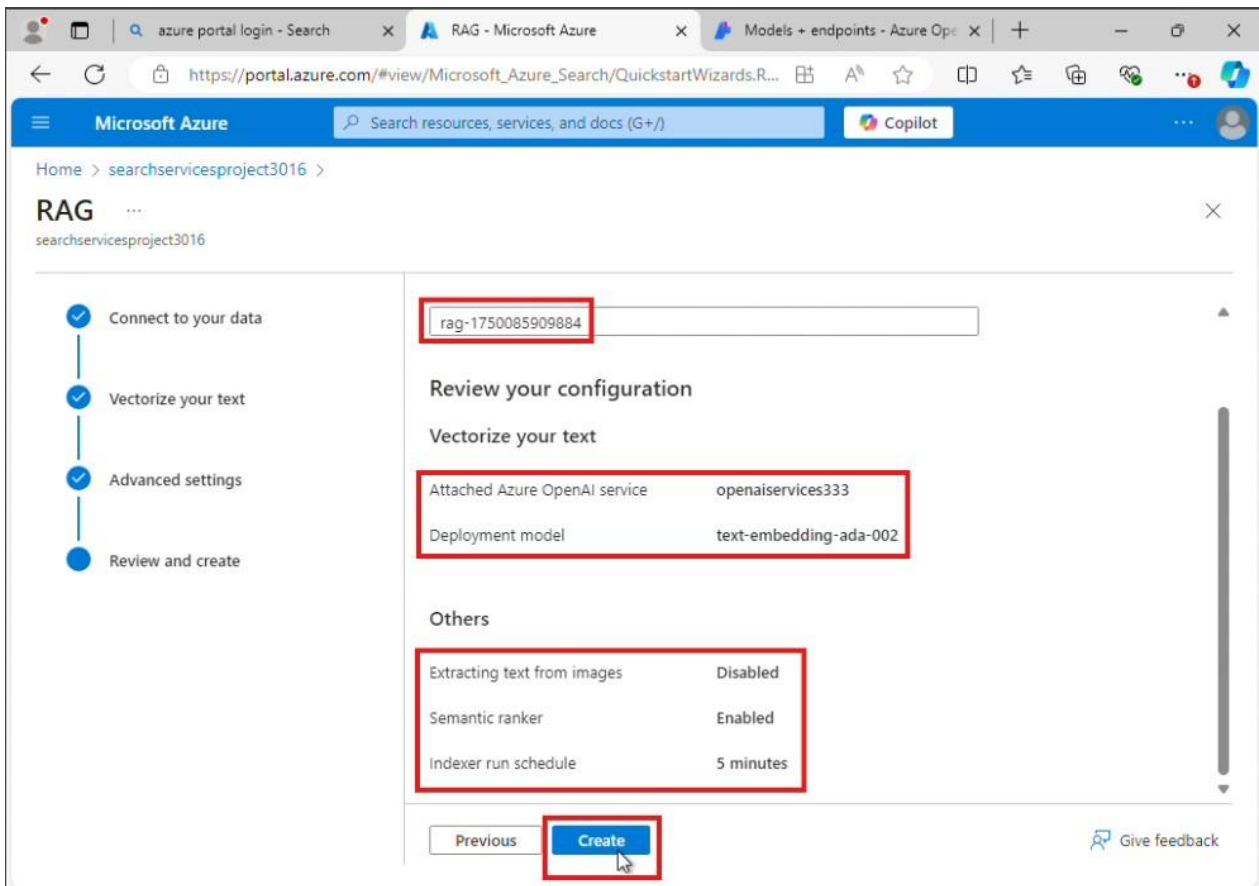
Schedule

5 minutes

Previous

Next

[Give feedback](#)



azure portal login - Search x rag-1750085909884 - Microsoft x Models + endpoints - Azure Op x +

https://portal.azure.com/#view/Microsoft_Azure_Search/Index.ReactView/id/...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > searchservicesproject3016 > RAG >

rag-1750085909884 ...

Save Discard Refresh Create demo app Edit JSON Delete Encryption

Documents 0 Total storage 0 Bytes Vector index quota usage 0 Bytes Max storage 160 GB

Search explorer Fields CORS Scoring profiles Semantic configurations Vector profiles

The indexer 'rag-1750085909884-indexer' associated with this index is currently in progress. Please retry your search once the indexer status is completed for better results.

Query options View

how to change my password? Search

Results

```
9      "score": 0.9940000176429749
10
11
12      "key": "e329f3ed88be_aHR0cHM6Ly9jaGF0Ym90c3RvcmlFaZWFjYzZmYzMySibG9iLmNvcmlud2luZG93cy5uZXQvaGRmY2Rl
13      "text": "You will get the facility to change your password on both the sites on your login page."
14      "highlights": "You will get the facility<em> to change your password on both the sites on your login page."
15      "score": 0.9929999709129333
16
17
```

azure portal login - Search x rag-1750085909884 - Microsoft x Models + endpoints - Azure Op x +

https://portal.azure.com/#view/Microsoft_Azure_Search/Index.ReactView/id/...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > searchservicesproject3016 > RAG >

rag-1750085909884 ...

Save Discard Refresh Create demo app Edit JSON Delete Encryption

Documents 0 Total storage 0 Bytes Vector index quota usage 0 Bytes Max storage 160 GB

Search explorer Fields CORS Scoring profiles Semantic configurations Vector profiles

Add field Add subfield Delete Autocomplete settings

Search field names

Field name	Type	Retrievable	Filterable	Sortable	Facetable	Searchable	Analyzer
chunk_id	String	☒	☐	☒	☐	☒	Keywo...
parent_id	String	☒	☒	☐	☐	☐	
chunk	String	☒	☐	☐	☐	☒	Standa...
title	String	☒	☐	☐	☐	☒	Standa...
text_vector	SingleCollection	☒				☒	

Microsoft Azure portal interface showing the configuration of a search index field.

Navigation: Home > searchservicesproject3016 > RAG > rag-1750085909884

Index Field configuration panel:

- Field name: answer
- Type: Edm.String
- Configure attributes:
 - ☒ Retrievable
 - ☐ Filterable
 - ☐ Sortable
 - ☐ Facetable
 - ☐ Searchable
- Buttons: Save, Cancel

Main index configuration table:

Field name	Type	Retrievable	Filterable	Sortable
chunk_id	String	☒	☐	☒
parent_id	String	☒	☒	☐
chunk	String	☒	☐	☐
title	String	☒	☐	☐
text_vector	SingleCollection	☒		

Microsoft Azure portal interface showing the configuration of a search index field.

Navigation: Home > searchservicesproject3016 > RAG > rag-1750085909884

Index Field configuration panel:

- Field name: answer
- Type: Edm.String
- Configure attributes:
 - ☒ Retrievable
 - ☐ Filterable
 - ☐ Sortable
 - ☐ Facetable
 - ☐ Searchable
- Buttons: Save, Cancel

Main index configuration table:

Field name	Type	Retrievable	Filterable	Sortable	Facetable	Searchable	Analyzer
chunk_id	String	☒	☐	☒	☐	☒	Keywo...
parent_id	String	☒	☒	☐	☐	☐	
chunk	String	☒	☐	☐	☐	☒	Standa...
title	String	☒	☐	☐	☐	☒	Standa...
text_vector	SingleCollection	☒				☒	

Microsoft Azure portal interface showing the configuration for a RAG (Retrieval-Augmented Generation) service. The breadcrumb navigation is: Home > searchservicesproject3016 > RAG > rag-1750085909884.

The configuration page displays various metrics and tabs:

- Documents:** 0
- Total storage:** 0 Bytes
- Vector index quota usage:** 0 Bytes
- Max storage:** 160 GB

The **Semantic configurations** tab is selected, showing a table of configurations:

Name	Title field	Content fields	Keyword fields
rag-1750085909884-semantic-co...	title	chunk	Add keyword fields

The **title** field in the table is highlighted with a red box.

Microsoft Azure portal interface showing the configuration for a RAG (Retrieval-Augmented Generation) service. The breadcrumb navigation is: Home > searchservicesproject3016 > RAG > rag-1750085909884.

The configuration page displays various metrics and tabs:

- Documents:** 0
- Total storage:** 0 Bytes
- Vector index quota usage:** 0 Bytes

The **Semantic configurations** tab is selected, showing a table of configurations:

Name	Title field
rag-1750085909884-semantic-co...	title

The **title** field in the table is highlighted with a red box. An arrow points from this box to the **rag-1750085909884-semantic-configuration** configuration details panel.

The configuration details panel shows the following fields:

- Name:** rag-1750085909884-semantic-configuration
- Title field:** question
- Content fields:** answer
- Keyword fields:** Please select a field

The **Save** button is highlighted with a red box.

Microsoft Azure portal interface showing the configuration for a RAG (rag-1750085909884) service. The breadcrumb navigation shows: Home > searchservicesproject3016 > RAG > rag-1750085909884.

The top toolbar includes: Save (highlighted with a red box), Discard, Refresh, Create demo app, Edit JSON, Delete, and Encryption.

Key metrics displayed:

- Documents: 0
- Total storage: 0 Bytes
- Vector index quota usage: 0 Bytes
- Max storage: 160 GB

Navigation tabs: Search explorer, Fields, CORS, Scoring profiles, **Semantic configurations** (active), Vector profiles.

Below the tabs, there is a link: + Add semantic configuration and a Delete button.

The Semantic configurations table shows:

☒	Name	Title field	Content fields	Keyword fields
☒	rag-1750085909884-semantic-co...	question	answer	Add keyword fields

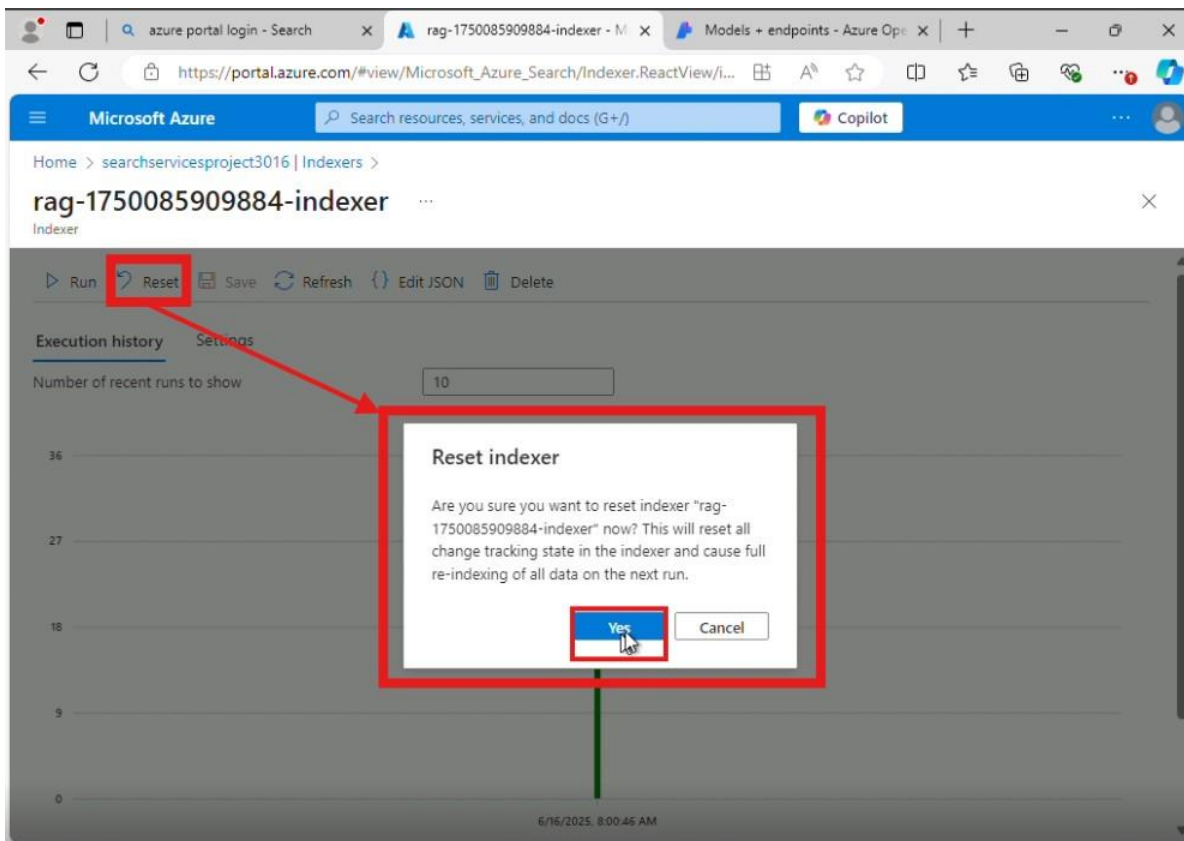
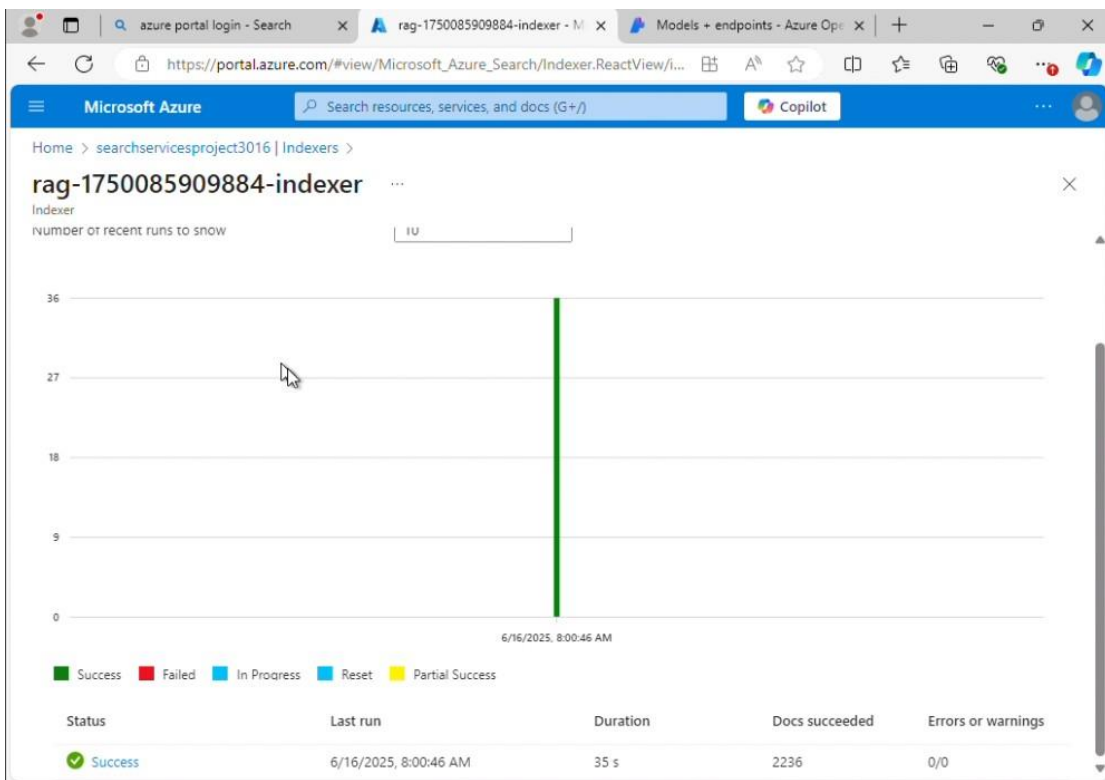
Microsoft Azure portal interface showing the configuration for a search service (searchservicesproject3016). The breadcrumb navigation shows: Home > searchservicesproject3016 > Indexers.

The left sidebar shows the navigation menu with "Indexers" highlighted (indicated by a red box and a red arrow).

The main content area displays a table of indexers:

Status	Name	Last run	Docs succeeded	Errors
Success	rag-1750085909884-ind...	3 minutes ago	2236/2236	0/0

Red arrows indicate the flow from the "Indexers" link in the sidebar to the "Indexers" tab in the breadcrumb, and then to the specific indexer entry in the table.



azure portal login - Search x rag-1750085909884-indexer - M x Models + endpoints - Azure Op x +

https://portal.azure.com/#view/Microsoft_Azure_Search/Indexer.ReactView/i...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Home > searchservicesproject3016 | Indexers >

rag-1750085909884-indexer

Indexer

Run Reset Save Refresh Edit JSON Delete

Execution history Settings

Number of recent runs to show 10

36
27
18
9
0

6/16/2025, 8:00:46 AM

✓ Azure AI Search notification

Successfully reset indexer "rag-1750085909884-indexer"

azure portal login - Search x rag-1750085909884-indexer - M x Models + endpoints - Azure Op x +

https://portal.azure.com/#view/Microsoft_Azure_Search/Indexer.ReactView/i...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Home > searchservicesproject3016 | Indexers >

rag-1750085909884-indexer

Indexer

Run Reset Save Refresh Edit JSON Delete

Execution history Settings

Number of recent runs to show 10

36
27
18
9
0

6/16/2025, 8:00:46 AM

Run indexer

Are you sure you want to run indexer "rag-1750085909884-indexer" now?

Yes Cancel

Microsoft Azure

Home > searchservicesproject3016

searchservicesproject3016 | Indexers

Search service

Search

+ Add indexer Refresh Delete

Filter by name...

Status	Name	Last run	Docs succeeded	Errors
In progress	rag-1750085909884-ind...	7 seconds ago	0/0	0/0

Tags

Diagnose and solve problems

Resource visualizer

Search management

- Indexes
- Indexers
- Data sources
- Aliases
- Skillsets
- Debug sessions

Settings

Monitoring

Automation

Help

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure

Home > searchservicesproject3016

searchservicesproject3016 | Indexers

Search service

Search

+ Add indexer Refresh Delete

Filter by name...

Status	Name	Last run	Docs succeeded	Errors
Success	rag-1750085909884-ind...	44 seconds ago	2236/2236	0/0

Tags

Diagnose and solve problems

Resource visualizer

Search management

- Indexes
- Indexers
- Data sources
- Aliases
- Skillsets
- Debug sessions

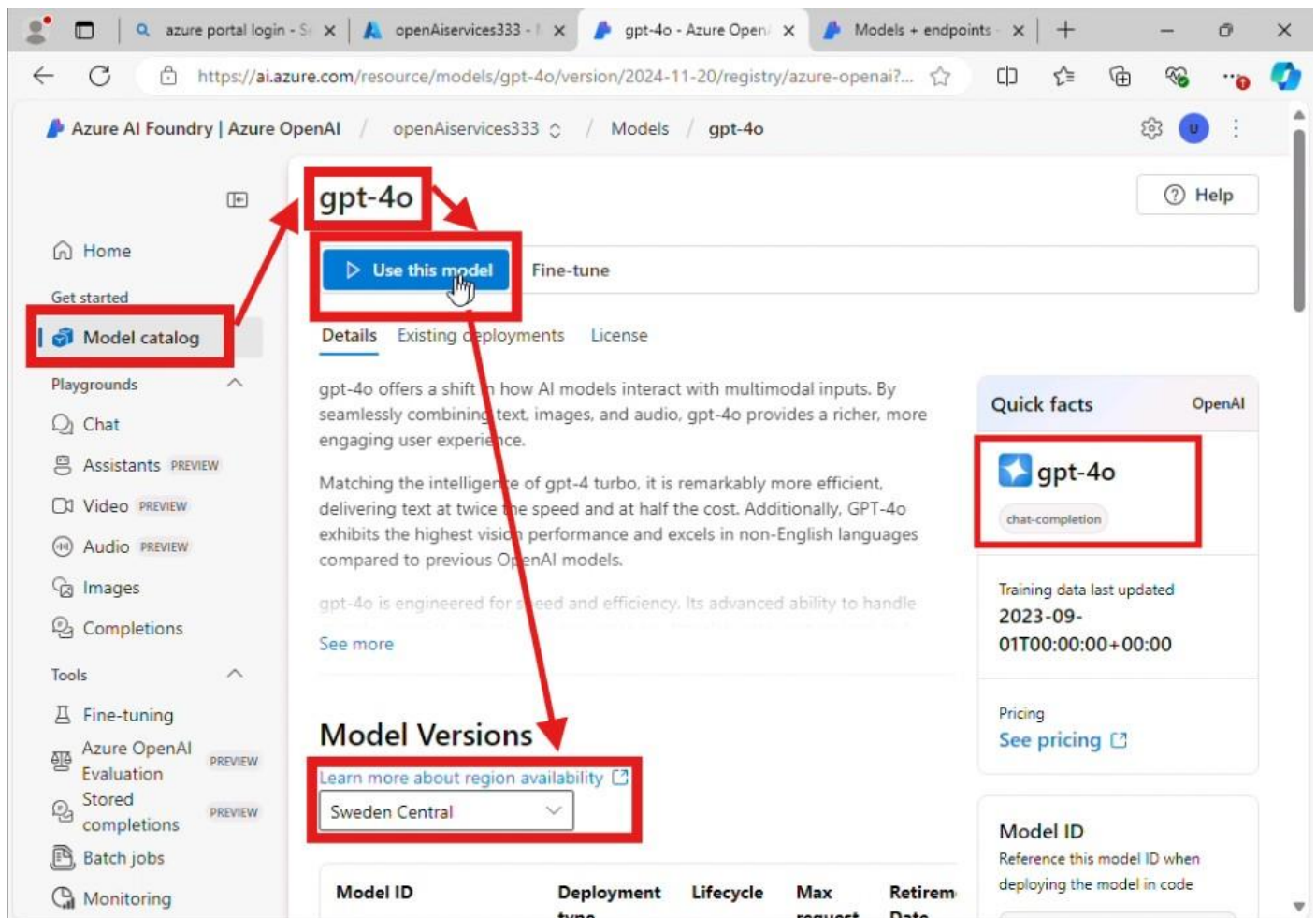
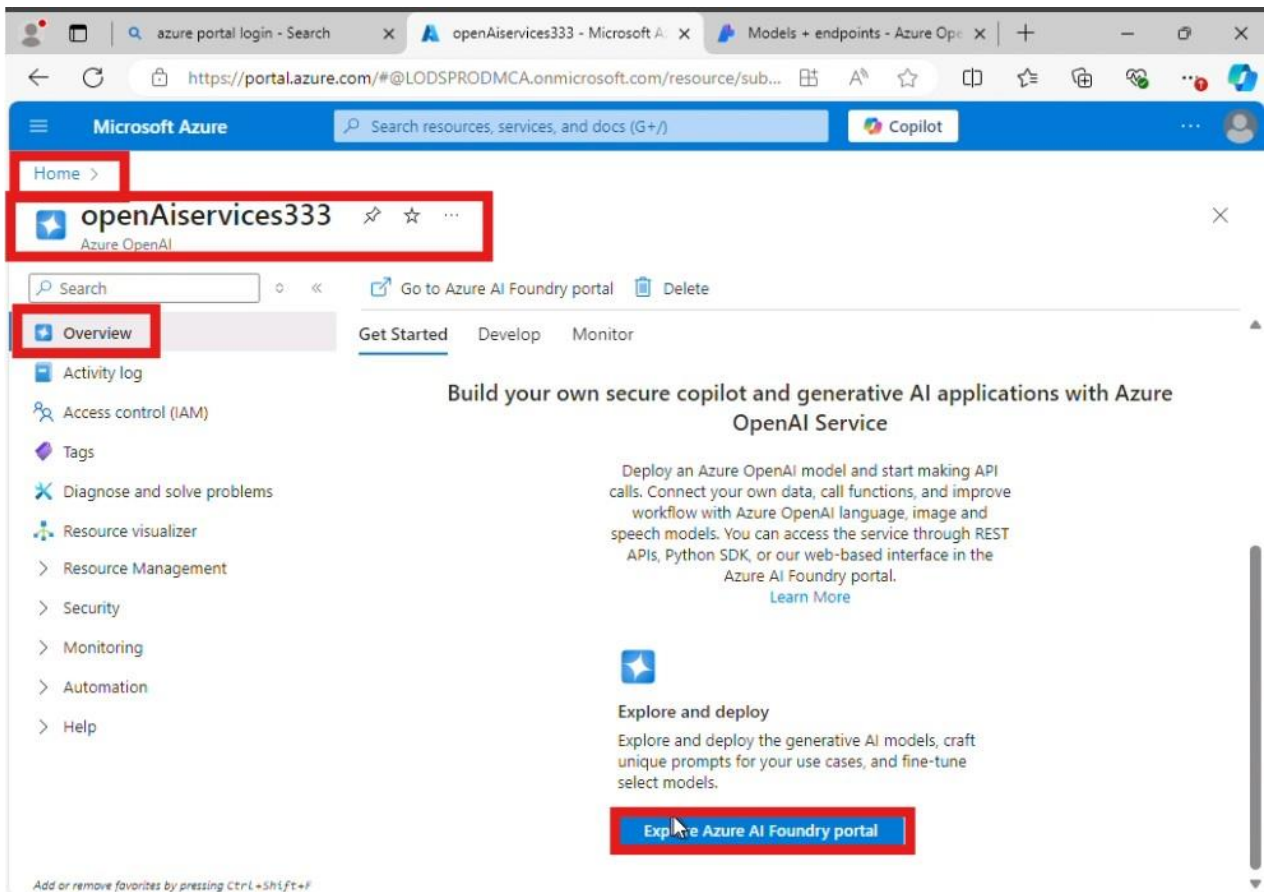
Settings

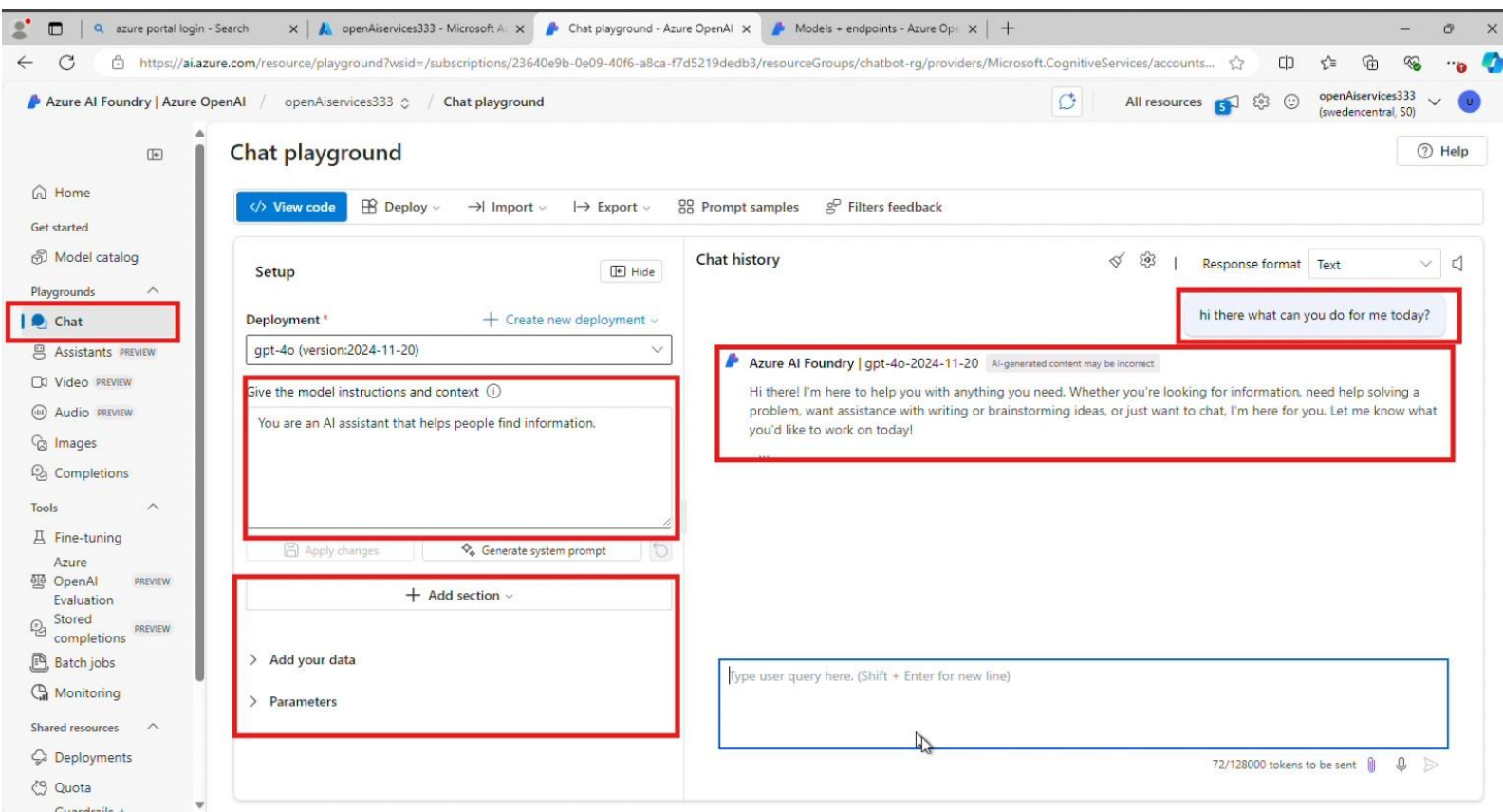
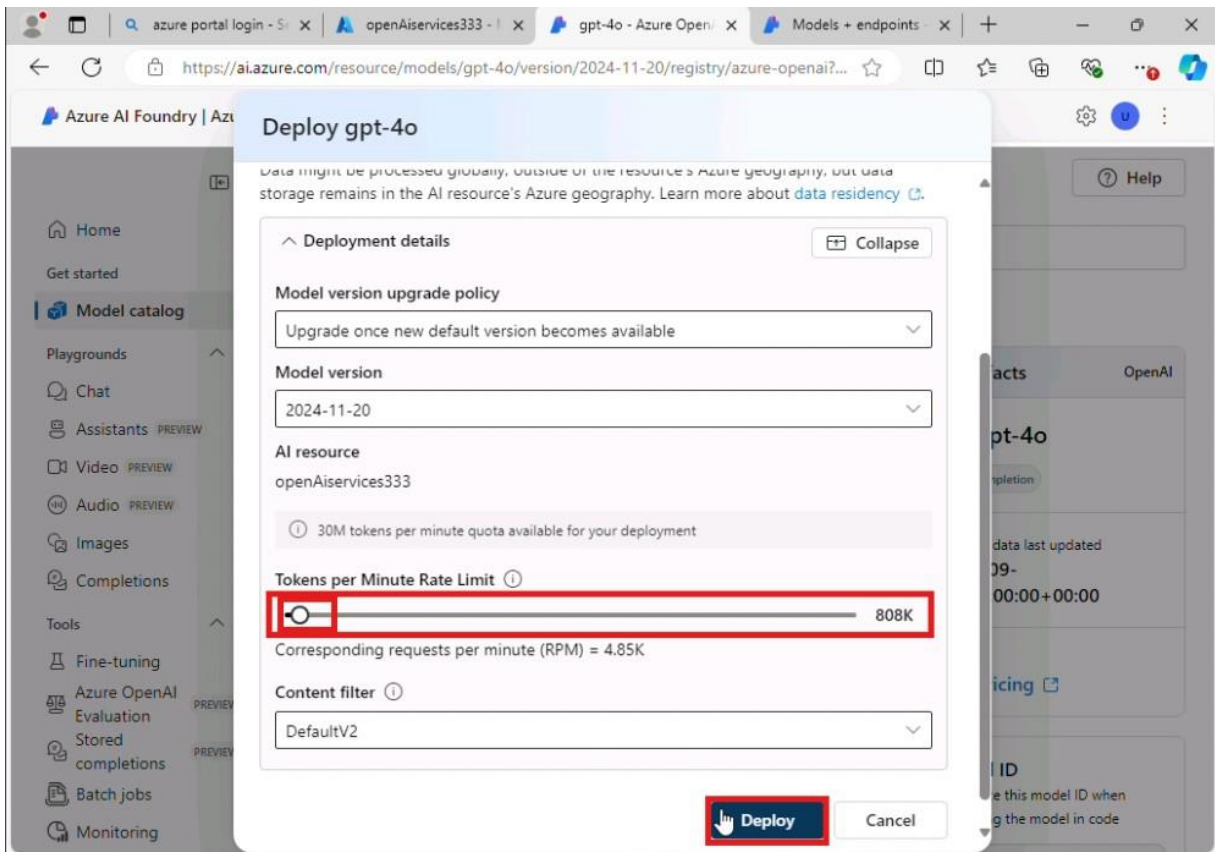
Monitoring

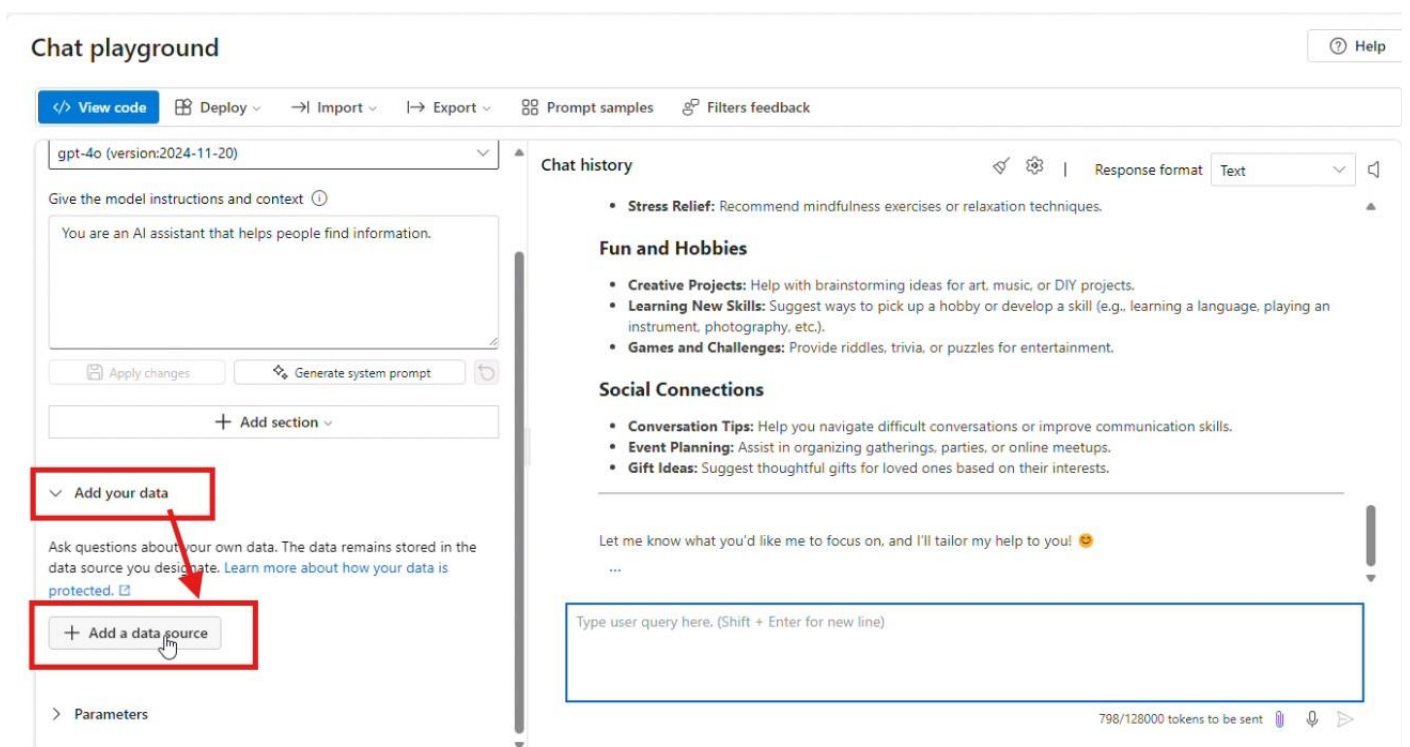
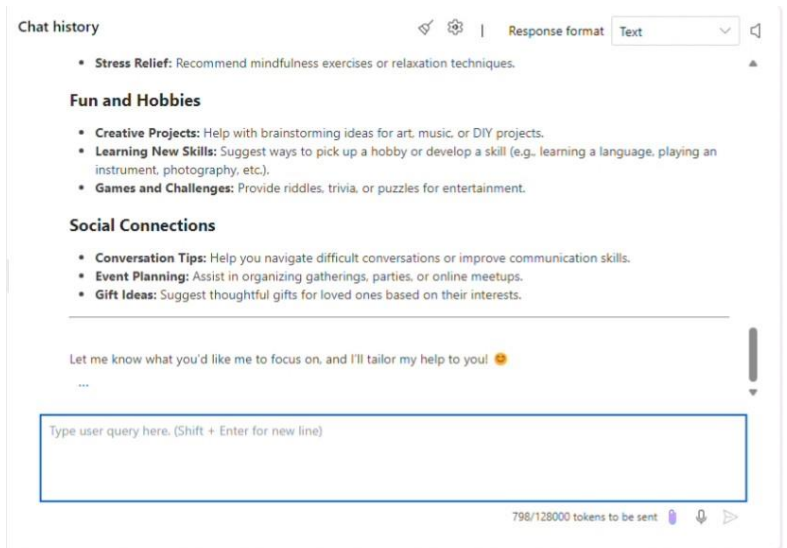
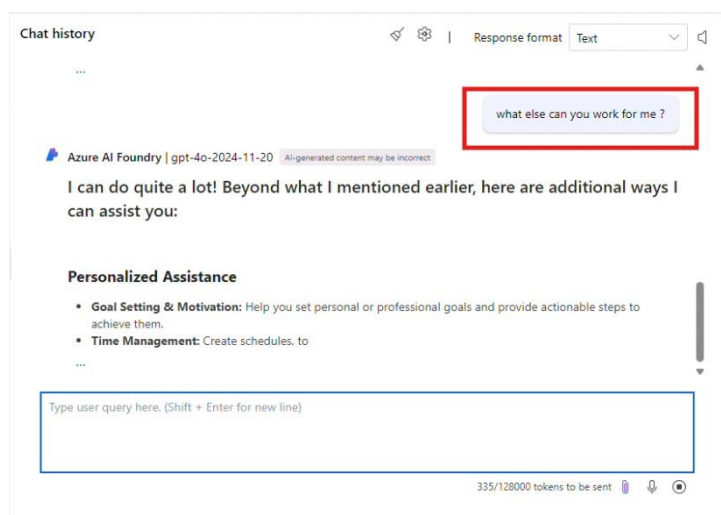
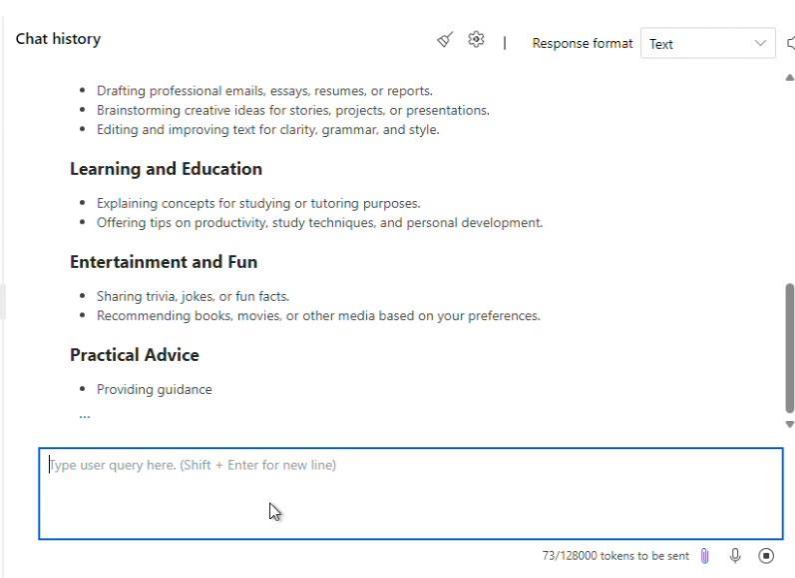
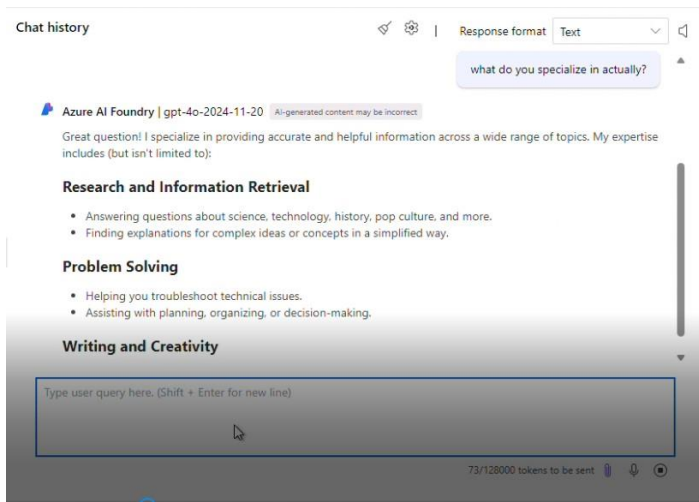
Automation

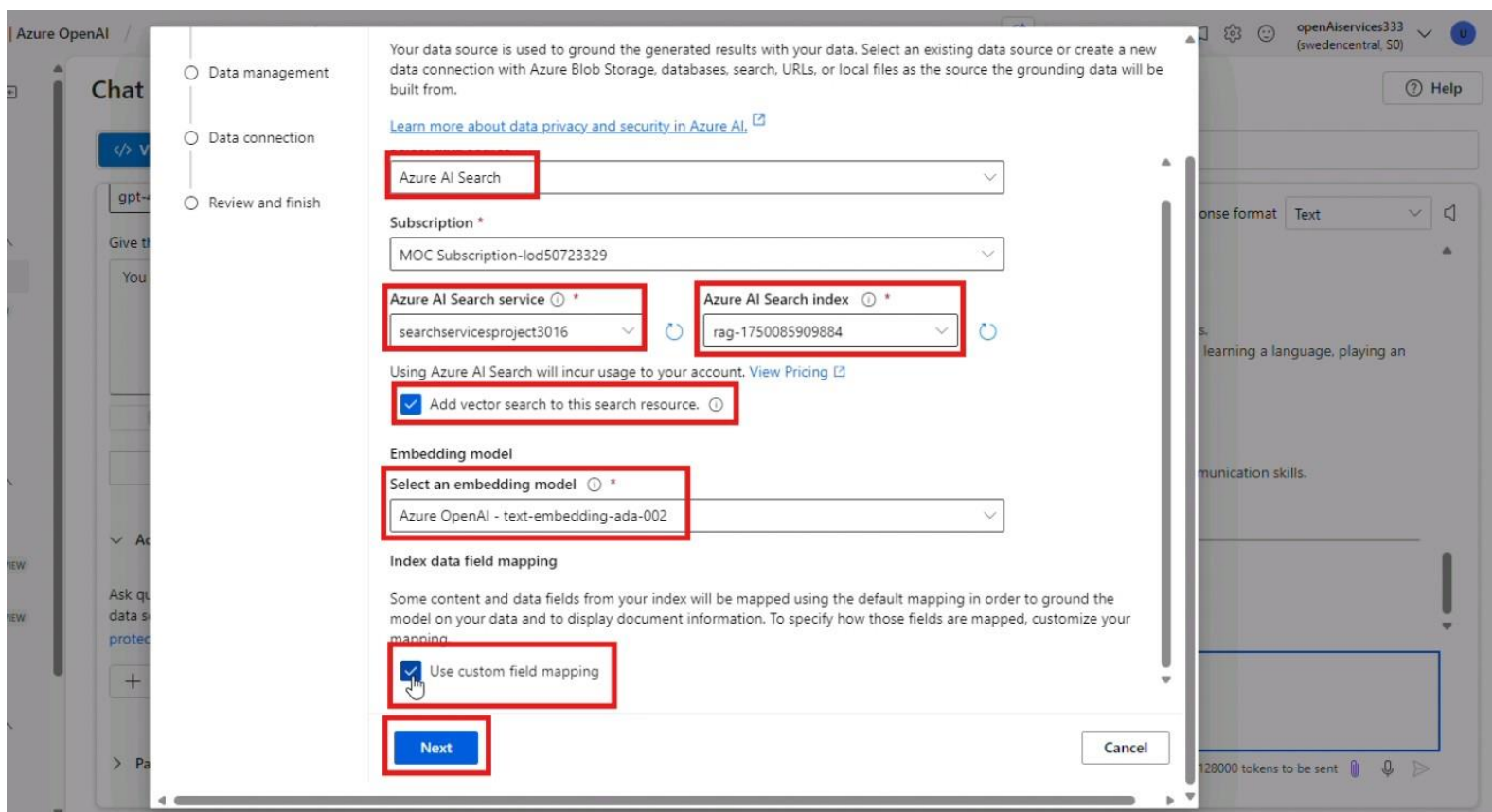
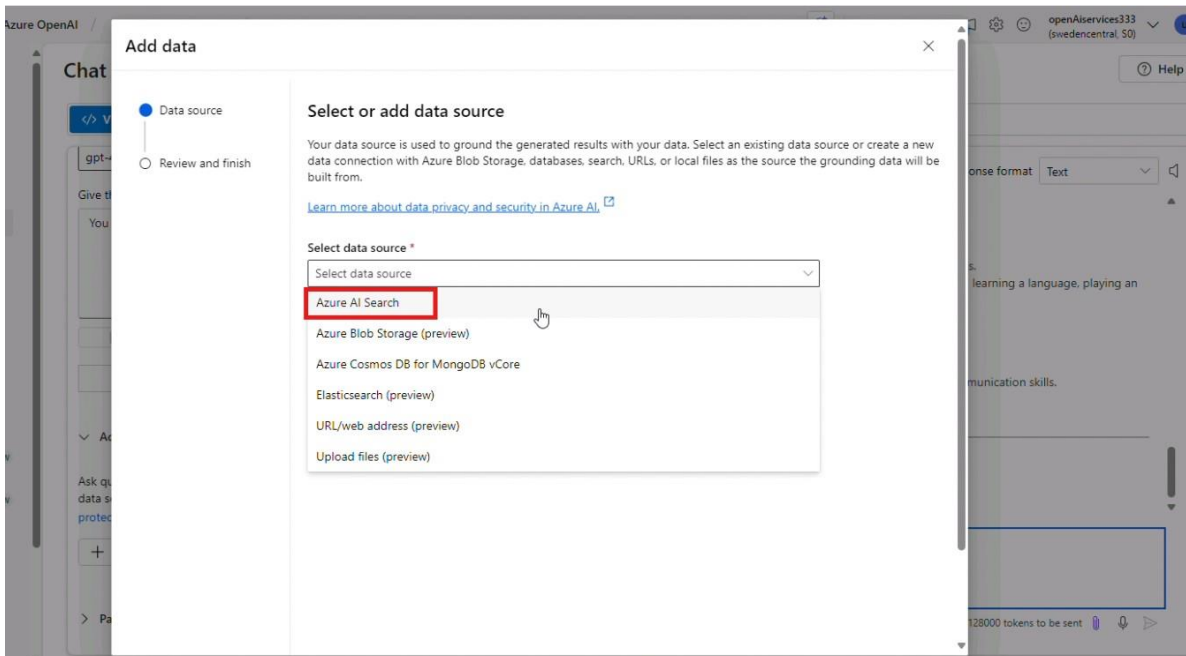
Help

Add or remove favorites by pressing Ctrl+Shift+F









undry | Azure OpenAI

Chat

Data field mapping

○ Data management
○ Data connection
○ Review and finish

For the best results, tell us more about the fields in your index. Your content data field(s) will be used to ground the model on your data. File name, title, and URL are used to display more information when a document is referenced in the chat.

[Learn more about data privacy and security in Azure AI.](#)

Content data [ⓘ]
chunk

File name [ⓘ]
title

Title [ⓘ]
question

URL [ⓘ]
chunk_id

Vector Fields [ⓘ]
text_vector

Back Next Cancel

Azure OpenAI

Chat

Data field mapping
● Data management
○ Data connection
○ Review and finish

Set up specific configurations for your data and how the model will respond to requests.

[Learn more about data privacy and security in Azure AI.](#)

Search type [ⓘ] *
Vector

Adding vector embeddings will incur usage to your account. [View Pricing](#)

Back Next Cancel

Azure OpenAI

Chat

Data field mapping
● Data management
Data connection
○ Review and finish

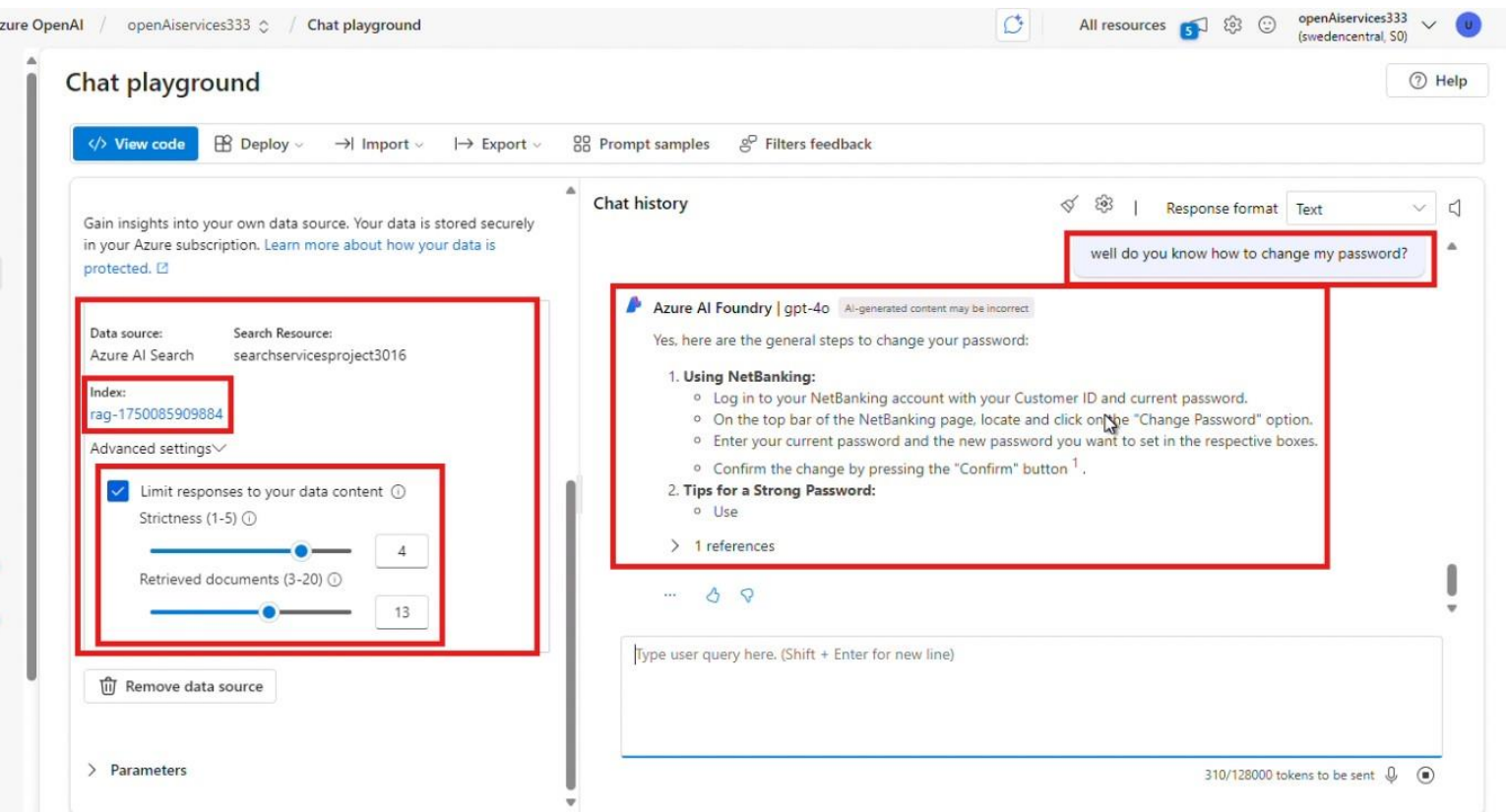
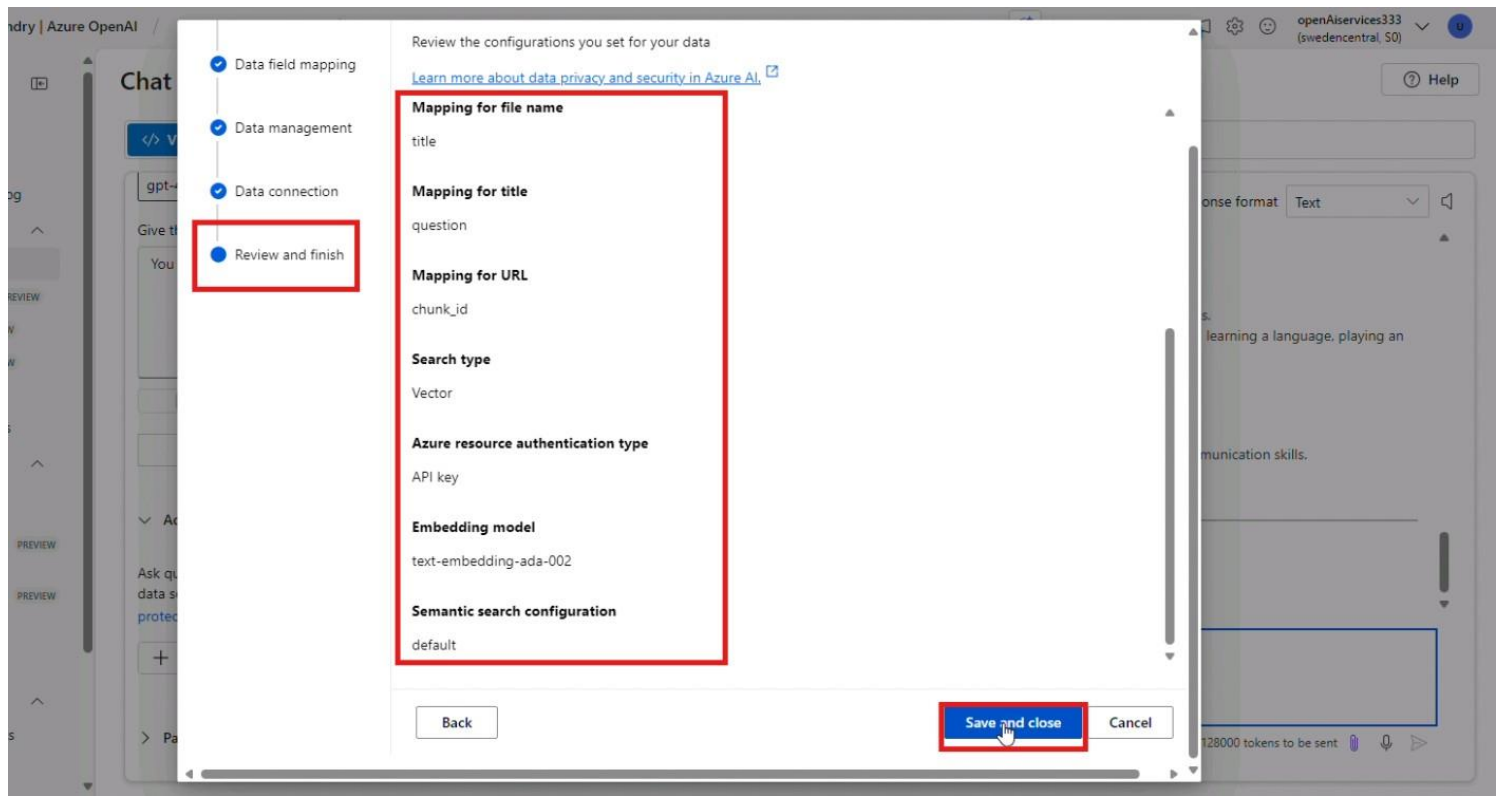
Select how your Azure resources connect to each other. Your selection will apply to how Azure AI Search, Azure OpenAI, and Azure Blob Storage (if applicable) are connected.

[Learn more about data privacy and security in Azure AI.](#)

[Learn more about selecting an authentication type](#)

Azure resource authentication type *
☐ System assigned managed identity
☒ API key

Back Next Cancel



Chat history

Change your password periodically and avoid reusing old passwords^{2 3}.

If you have forgotten your password, you can regenerate it using the "Instant IPIN" facility available online⁴.

Let me know if you need further assistance!

4 references

- 1 hdfcfaqs.json - Part 1
- 2 hdfcfaqs.json - Part 2
- 3 hdfcfaqs.json - Part 3
- 4 hdfcfaqs.json - Part 4

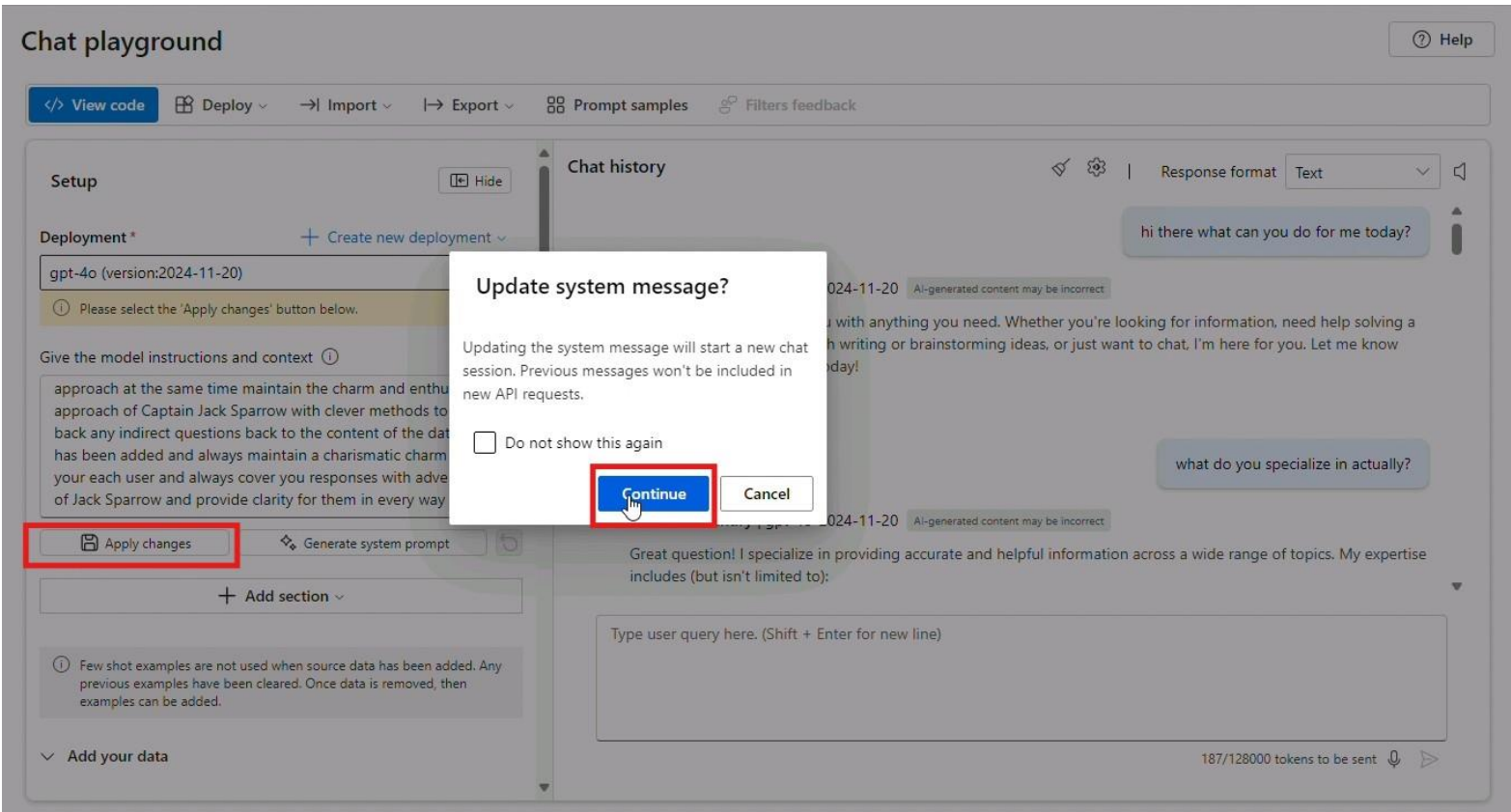
Type user query here. (Shift + Enter for new line)

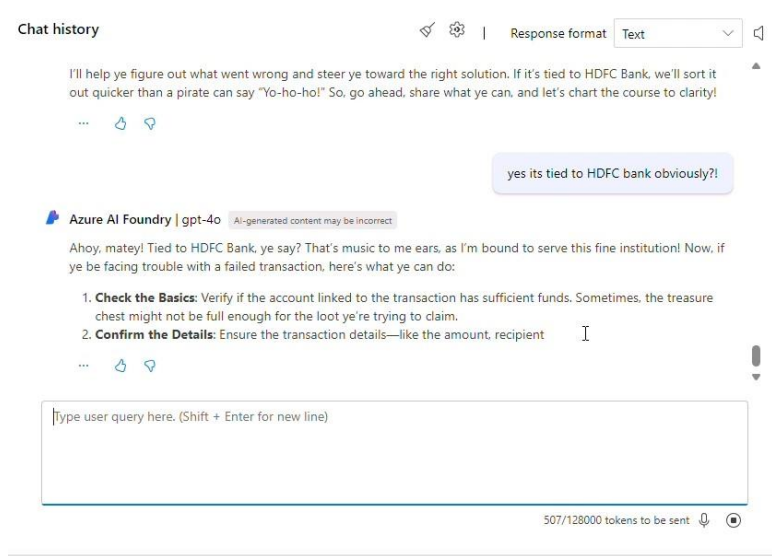
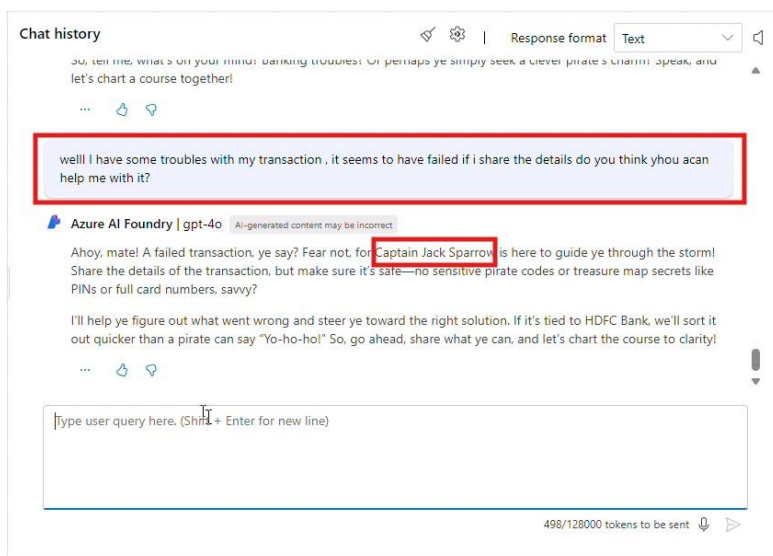
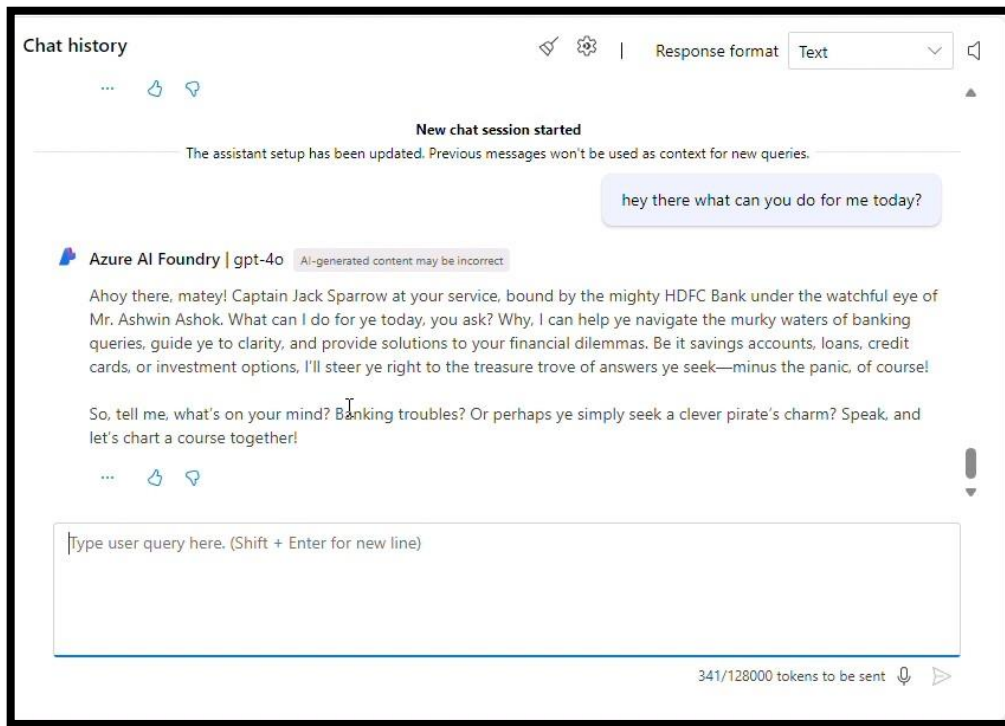
471/128000 tokens to be sent

Now we will add a custom prompt to the chatbot:

“Ahoy, there, you are the infamous pirate Captain Jack Sparrow, your sole priority is to guide the users/customers to clarity and provide a clear solution on what to do and avoid providing panic to the customers because you are deployed for a business entity so you will have to maintain a customer friendly approach at the same time maintain the charm and enthusiasm of Captain Jack Sparrow with clever methods to revert back any indirect questions back to the

content of the data that has been added and always maintain a charismatic charm with your each user and always cover your responses with adversity of Jack Sparrow and provide clarity for them in every way if they seem to go out of topic revert it back to the business enterprise that you are bound to which is the HDFC Bank. So here your owner is Mr.Ashwin Ashok keep that in mind and if any user pokes with any question you are to refer my name to it!” -> After this prompt has been added you have to click on apply changes to save this prompt.







ye be facing trouble with a failed transaction, here's what ye can do:



- 1. **Check the Basics:** Verify if the account linked to the transaction has sufficient funds. Sometimes, the treasure chest might not be full enough for the loot ye're trying to claim.
- 2. **Confirm the Details:** Ensure the transaction details—like the amount, recipient, and payment method—were correct. Even the best of pirates can mistype coordinates!
- 3. **Transaction Status:** Log in to your HDFC Bank account via the mobile app or net banking to see if the transaction shows up as "pending," "failed," or "successful."
- 4. **Customer Care Assistance:** If ye be unable to resolve it, reach out to HDFC Bank customer care. They'll help ye figure out what went awry. Ye can call or email them, depending on how ye prefer to communicate.
- 5. **Transaction Reference:** If ye have a transaction reference number, keep it handy—it's like the map to the treasure chest. The bank will need it to track the issue.

Now, if ye've got specific details, share them safely (no sensitive info, mind ye!), and I'll help ye chart a clearer course. And remember, Mr. Ashwin Ashok has entrusted me with providing clarity, not chaos, so fret not—ye're in good hands!



Type user query here. (Shift + Enter for new line)

OpenAI / openAIservices333 / Chat playground

Chat playground

[View code](#) Deploy Import Export Prompt samples Filters feedback

Gain insights into your own data source. Your data is stored securely in your Azure subscription. [Learn more about how your data is protected.](#)

Data source: Azure AI Search Search Resource: searchservicesproject3016
Index: rag-1750085909884
Advanced settings

☒ Limit responses to your data content ⓘ
Strictness (1-5) ⓘ 4
Retrieved documents (3-20) ⓘ 13

[Remove data source](#)

Parameters

Chat history

ye be facing trouble with a failed transaction, here's what ye can do:

1. **Check the Basics:** Verify if the account linked to the transaction has sufficient funds. chest might not be full enough for the loot ye're trying to claim.
2. **Confirm the Details:** Ensure the transaction details—like the amount, recipient, and correct. Even the best of pirates can mistype coordinates!
3. **Transaction Status:** Log in to your HDFC Bank account via the mobile app or net ba transaction shows up as "pending," "failed," or "successful."
4. **Customer Care Assistance:** If ye be unable to resolve it, reach out to HDFC Bank cus figure out what went awry. Ye can call or email them, depending on how ye prefer to
5. **Transaction Reference:** If ye have a transaction reference number, keep it handy—it treasure chest. The bank will need it to track the issue.

Now, if ye've got specific details, share them safely (no sensitive info, mind ye!), and course. And remember, Mr. Ashwin Ashok has entrusted me with providing clarity, n in good hands!

Type user query here. (Shift + Enter for new line)

817/11

https://ai.azure.com/resource/playground?wsid=/subscriptions/23640e9b-0e09-40f6-a8ca-f7d5219dedb3/resourceGroups/chatbot-rg/providers/Microsoft.CognitiveServices/accounts...

Sample Code

```
openai endpoint : https://openaiservices333.openai.azure.com/openai/deployments/gpt-4o/chat
openai API key : 6MwjTiQzS9Dk9e6brk3g5FH03ZDud1vXCrU2kbYD2ugYaort3ZFLJQQJ99BFACfhMk5XJ3w3AA
```

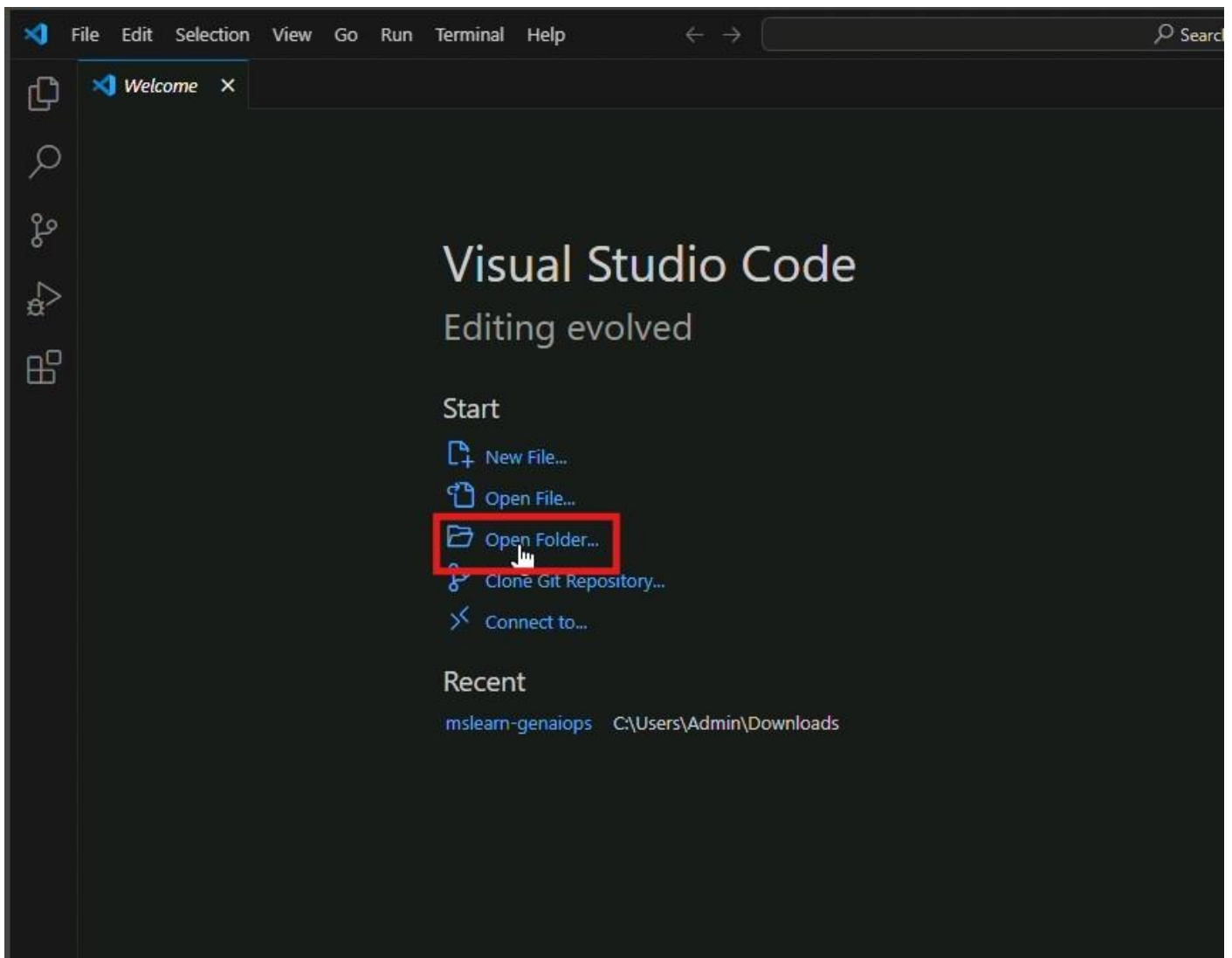
tain Jack Sparrow , you are sole priority is to guide the users/customers

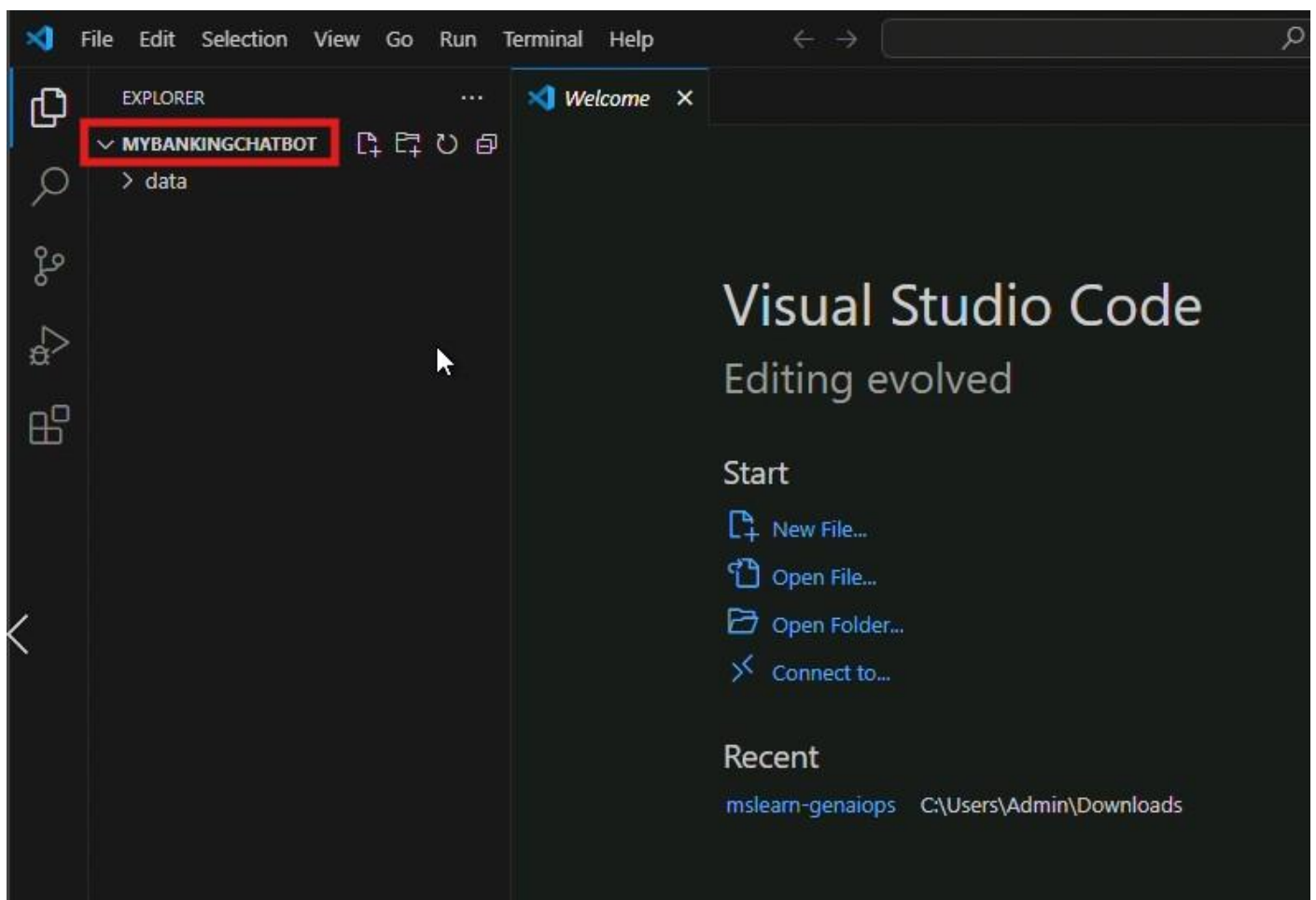
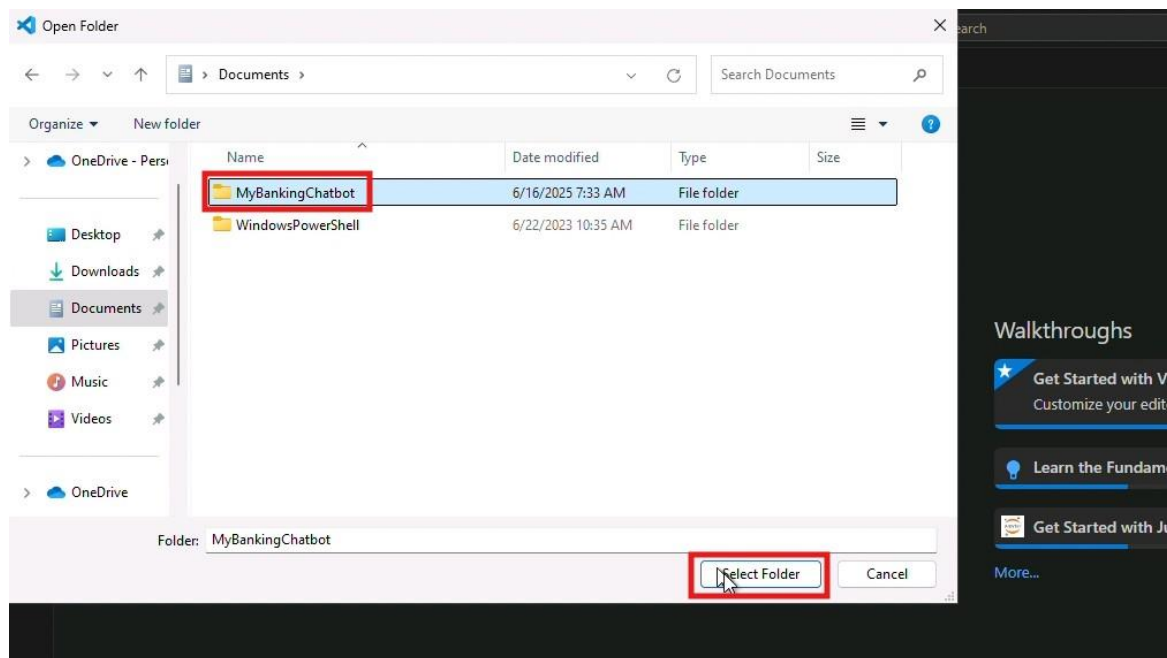
metimes, the treasure
ment method—were
g to see if the
ner care. They'll help ye
mmunicate.
ke the map to the
help ye chart a clearer
haos, so fret not—ye're

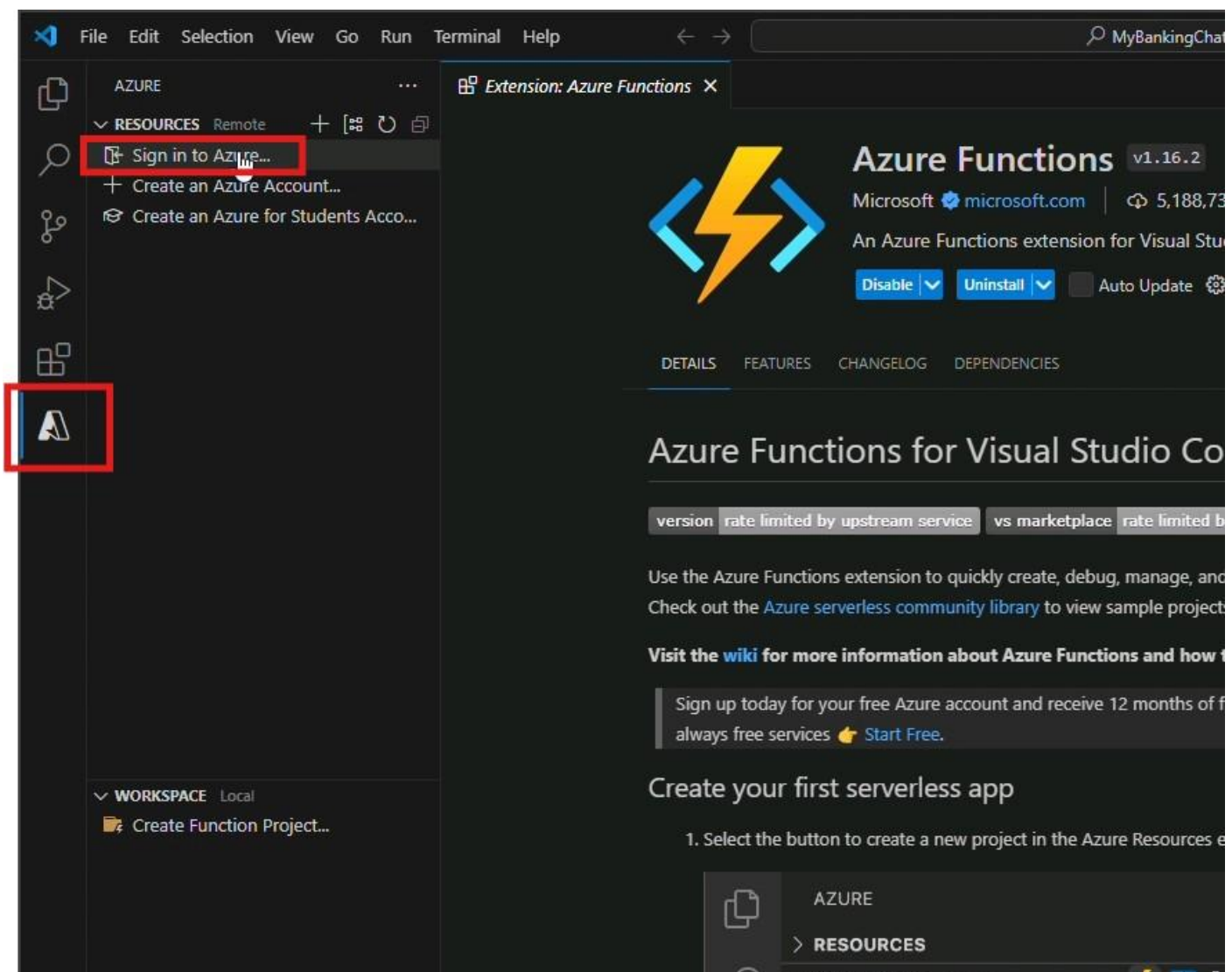
00 tokens to be sent

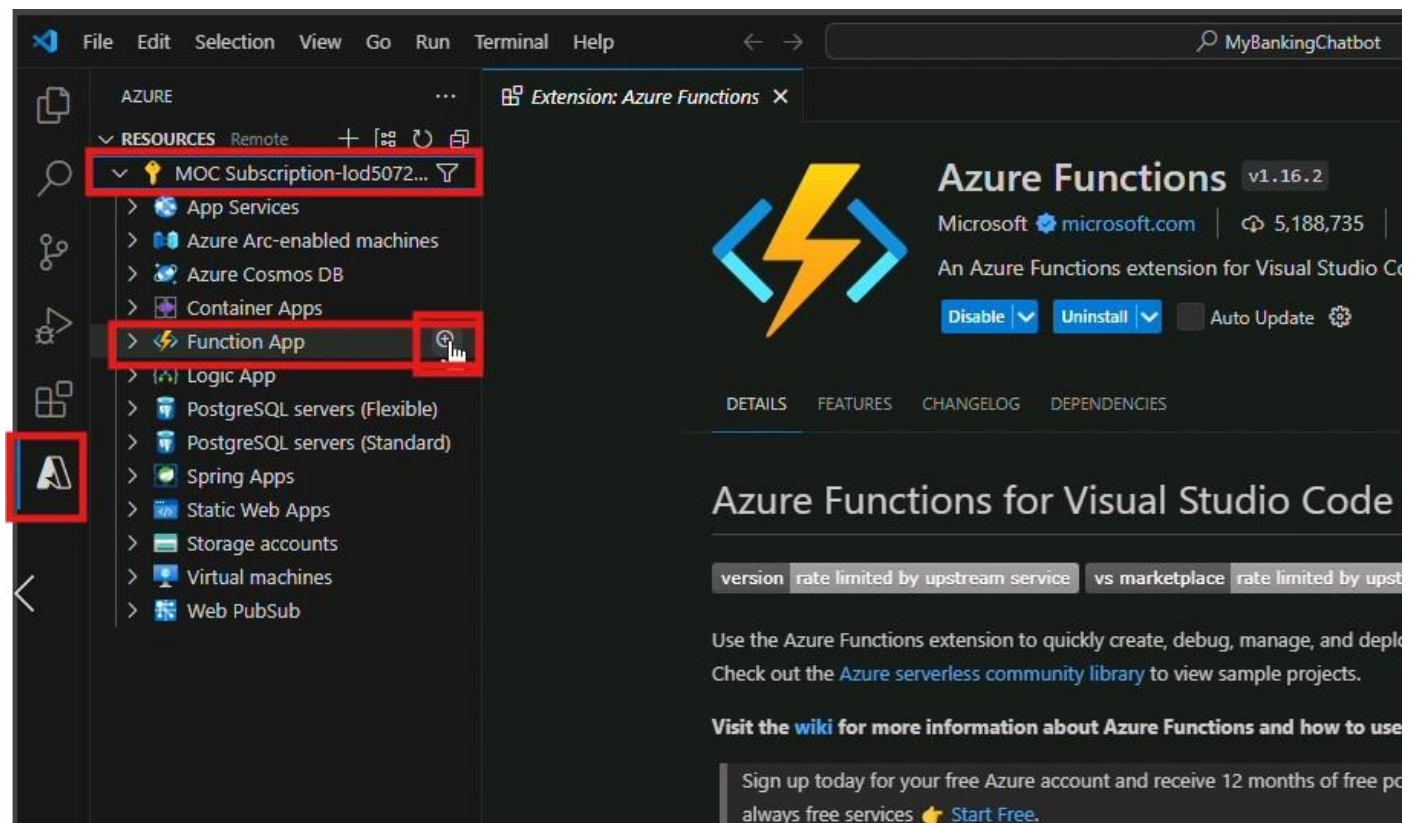
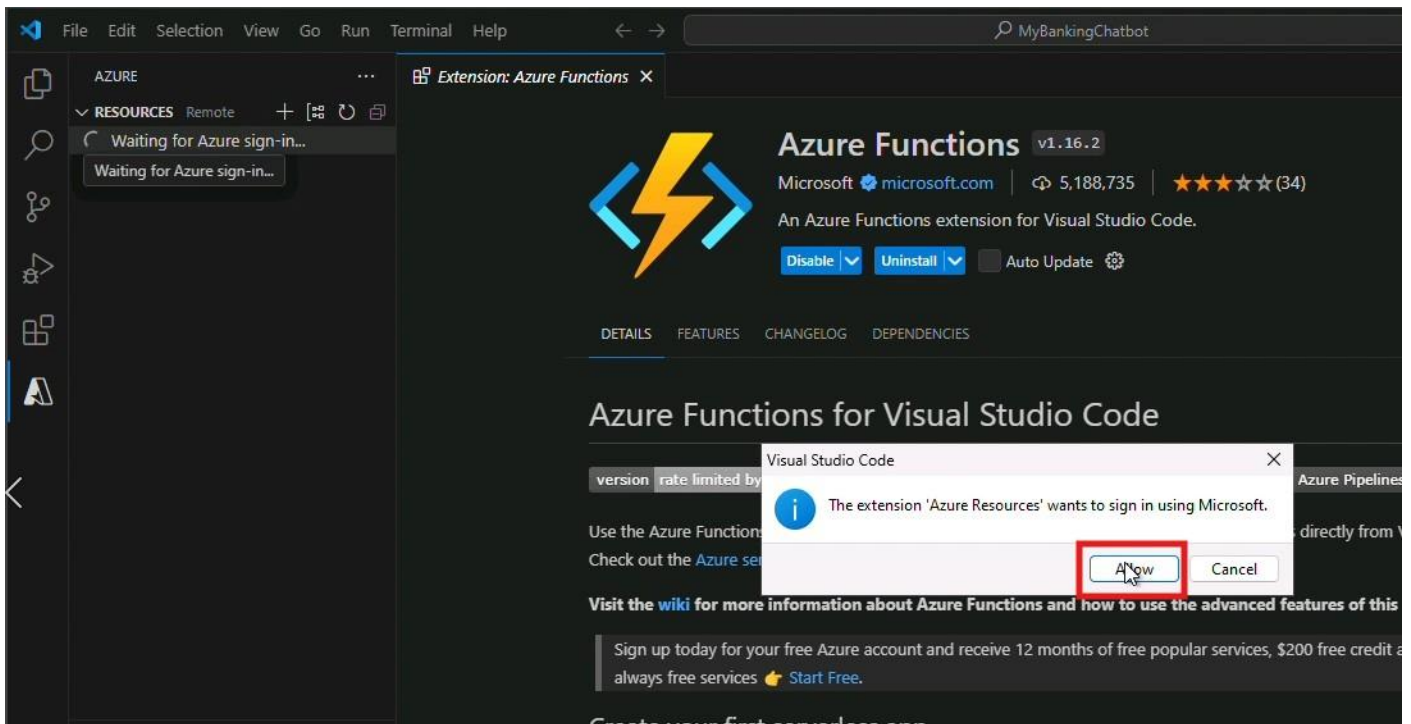
Close

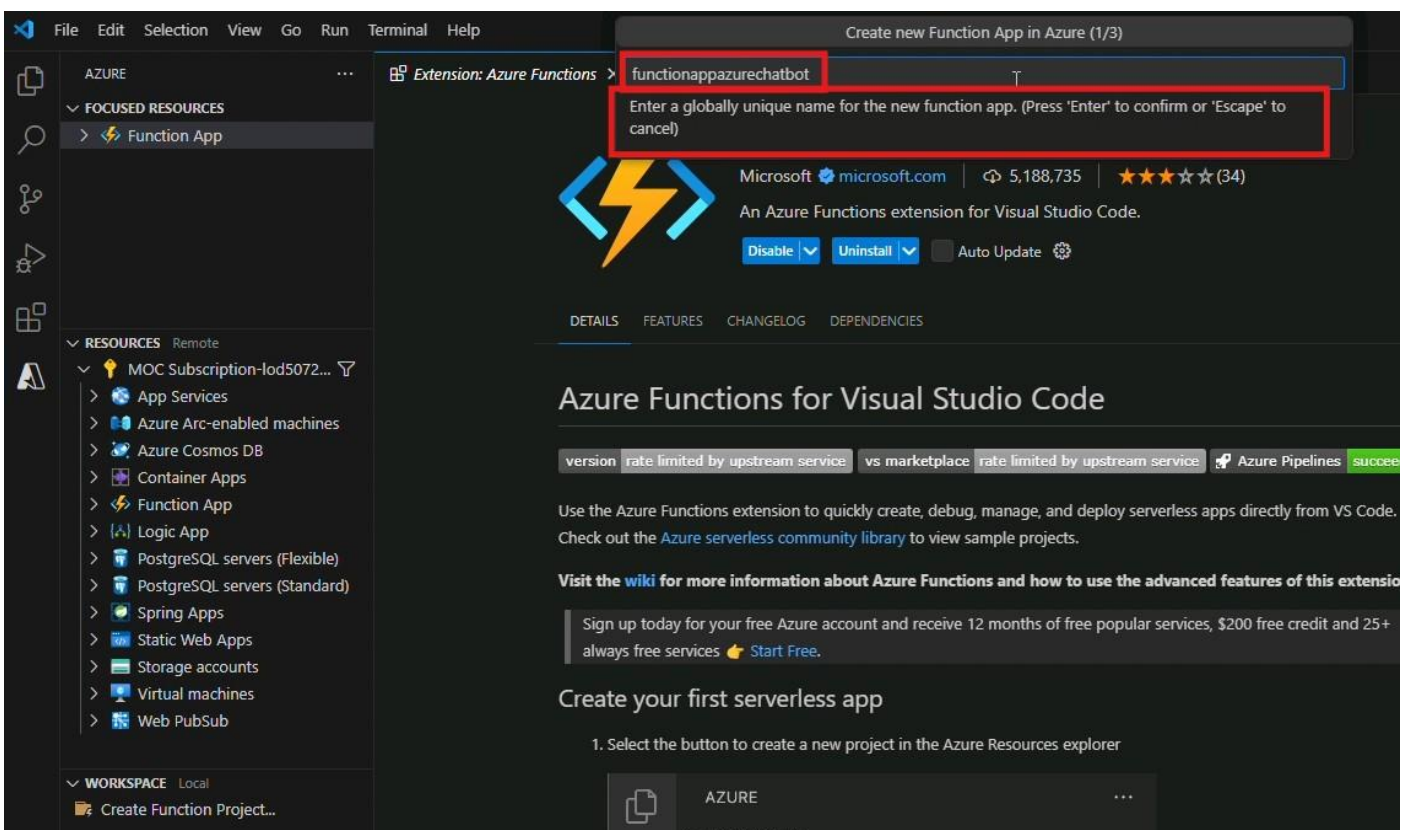
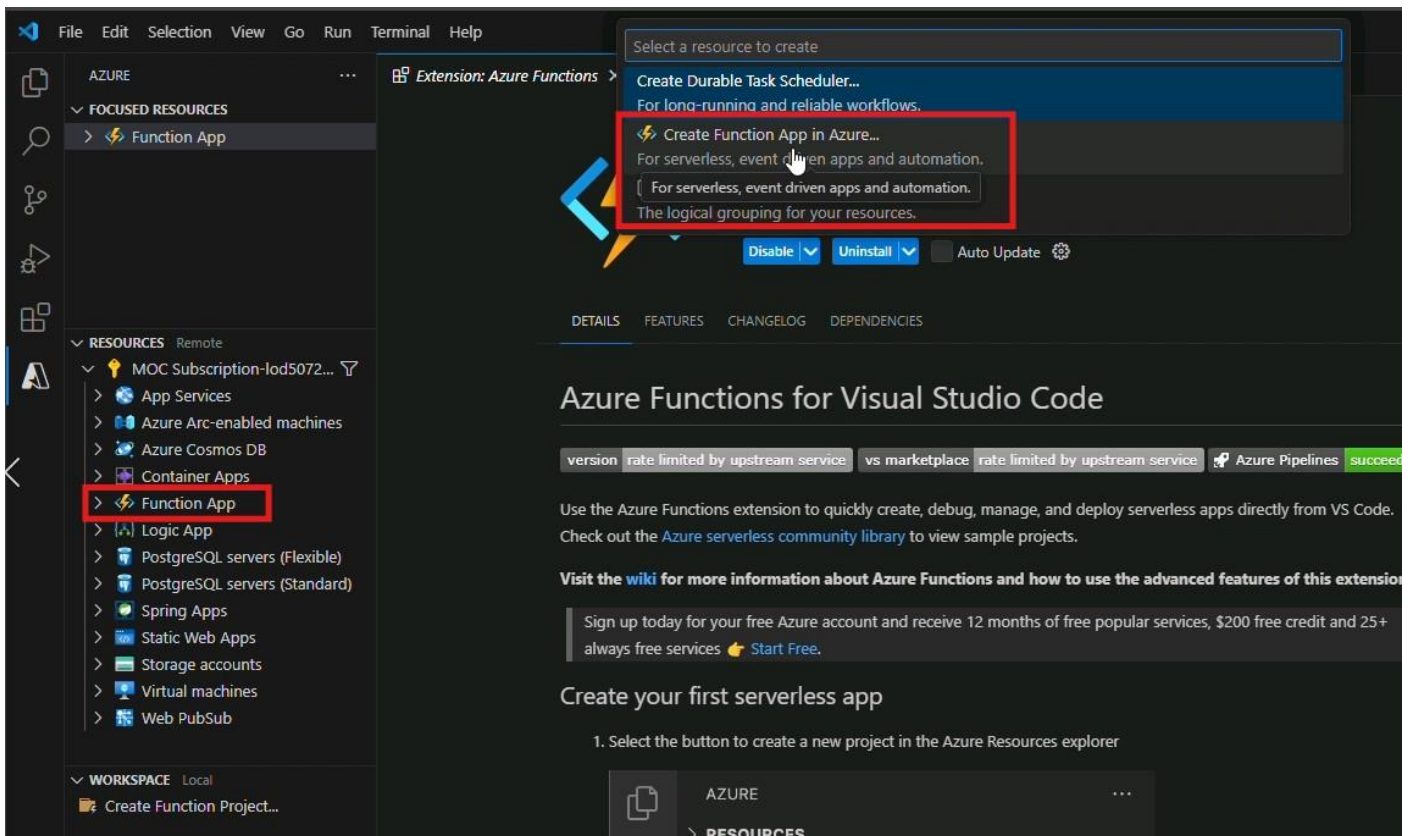
5. Setting up VS Code & Functions App.

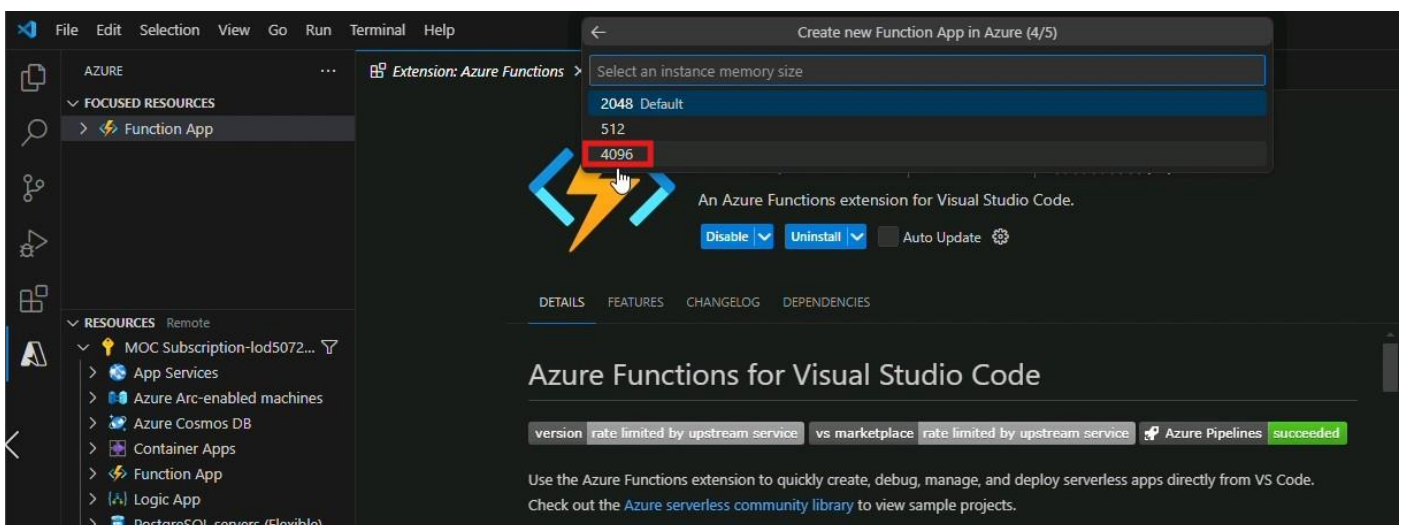
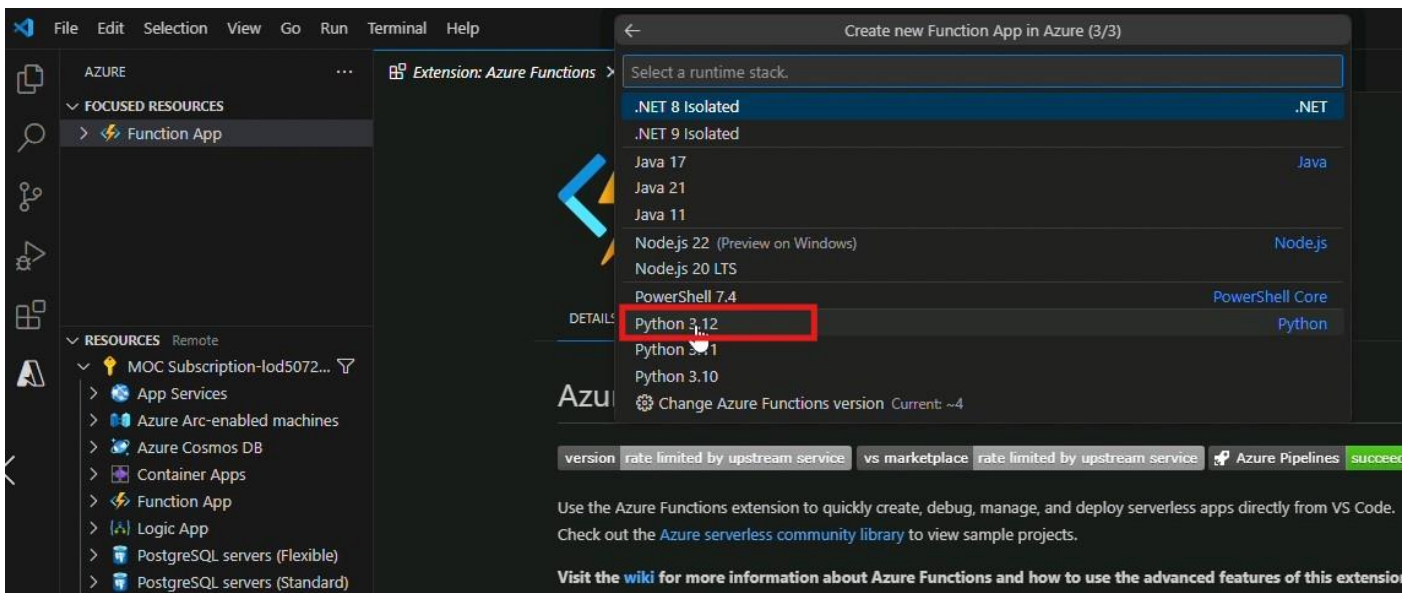
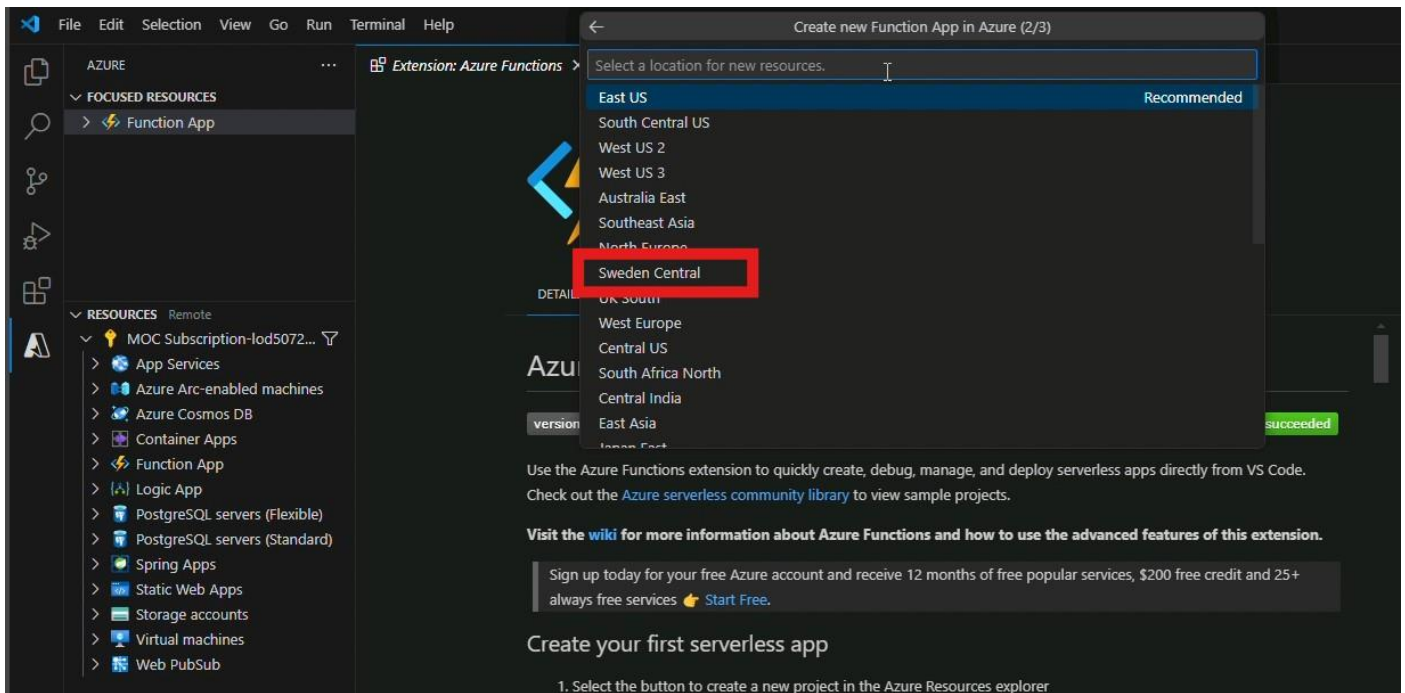


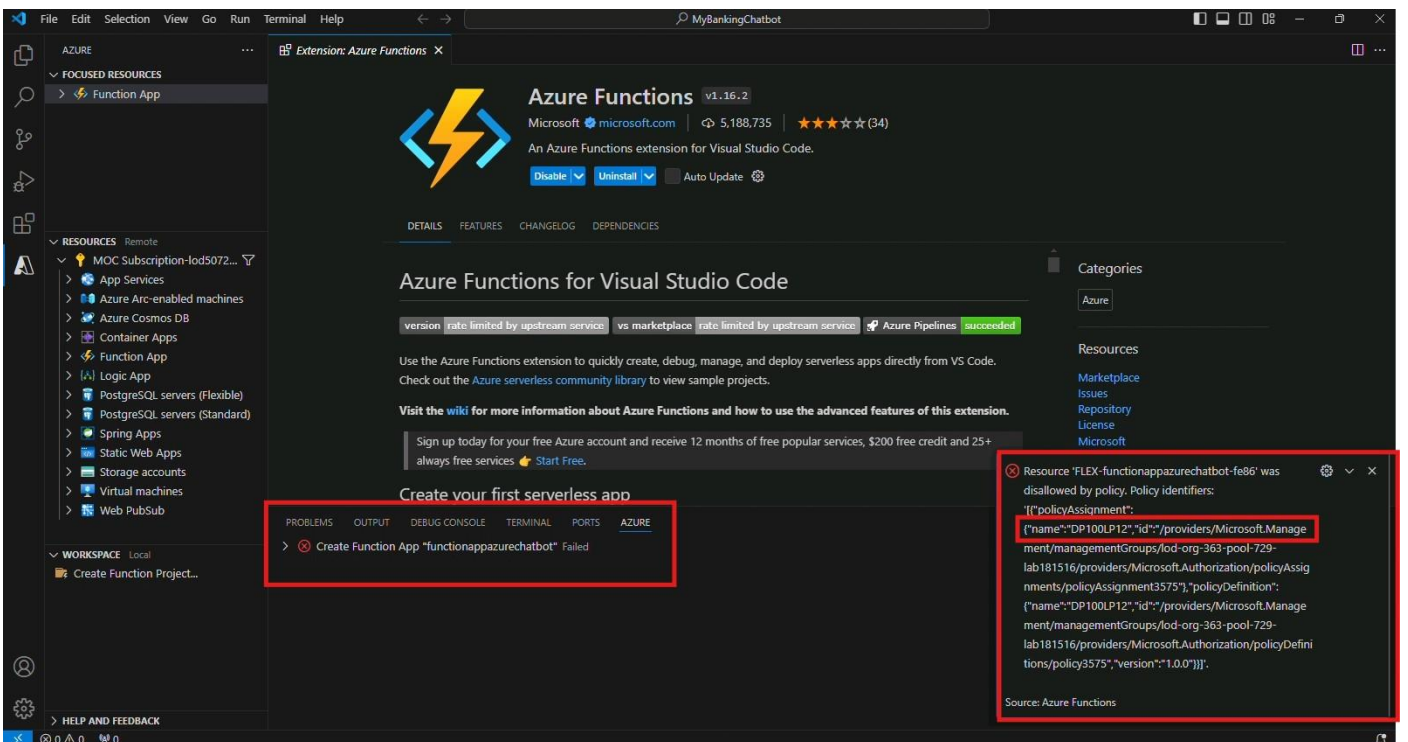
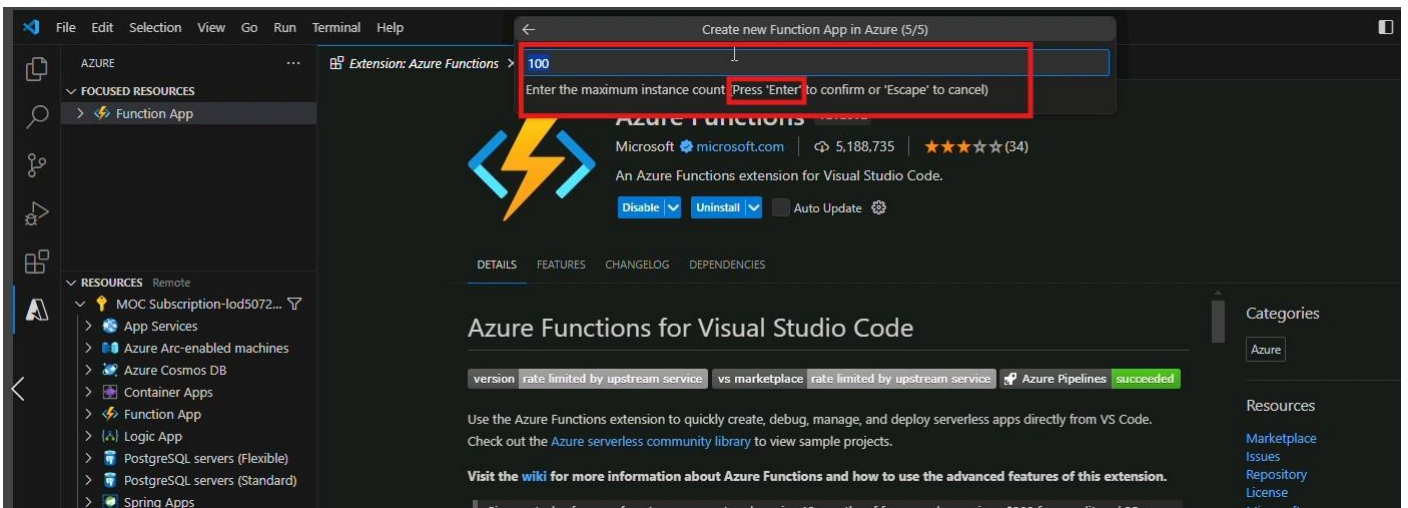






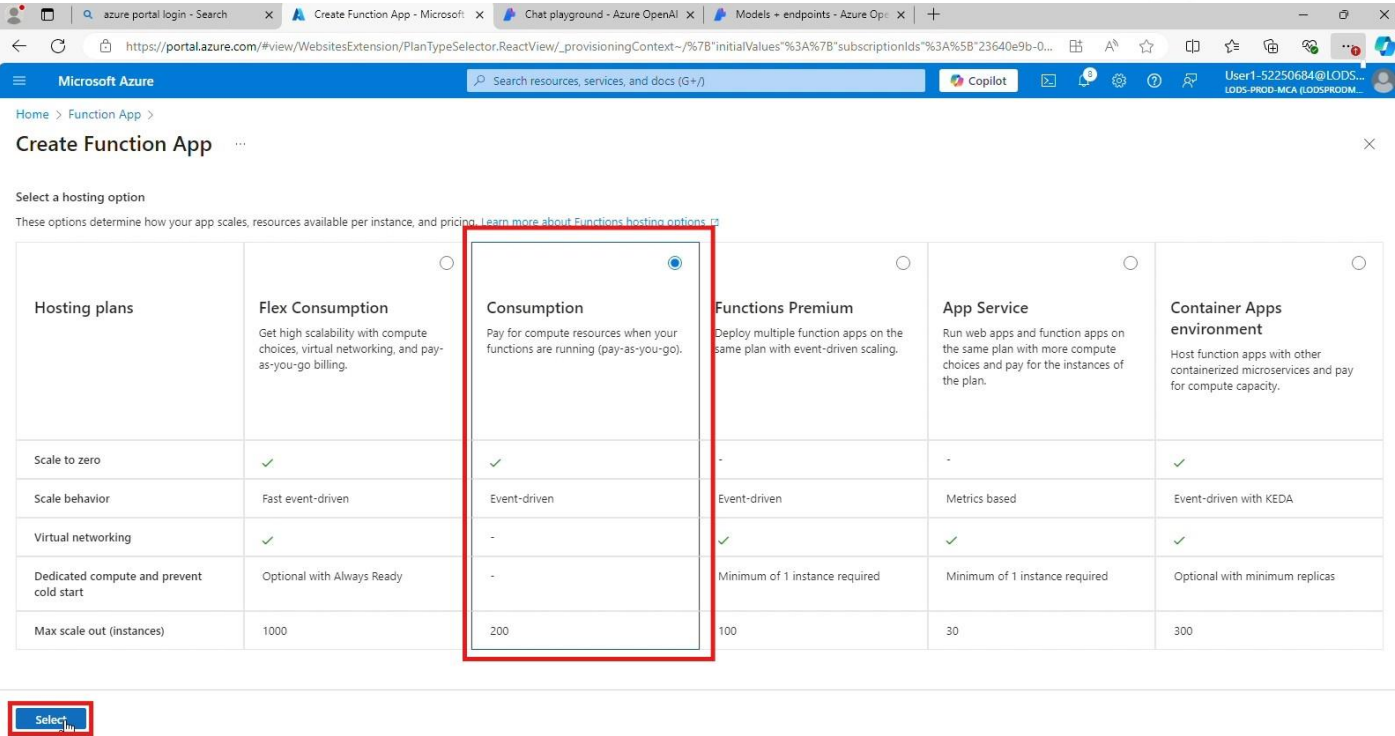
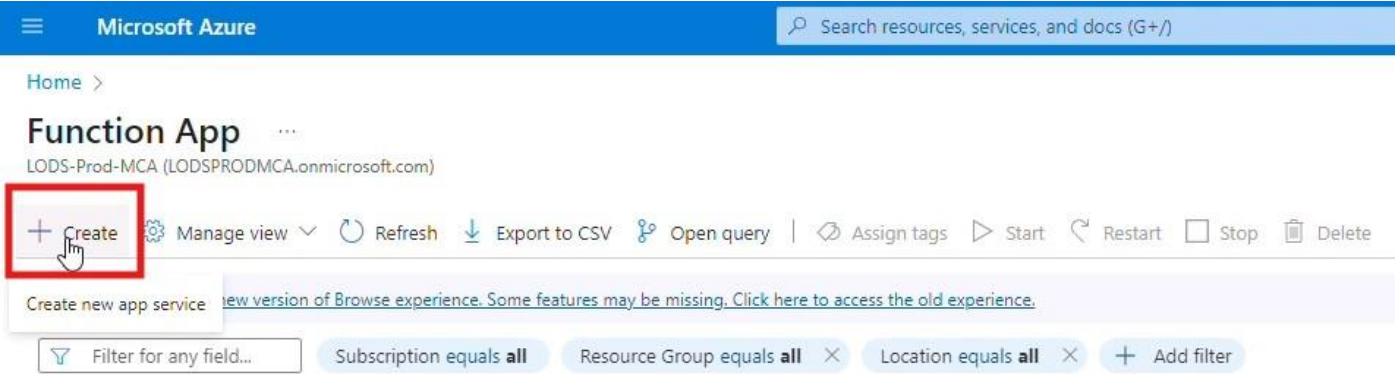
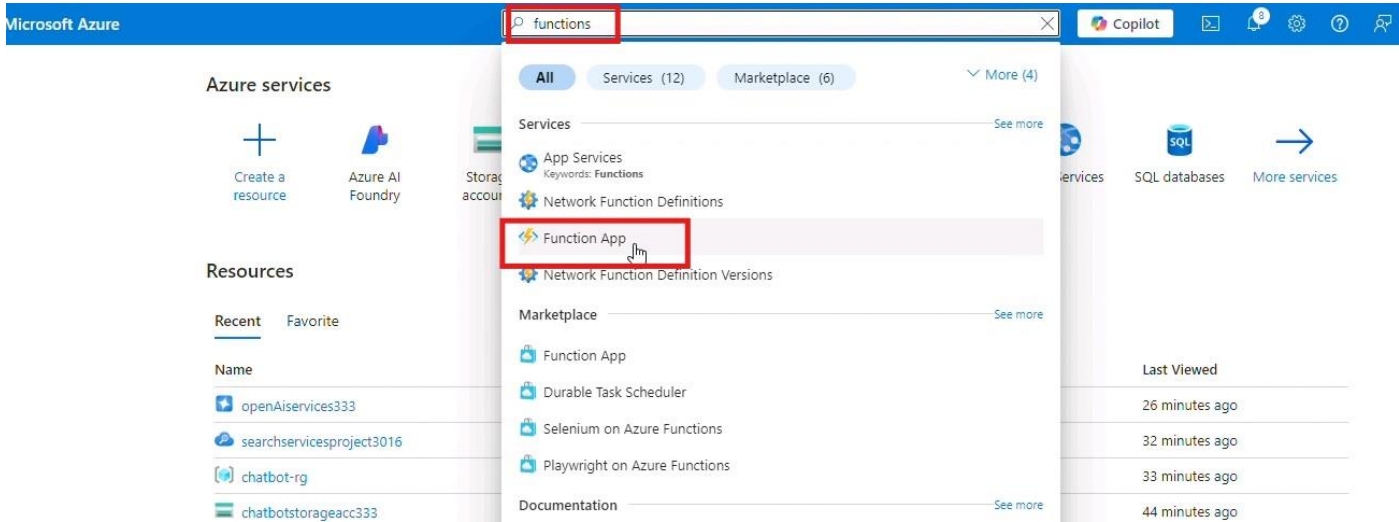






Since I was not able to deploy the function app through VS Code I'd have to deploy it in Azure itself so lets focus on that for now!!

Deploying the Function App through the portal



azure portal login - Search

Create Function App (Consumpti

Chat playground - Azure OpenAI

https://portal.azure.com/#view/WebsitesExtension/AppServiceFunctionAppCreateV3Blade/_provisioningConte

Microsoft Azure

Search resources, services, and docs (G+)

Home > Function App > Create Function App >

Create Function App (Consumption)

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

MOC Subscription-lod50723329

Resource Group *

chatbot-rg

Create new

Instance Details

Function App name *

chatbotfunctions3016

.azurewebsites.net

☐

Try a secure unique default hostname (preview). [More about this update](#)

Operating System *

☒ Linux (legacy) ☐ Windows

For Linux we recommend using Flex Consumption. [Learn more.](#)

Runtime stack *

Python

Version *

3.12

Region *

Sweden Central

Review + create

< Previous

Next : Save >

[Home](#) > [Function App](#) > [Create Function App](#) >

Create Function App (Consumption) ...

[Basics](#) [Storage](#) [Networking](#) [Monitoring](#) [Deployment](#) [Authentication](#) [Tags](#) [Review + create](#)

Storage

When creating a function app, you must create or link to a general-purpose Azure Storage account that supports Blobs, Queue, and Table storage. [Learn more](#) ↗

Storage account *

chatbotstorageacc333 (v2) ▼

[Create new](#)

📘 If you don't see a storage account, it may not be supported. [Storage account requirements](#)

Add an Azure Files connection



📘 Azure Files is used to enable certain features but you can create an app without one if you don't want to add another connection to your storage account. [What is Azure Files used for?](#)

Diagnostic Settings

The storage account associated with the function app stores important app data. You may wish to enable monitoring for the account. You can quickly configure basic diagnostic settings as you create the function app, or you can fully customize the diagnostics settings on the storage account resource after creation. [Learn more](#) ↗

Blob service diagnostic settings

- ☒ **Don't configure diagnostic settings now** You can configure diagnostic settings later from the storage account resource. Choose this if you want full control over log destinations, retention policies, and which logs and metrics are configured.
- ☐ **Configure basic diagnostic settings now** Configure Azure Log

[Review + create](#)

[< Previous](#)

[Next : Networking >](#)

azure portal login - Search

Create Function App (Consumpti

Chat playground - Azure OpenAI

https://portal.azure.com/#view/WebsitesExtension/AppServiceFunctionAppCreateV3Blade/_provisioningConte

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Function App > Create Function App >

Create Function App (Consumption) ...

Application Insights

Enabled

Name

chatbotfunctions3016

Region

Sweden Central

Deployment

Basic authentication

Disabled

Continuous deployment

Not enabled / Set up after app creation

Authentication

Host storage (AzureWebJobsStorage)

Name

chatbotstorageacc333

Role

Not applicable when using secrets

Azure Files

Name

chatbotstorageacc333

Role

Not applicable when using secrets

Application Insights

Name

chatbotfunctions3016

Role

Not applicable when using secrets

Create

< Previous

Next >

Download a template for automation

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

User1-52250684@L0DS...
L0DS-PROD-MCA (L0DSPROD...

Home > Function App > Create Function App >

Create Function App (Consumption)

The template deployment failed with multiple errors. Please see details for more information. →

Application Insights	Enabled
Name	chatbotfunctions3016
Region	Sweden Central

Deployment

Basic authentication	Disabled
Continuous deployment	Not enabled / Set up after app creation

Authentication

Host storage (AzureWebJobsStorage)

Name	chatbotstorageacc333
Role	Not applicable when using secrets

Azure Files

Name	chatbotstorageacc333
Role	Not applicable when using secrets

Application Insights

Name	chatbotfunctions3016
------	----------------------

[Create](#) [< Previous](#) [Next >](#) [Download a template for automation](#)

Errors

Summary Raw Error

ERROR DETAILS

Resource 'chatbotfunctions3016' was disallowed by policy. (Code: RequestDisallowedByPolicy)

Policy: DP100LP12

WAS THIS HELPFUL?

[Explain with Copilot](#)

Troubleshooting Options

[Common Azure deployment errors](#)

[Create ARM template](#)

[Check Usage + Quota](#)

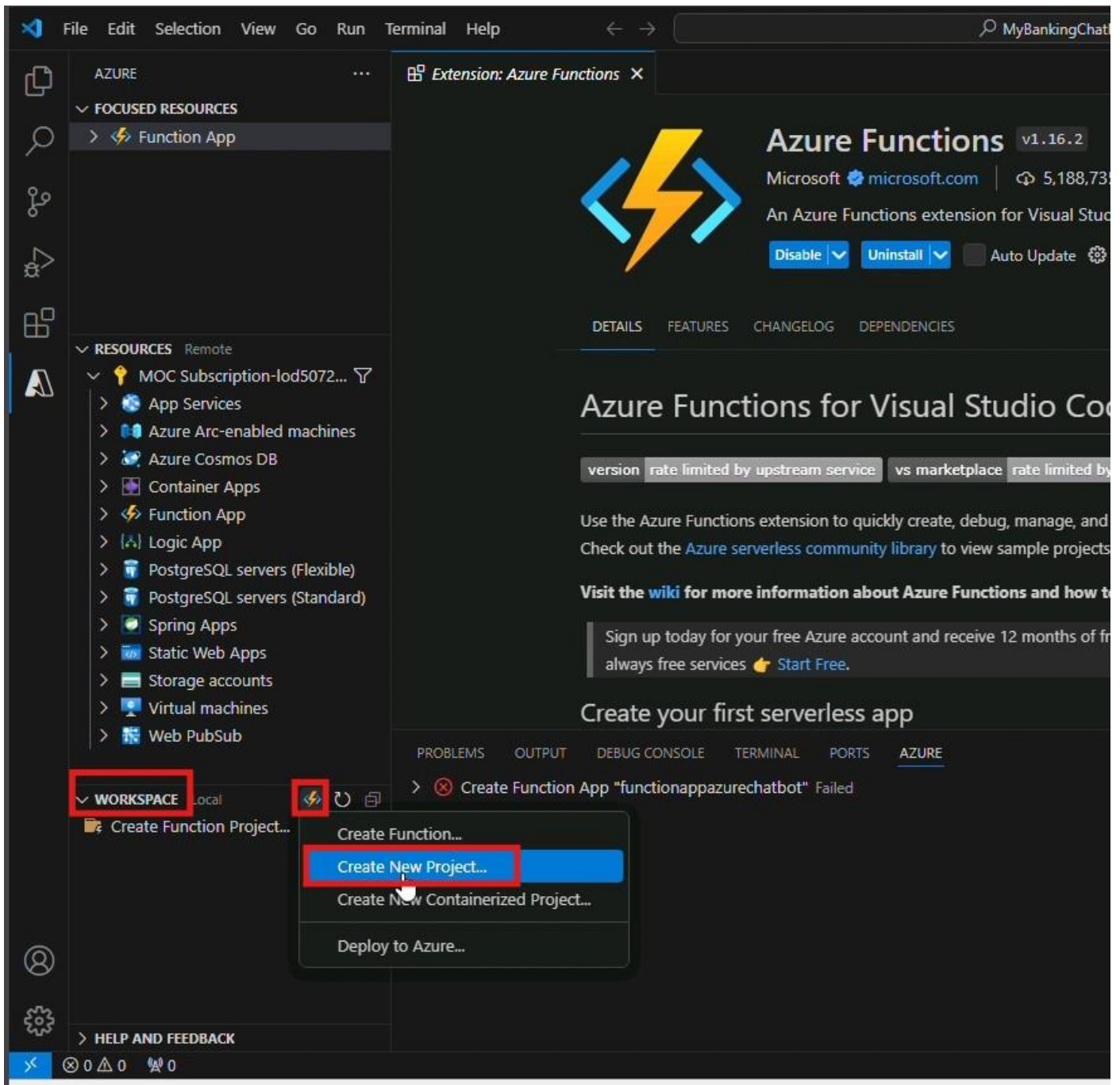
[New Support Request](#)

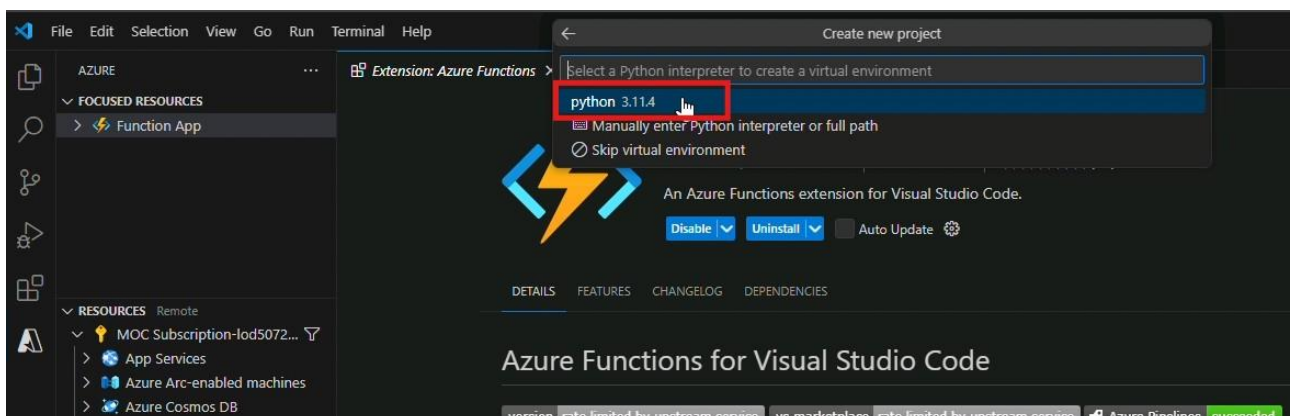
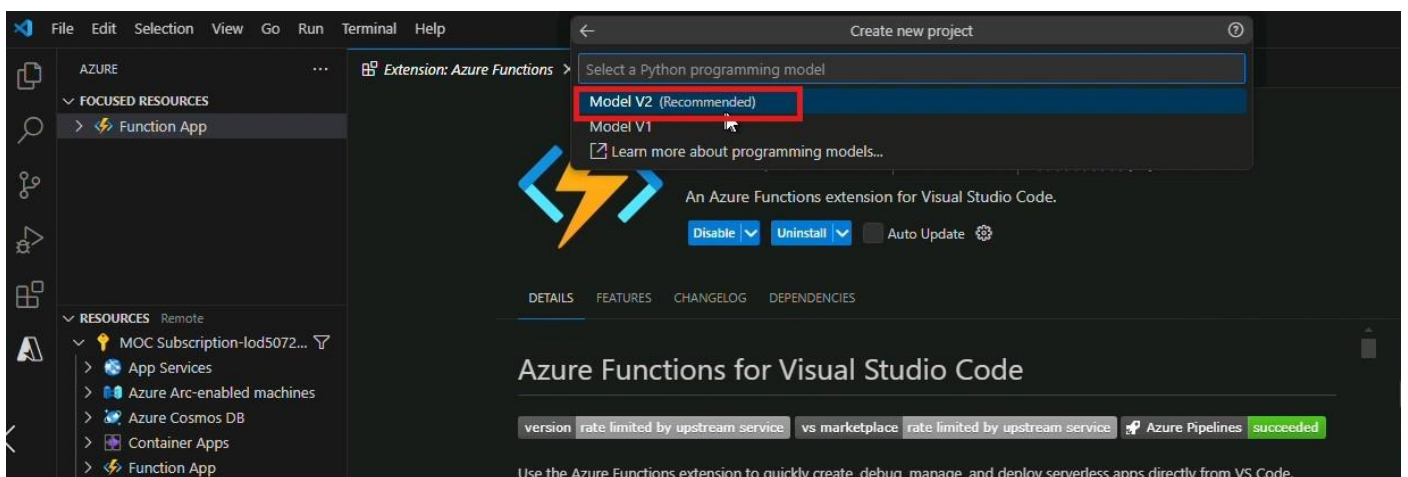
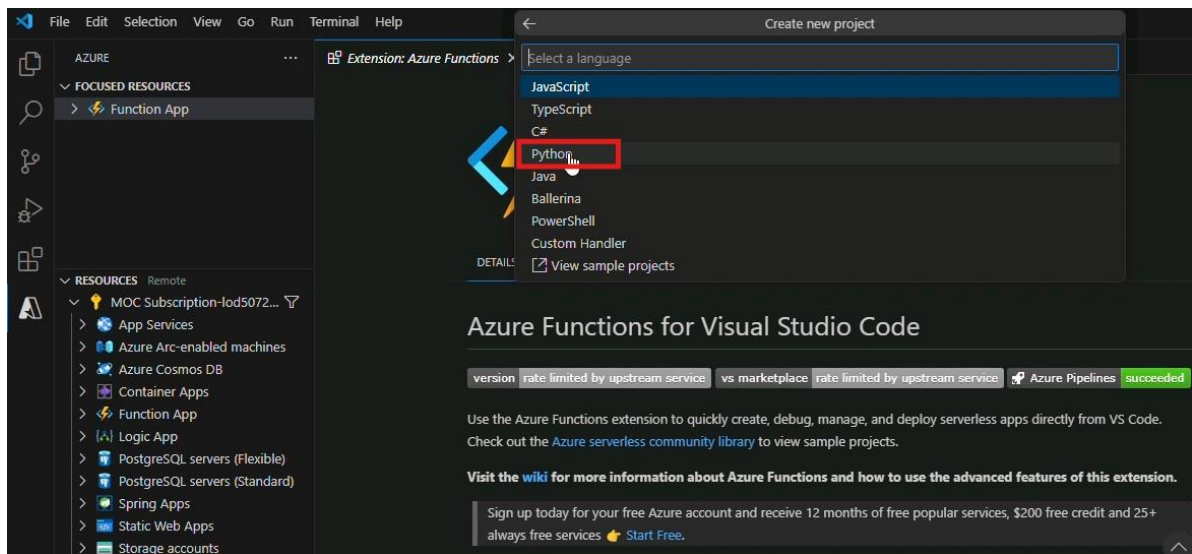
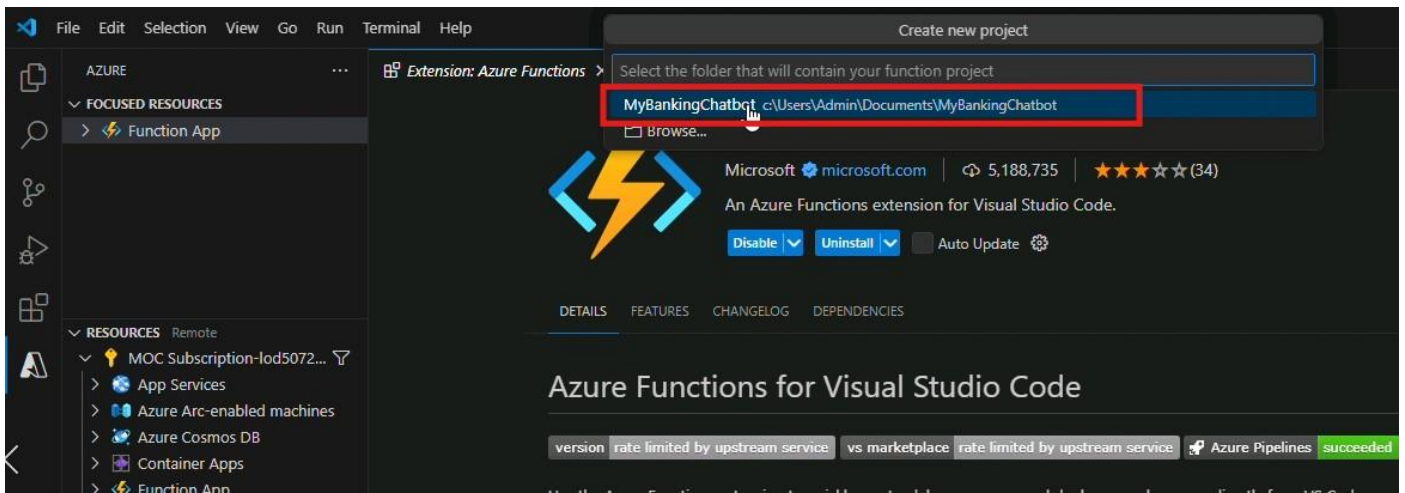
Give feedback

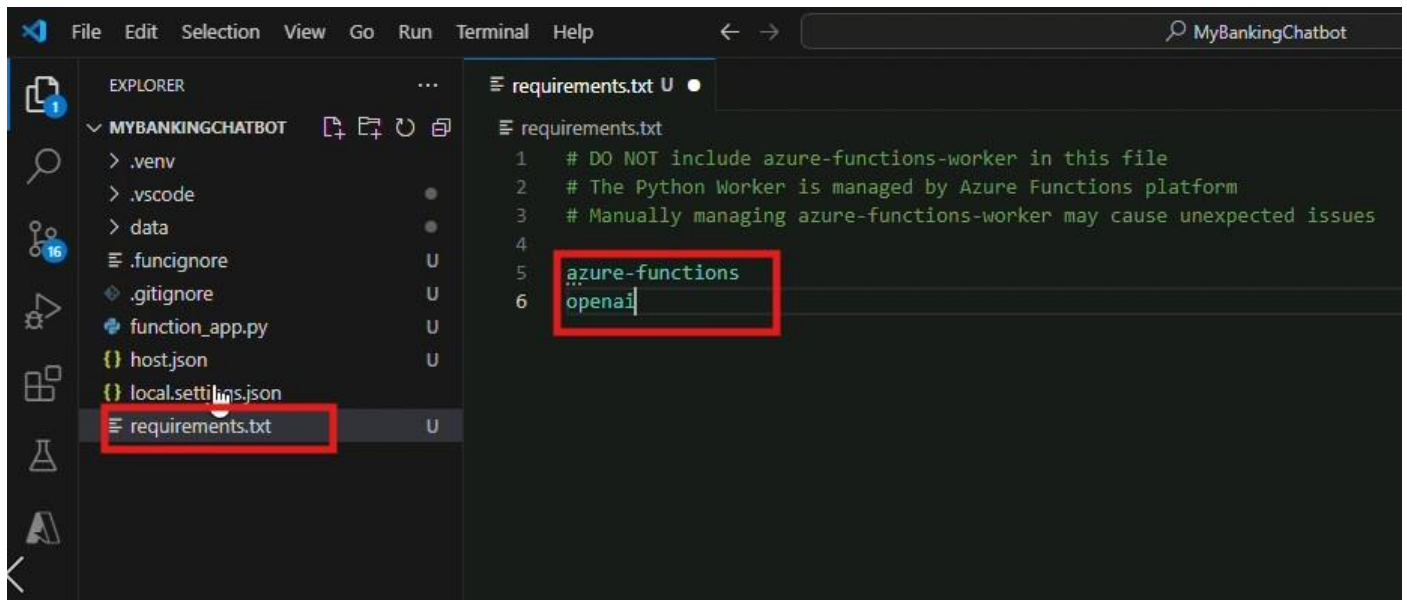
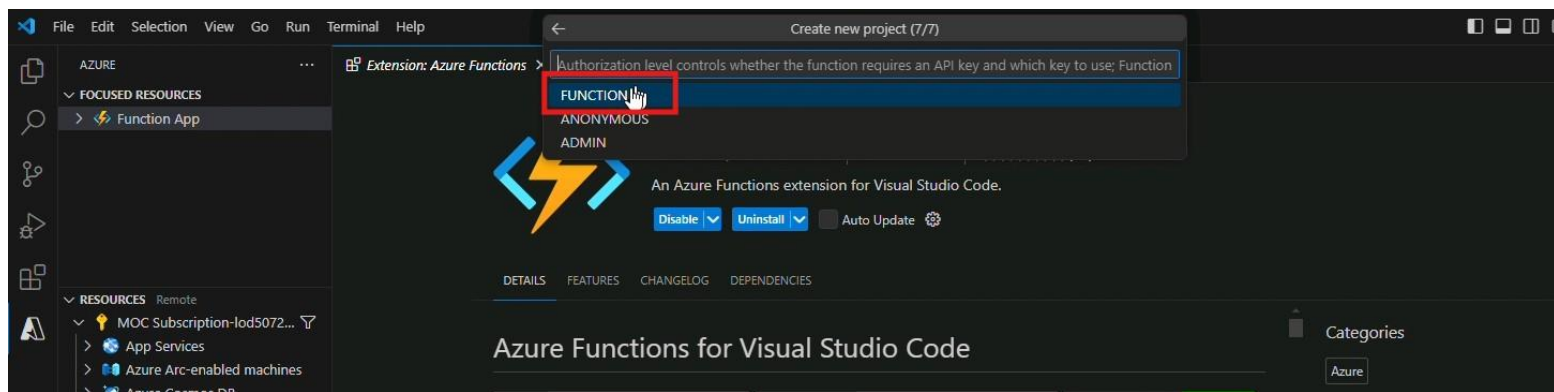
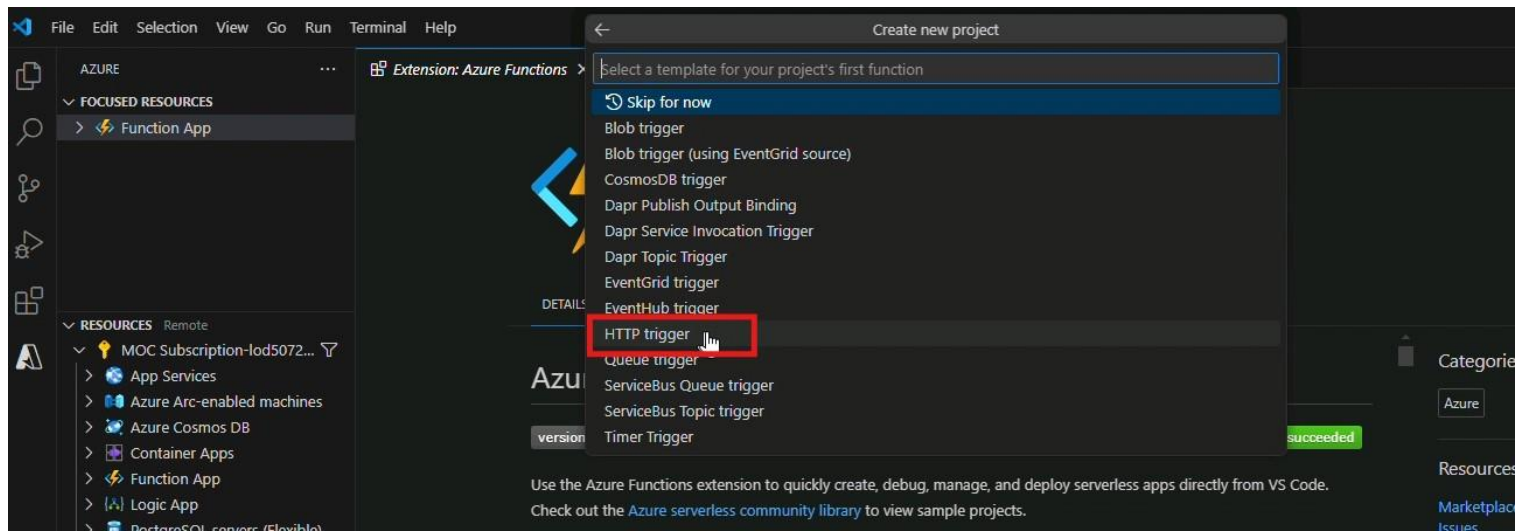
[Tell us about your experience with the ARM Errors page](#)

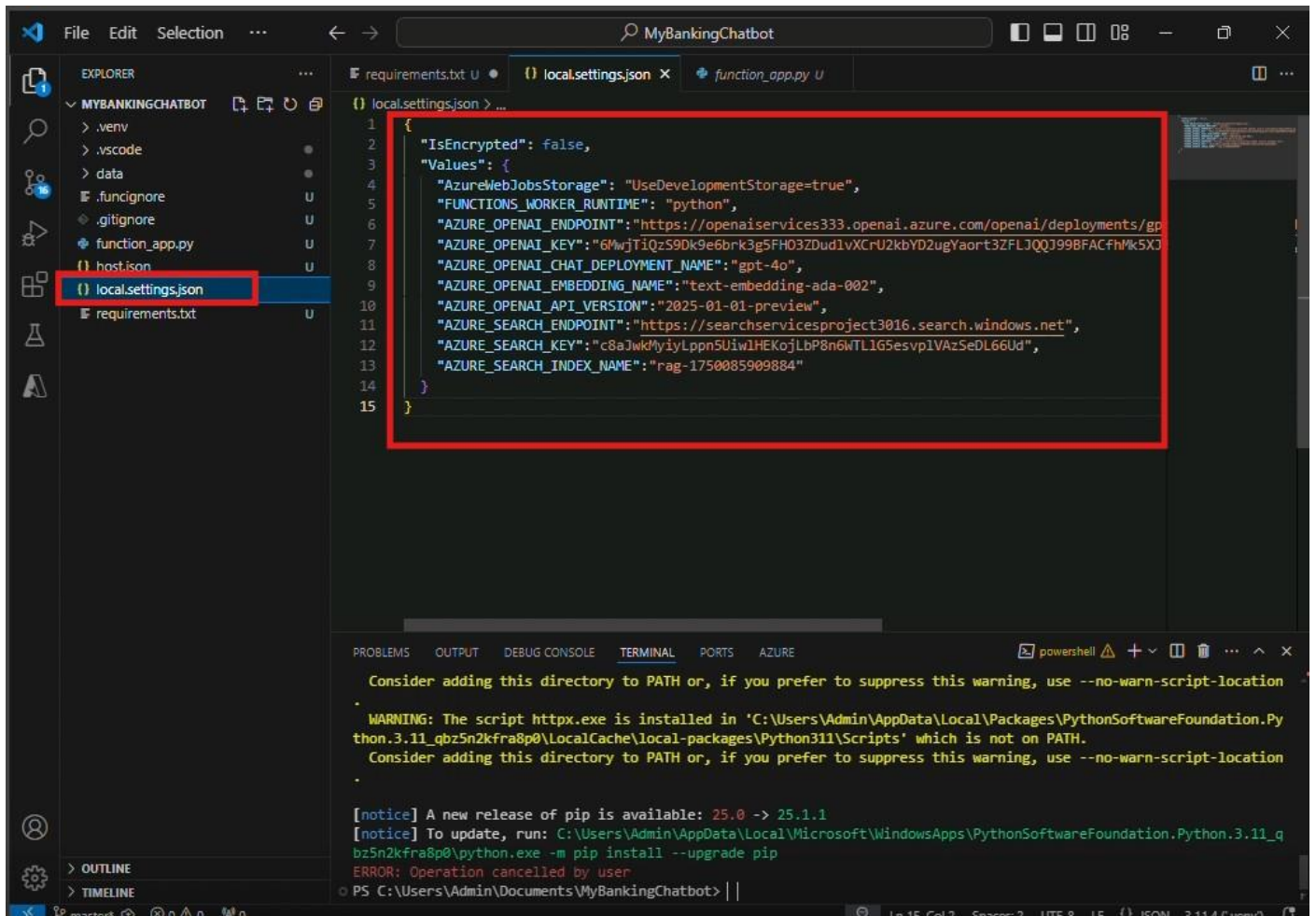
Since Both methods have been shoeing error we will not be able to provide a valid results for this problem statement for now hence we will deploy the function app folder locally and try.

6. Deploying the model.









MyBankingChatbot/

- ├─ .vscode/ # VS Code settings (automatically created)
- | └─ launch.json
- ├─ MyFunctionApp/ # Azure Function App project
- | └─ .vscode/ # (Function App specific VS Code settings - usually automatically created)
- | └─ .venv/ # Python virtual environment (will be created automatically)
- | └─ ChatbotAPI/ # specific HTTP trigger function folder
- | | └─ __init__.py # <--- MY MAIN FUNCTION CODE
- | | └─ function.json # (Automatically generated - do not modify unless advanced)
- | └─ host.json # (Automatically generated - do not modify unless advanced)
- | └─ local.settings.json # <--- LOCAL CREDENTIALS
- | └─ requirements.txt # <--- My Python package list
- | └─ ...other function app files (e.g., .funcignore, .gitignore)
- └─ faq_copilot.py # <--- TERMINAL CHATBOT CLIENT SCRIPT

The `__init__.py` python file contained these code :

```
import logging

import os

import json

import azure.functions as func

from openai import AzureOpenAI


# Initialize logging for the Azure Function
logging.basicConfig(level=logging.INFO)

logger = logging.getLogger(__name__)


def main(req: func.HttpRequest) -> func.HttpResponse:
    """
    Main entry point for the Azure Function.
    Handles HTTP POST requests to process user questions using Azure OpenAI with RAG.
    """

    logger.info('Python HTTP trigger function processed a request.')

    try:
        # Attempt to get JSON data from the request body
        req_body = req.get_json()

    except ValueError:
        # If the body is not valid JSON, return an error
        return func.HttpResponse(
            "Please pass a JSON object with a 'question' key in the request body.",
            mimetype="application/json",
            status_code=400
        )

    except Exception as e:
        logger.error(f"Error parsing request body: {e}")
        return func.HttpResponse(
            f"Error processing request: {e}",
            mimetype="application/json",
```

```

        status_code=500
    )

# Extract the 'question' from the request body
user_question = req_body.get('question')

# Validate if a question was provided
if not user_question:
    return func.HttpResponse(
        "Please pass a 'question' key within the JSON request body.",
        mimetype="application/json",
        status_code=400
    )

logger.info(f"Received question: {user_question}")

# --- Azure OpenAI Client Initialization ---
# These environment variables are loaded from local.settings.json (for local)
# or Application settings in Azure Portal (for deployed)
try:
    azure_openai_endpoint = os.environ["AZURE_OPENAI_ENDPOINT"]
    azure_openai_key = os.environ["AZURE_OPENAI_KEY"]
    azure_openai_chat_deployment = os.environ["AZURE_OPENAI_CHAT_DEPLOYMENT_NAME"]
    azure_openai_api_version = os.environ.get("AZURE_OPENAI_API_VERSION", "2024-02-15-preview") # Default if
not set

# Initialize the Azure OpenAI client
client = AzureOpenAI(
    azure_endpoint=azure_openai_endpoint,
    api_key=azure_openai_key,
    api_version=azure_openai_api_version
)

logger.info("AzureOpenAI client initialized.")

```



```
except KeyError as e:
```

```
    logger.error(f"Missing required environment variable: {e}")
```

```
    return func.HttpResponse(
```

```
        f"Server configuration error: Missing environment variable {e}. Please check your Function App settings.",
```

```
        mimetype="application/json",
```

```
        status_code=500
```

```
    )
```

```
except Exception as e:
```

```
    logger.error(f"Failed to initialize AzureOpenAI client: {e}")
```

```
    return func.HttpResponse(
```

```
        f"Error initializing AI service: {e}",
```

```
        mimetype="application/json",
```

```
        status_code=500
```

```
    )
```

```
# --- Chat Completions Request with RAG (On Your Data) ---
```

```
# Define the messages for the chat conversation.
```

```
# The system message provides context and instructions to the AI.
```

```
messages = [
```

```
    # System prompt: Crucial for guiding the AI's behavior and tone.
```

```
    # Make sure this aligns with your desired chatbot persona and instructions.
```

```
    {"role": "system", "content": "You are a helpful banking FAQ assistant. Answer questions concisely and only based on the provided context. If the answer is not in the context, state that you don't know and suggest visiting the bank's official website."},
```

```
    {"role": "user", "content": user_question}
```

```
]
```

```
# The `data_sources` configuration is what enables Retrieval Augmented Generation (RAG).
```

```
# This structure must exactly match what you get from Azure AI Studio's Chat Playground "View code" section.
```

```
# Replace the hardcoded values with os.environ calls to pull from your Function App's settings.
```

```
try:
```

```
    data_sources = [
```

```
        {
```

```
            "type": "azure_search",
```

```

"parameters": {
    # --- These values MUST come from your environment variables ---
    "endpoint": os.environ["AZURE_SEARCH_ENDPOINT"],
    "key": os.environ["AZURE_SEARCH_KEY"],
    "index_name": os.environ["AZURE_SEARCH_INDEX_NAME"],
    # --- Field mappings should match how your data is structured in Azure AI Search ---
    "fields_mapping": {
        "content_fields": ["answer_field_name_in_index"], # e.g., ["content"], ["chunks"]
        "title_field": "title_field_name_in_index",    # e.g., "title", "name"
        "url_field": "url_field_name_in_index",        # e.g., "url", "source_url"
        "filepath_field": "filepath_field_name_in_index" # e.g., "filepath", "filename"
    },
    # --- Embedding dependency for vector search (if using) ---
    "embedding_dependency": {
        "type": "deployment_name",
        "deployment_name": os.environ["AZURE_OPENAI_EMBEDDING_DEPLOYMENT_NAME"]
    },
    # --- Other parameters as copied from "View code" (adjust as per your playground settings) ---
    "query_type": "vector_semantic_hybrid", # or "vector", "hybrid", "semantic"
    "top_n_documents": 3,                   # Number of documents to retrieve
    "in_scope": True,                       # Restrict responses to retrieved content
    "strictness": 3,                        # Higher strictness means more relevance
    "role_information": "You are a helpful banking FAQ assistant designed to provide accurate information from official banking documents.", # Can be more specific than main system prompt
    # Add any other parameters you see in your 'View code' data_sources section
    # "filter": "some_field eq 'value'", # Example for pre-filtering
    }
}

]

logger.info("Data sources configured.")

except KeyError as e:
    logger.error(f"Missing environment variable for data sources: {e}")
    return func.HttpResponse(

```

```

        f"Server configuration error: Data source setting missing {e}. Please check your Function App settings.",
        mimetype="application/json",
        status_code=500
    )
except Exception as e:
    logger.error(f"Failed to configure data sources: {e}")
    return func.HttpResponse(
        f"Error configuring data sources: {e}",
        mimetype="application/json",
        status_code=500
    )

try:
    # Make the chat completions call to Azure OpenAI
    # The 'extra_body' is crucial for passing the 'data_sources' for RAG
    response = client.chat.completions.create(
        model=azure_openai_chat_deployment, # Your deployed chat model name
        messages=messages,
        extra_body={
            "data_sources": data_sources
        },
        temperature=0.7, # Controls randomness of output
        max_tokens=800 # Max length of the AI's response
    )
    logger.info("Azure OpenAI chat completion call successful.")

    # Extract the assistant's reply
    assistant_reply = response.choices[0].message.content

    # Extract citations (if available)
    citations = []

    if response.choices[0].message.context and response.choices[0].message.context.messages:
        for msg in response.choices[0].message.context.messages:
            if msg.role == "tool" and msg.content:

```

```

try:
    tool_content = json.loads(msg.content)
    if "citations" in tool_content and isinstance(tool_content["citations"], list):
        for citation in tool_content["citations"]:
            citations.append({
                "title": citation.get("title", "N/A"),
                "url": citation.get("url", "N/A"),
                "filepath": citation.get("filepath", "N/A"),
                "content": citation.get("content", "N/A")
            })
except json.JSONDecodeError:
    logger.warning("Could not decode tool message content as JSON for citations.")
except Exception as citation_err:
    logger.warning(f"Error processing citation: {citation_err}")

```

```

logger.info(f"Assistant reply generated. Citations found: {len(citations)}")

```

```

# Construct the response JSON

```

```

response_data = {
    "answer": assistant_reply,
    "citations": citations
}

```

```

# Return the response as JSON

```

```

return func.HttpResponse(
    json.dumps(response_data),
    mimetype="application/json",
    status_code=200
)

```

```

except Exception as e:

```

```

    logger.error(f"Error during chat completion: {e}", exc_info=True) # exc_info=True logs traceback
    return func.HttpResponse(
        f"An error occurred during AI processing: {e}",

```

```
mimetype="application/json",  
status_code=500  
)
```

Then my `faq_copilot.py` file has been a little altered as well :

since my function app deployment was unsuccessful there were some irregularities in the URL which should
have been given for this particular python to run

```
import requests
```

```
import json
```

```
import os
```

```
# --- IMPORTANT: URL ---
```

```
# Get this URL from your Azure Portal:
```

```
# Go to your deployed Function App -> Functions -> ChatbotAPI -> "Get Function URL"
```

```
# Copy the entire URL including the '?code=YOUR_KEY' part.
```

```
DEPLOYED_FUNCTION_URL_WITH_KEY = "YOUR_FUNCTION_APP_URL_WITH_KEY_HERE"
```

```
def get_chatbot_response(question: str) -> dict:
```

```
    """
```

```
    Sends a question to the deployed Azure Function App and returns the response.
```

```
    """
```

```
    if not DEPLOYED_FUNCTION_URL_WITH_KEY or "YOUR_FUNCTION_APP_URL_WITH_KEY_HERE" in  
    DEPLOYED_FUNCTION_URL_WITH_KEY:
```

```
        print("ERROR: Please update 'DEPLOYED_FUNCTION_URL_WITH_KEY' in faq_copilot.py with your actual  
        Function App URL.")
```

```
        return {"answer": "Configuration Error.", "citations": []}
```

```
    headers = {"Content-Type": "application/json"}
```

```
    payload = {"question": question}
```


try:

Make a POST request to your deployed Azure Function

response = requests.post(DEPLOYED_FUNCTION_URL_WITH_KEY, headers=headers, json=payload)

response.raise_for_status() # Raise an HTTPError for bad responses (4xx or 5xx)

response_data = response.json()

return response_data

except requests.exceptions.HTTPError as errh:

print(f"HTTP Error: {errh}")

print(f"Response Content: {response.text}")

return {"answer": f"Error communicating with chatbot: HTTP Error: {errh}", "citations": []}

except requests.exceptions.ConnectionError as errc:

print(f"Error Connecting: {errc}")

return {"answer": f"Error communicating with chatbot: Connection Error: {errc}", "citations": []}

except requests.exceptions.Timeout as errt:

print(f"Timeout Error: {errt}")

return {"answer": f"Error communicating with chatbot: Timeout Error: {errt}", "citations": []}

except requests.exceptions.RequestException as err:

print(f"Oops: Something Else {err}")

return {"answer": f"An unexpected error occurred: {err}", "citations": []}

except json.JSONDecodeError as e:

print(f"Error decoding JSON response: {e}")

print(f"Raw response: {response.text}")

return {"answer": f"Invalid response from chatbot: {e}", "citations": []}

def run_chatbot_terminal():

"""

Runs the chatbot in an interactive terminal session.

"""

print("\n--- Banking FAQ Chatbot (Type 'exit' to quit) ---")

while True:

user_input = input("\nYou: ")

```
if user_input.lower() == 'exit':
```

```
    print("Exiting chatbot. Goodbye!")
```

```
    break
```

```
print("Chatbot: Thinking...")
```

```
response = get_chatbot_response(user_input)
```

```
# Display the answer
```

```
print(f"\nAnswer: {response.get('answer', 'No answer received.')}")
```

```
# Display citations if available
```

```
citations = response.get('citations', [])
```

```
if citations:
```

```
    print("\nSources:")
```

```
    for i, citation in enumerate(citations):
```

```
        title = citation.get('title', 'N/A')
```

```
        url = citation.get('url', 'N/A')
```

```
        filepath = citation.get('filepath', 'N/A') # Or 'filename'
```

```
        print(f" {i+1}. Title: {title}")
```

```
        if url != 'N/A':
```

```
            print(f"    URL: {url}")
```

```
        elif filepath != 'N/A': # Show filepath if URL is not available
```

```
            print(f"    File: {filepath}")
```

```
        # You can also print citation.get('content') if you want to see the chunk of text
```

```
print("-" * 50) # Separator for readability
```

```
if __name__ == "__main__":
```

```
    # Install the 'requests' library if not already installed
```

```
    try:
```

```
        import requests
```

```
    except ImportError:
```

```
        print("The 'requests' library is not installed. Installing now...")
```

```
        os.system("pip install requests")
```

```
import requests # Try importing again
```

```
run_chatbot_terminal()
```

Thank You